

Virtual Network Embedding: Literature Assessment, Recent Advancements, Opportunities, and Challenges

Anurag Satpathy, Manmath Narayan Sahoo^{ID}, *Member, IEEE*, Chittaranjan Swain^{ID},
Paolo Bellavista^{ID}, *Senior Member, IEEE*, Mohsen Guizani^{ID}, *Fellow, IEEE*,
Khan Muhammad^{ID}, *Senior Member, IEEE*, and Sambit Bakshi^{ID}, *Senior Member, IEEE*

Abstract—Network virtualization (NV) allows service providers (SPs) to instantiate logically isolated entities called virtual networks (VNs) on top of a substrate network (SN). Though VNs bring about multiple benefits, particularly in terms of economic costs and elasticity, they also force various technical challenges to be addressed. The primary one is the issue of optimally allocating resources to VNs, also termed virtual network embedding (VNE). This paper presents an exhaustive survey of VNE by extensively covering the state-of-the-art research field in this very active field and focusing on the emerging research trends in industry and academia over the last decade. In addition, this survey originally contributes to the literature by proposing a novel taxonomy for existing VNE solutions and providing a thorough comparative study of their strategies.

Index Terms—Virtual networks, embedding strategies, virtual machines, virtual links.

I. INTRODUCTION

TRADITIONALLY, substrate networks (SNs), e.g., the ones used within traditional cloud data centers (DCs), have provided the cost-effective networking infrastructure for processing, storing considerable volumes of data, and hosting large-scale services [1], [2]. Multiple service providers (SPs),

such as Amazon [3], Google [4], Microsoft Azure [5], and Facebook [6], have set up multi-domain SNs to support different user services such as storage, web search, large-scale computations, and social networking. In its infancy, the DCs used dedicated physical servers to host user applications, resulting in poor resource utilization and higher operational expenditures for the SPs. However, server virtualization technologies such as VMware [7] and Xen [8] acted as game-changers. Server virtualization allowed multiple independent executable entities called virtual machines (VMs), hosting different services to be co-located on the same physical server host [9], [10], [11].

A virtualized server can host multiple VMs, execute separate applications, and have a dedicated operating system. The virtual machine monitor (VMM), or hypervisor, facilitates the creation and operation of VMs over physical hardware. Though VM-based services overcame the pitfalls of traditional DC architectures, they had the following pitfalls.

- *Limited performance isolation:* Modern applications such as search engines, online streaming, and REST-accessible services often have stringent requirements on acceptable latency and throughput. However, VM-based services provide best-effort delivery with no guarantee of performance isolation. Therefore, delivering quality-of-service (QoS) guarantees for applications based on such delivery models is challenging [12].
- *Increased security risks:* VM-based services do not restrict the communication patterns and bandwidth between applications hosted on VMs. This renders the underlying DC network vulnerable to insider attacks such as performance interference and denial of service (DoS) attacks [13].
- *Limited management flexibility:* Applications hosted on VMs require computation and networking resources of the underlying network. However, the VM-based services do not provide any authority to application owners to control the network fabric that application owners often use for load balancing, fault diagnosis, and security protection.
- *No support for network innovation:* VM-based services do not give application owners any network control. This restriction prohibits the owners from testing, deploying, and upgrading network protocols and introducing new networking services.

Received 16 July 2024; revised 27 November 2024; accepted 11 January 2025. Date of publication 20 January 2025; date of current version 19 December 2025. This work was supported in part by the Vishlesan I-Hub Foundation – Indian Institute of Technology Patna under NMICPS, Government of India, and in part by the Italian National PRIN Project JOULE. (Corresponding authors: Khan Muhammad; Paolo Bellavista.)

Anurag Satpathy is with the Department of Computer Science, Missouri University of Science and Technology, Rolla, MO 65409 USA (e-mail: anurag.satpathy@mst.edu).

Manmath Narayan Sahoo and Sambit Bakshi are with the Visual Surveillance Laboratory, Department of Computer Science and Engineering, National Institute of Technology at Rourkela, Rourkela 769008, India (e-mail: sahoom@nitrrkl.ac.in; bakshisambit@ieee.org).

Chittaranjan Swain is with the Department of Computer Science and Engineering, Atal Bihari Vajpayee Indian Institute of Information Technology and Management, Gwalior 474015, India (e-mail: cswain@iiitm.ac.in).

Paolo Bellavista is with the Department of Computer Science and Engineering, University of Bologna, 40136 Bologna, Italy (e-mail: paolo.bellavista@unibo.it).

Mohsen Guizani is with the Machine Learning Department, Mohamed Bin Zayed University of Artificial Intelligence, Abu Dhabi, UAE (e-mail: mguizani@ieee.org).

Khan Muhammad is with the Visual Analytics for Knowledge Laboratory, Department of Applied Artificial Intelligence, School of Convergence, College of Computing and Informatics, Sungkyunkwan University, Seoul 03063, Republic of Korea (e-mail: khan.muhammad@ieee.org).

Digital Object Identifier 10.1109/COMST.2025.3531724

TABLE I
COMPARISON OF EXISTING SURVEYS ON VNE BASED ON SOLUTION TYPES AND DEPLOYMENT ENVIRONMENTS

References	Solution Type										Deployment Environment				
	Branch and Bound	Exact		Approximate						ML & DL	DC Networks	Wireless Networks	Elastic Optical Networks	Satellite Networks	IoT's and IoVs
		Exact Exponentiation	Game Theoretic	Parameter Driven Heuristic	Greedy	Local Search	Randomized Rounding	Meta Heuristic	Primal Dual						
Belbekkouche <i>et al.</i> (2012) [14]	×	×	×	×	×	×	×	×	×	×	✓	✓	×	×	×
Fischer <i>et al.</i> (2013) [15]	×	✓	×	✓	×	×	×	✓	×	×	✓	✓	✓	×	×
Cao <i>et al.</i> (2016) [16]	✓	✓	×	×	×	×	×	×	×	×	✓	×	✓	×	×
Yu <i>et al.</i> (2016) [17]	×	×	×	×	×	×	×	×	×	×	×	×	✓	×	×
Cao <i>et al.</i> (2018) [18]	×	×	×	✓	×	×	×	×	×	×	✓	×	×	×	×
Cao <i>et al.</i> (2019) [19]	×	×	×	×	×	×	×	×	×	×	✓	✓	✓	×	×
Jiang <i>et al.</i> (2024) [20]	×	×	✓	×	×	×	×	×	×	×	×	×	×	✓	×
Wu <i>et al.</i> (2024) [21]	×	×	×	×	×	×	×	×	×	✓	✓	✓	×	×	×
This Work (2025)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

These significant gaps have forced DC owners to look at network virtualization (NV) in conjunction and synergy with server virtualization [22], [23], [24], [25] to provide flexible, secure, and innovative services for modern, distributed, and large-scale applications [1], [26], [27], [28], [29]. NV promises to allow the SPs to create multiple virtual networks (VNs) on top of multi-domain SNs [30], [31], [32], [33], [34], [35], [36]; it overcomes the shortcomings of VM-based services by:

- Providing complete *service isolation* in the form of VN.
- Logically separating VNs and facilitating the SPs in achieving the *QoS* demands in terms of *latency* and *network throughput*.
- *Controlling and managing* the communication between the VMs, thereby reducing security threats.
- Logically isolating the networking resources and assisting the users in implementing their customized security protocols, thereby aiding *innovation*.

Typically, a VNR, as illustrated in Figure 1, comprises multiple interactive VMs. The interaction between the VMs is captured in the form of VLs [37], [38], [39], [40]. Virtualizing physical servers and links to multiple applications with a wide variety of topological configurations can be co-hosted on the same SN. In this regard, the main challenge for the SPs is to effectively assign resources to the virtual components of VNRs, i.e., VMs and VLs, as illustrated in Figure 2 is referred to as virtual network embedding (VNE) [41]. Although NV can effectively utilize the SN resources, optimal resource allocation is crucial in providing end-to-end user services. Resource allocation can be performed from different aspects such as QoS, energy efficiency, resource utilization, sustainability, economic gains, survivability, and security. The complex and articulated VNE problem can be structured into two closely related sub-problems. Firstly, the VMs corresponding to the VN will be assigned substrate servers called virtual machine embedding (VME). Secondly, the VLs are assigned to substrate paths connecting the corresponding VMs, termed virtual link embedding (VLE) [42], [43]. The optimization of VME and VLE has proven to be a computationally intractable problem [44], [45], [46], [47], [48].

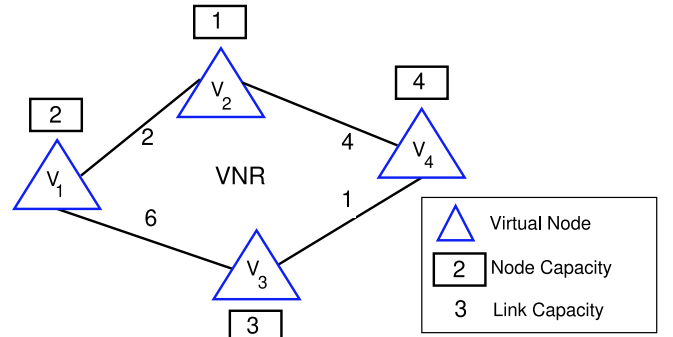


Fig. 1. A virtual network request (VNR) with VMs and virtual links (VLs) and their respective resource demands. The value 2 of v_1 represents the resource demand of the VM, whereas the number 2 between v_1 and v_2 indicates the bandwidth demands for a VNR.

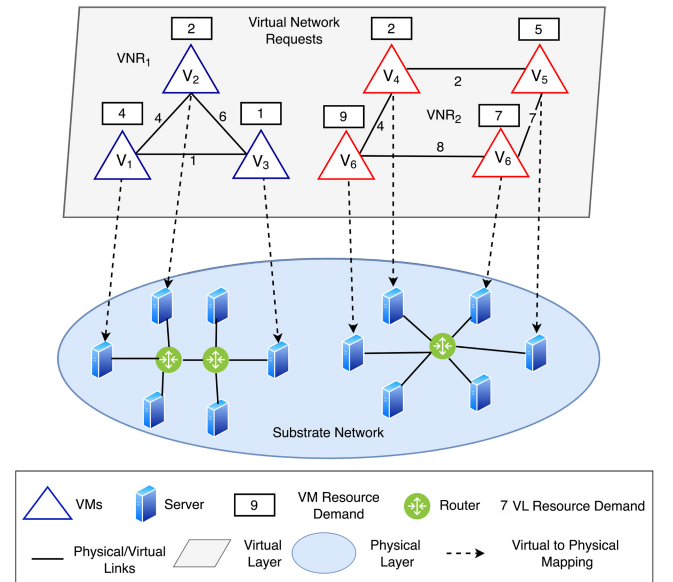


Fig. 2. An illustration of VNE.

A. Comparison With Existing Surveys

Table I presents various VNE surveys conducted in the past decade, highlighting the novelty of our survey. In this

TABLE II
DIFFERENCE BETWEEN SURVEYS FOCUSED ON SDN, VNF, AND NV

Technologies	Purpose	Focus Area	Research Challenges
NV	Resource Sharing over SN	Resource optimization	Resource management and performance overhead
SDN	Network control and programmability	Control plane separation	Complexity, scalability, and integration
VNF	Network functions as dedicated hardware	Virtualize network functions	Orchestration complexity

regard, this survey presents a novel classification based on the adopted solution strategy, with one approach being *exact* methods, which offer optimal solutions for small-scale problem instances but are compute-intensive, time-consuming, and non-scalable for more extensive cases. The alternative category is based on *approximation* techniques to overcome the abovementioned limitations to generate satisfactory solutions within reasonable time constraints. Belbekkouche et al. [14] offered an early survey on resource discovery and allocation mechanisms in virtualized environments. Surveys by Cao et al. [16] and Cao et al. [18] focused on exact methods and heuristics for VNE. Yu et al. [17] and Jiang et al. [20], respectively explored VNE strategies in optical and satellite networks. While Fischer et al. [15] and Correa et al. [49] conducted thorough reviews, their findings did not encompass the most recent advancements in the field. A survey by Cao et al. [19] addressed some of these new embedding strategies. Additionally, a recent study by Wu et al. [21] focused exclusively on AI-powered VNE schemes, overlooking significant developments in non-AI-powered solutions, which remain crucial in VNE.

Although initially similar in functionality, technologies such as NV, software-defined networking (SDN), and network function virtualization (NFV) play disparate roles in network management. SDN enhances network control and management through centralized, software-based control, facilitating dynamic and programmable networking. In contrast, NV efficiently allocates SN resources to user requests represented as VNRs, optimizing resource utilization in shared multi-tenant environments. On the other hand, NFV virtualizes network functions to operate on standard hardware, enabling SPs to deploy and manage network services more flexibly and efficiently. While these technologies share foundational principles, they diverge significantly in practice, as Table II summarizes. Surveys on SDN [50], [51], [52] and NFV [53], [54], [55] have investigated their respective research challenges, proposed solutions, and identified critical areas for further exploration, highlighting their complementary nature. However, these surveys primarily focus on challenges specific to their respective domains and do not extensively explore the intricacies of VNE, specifically optimizing resource assignments and re-assignments.

This survey paper has the ambition of going largely beyond all the existing surveys by providing the readers with a comprehensive view of the up-to-date state-of-the-art strategies, including recent significant strategies based on machine learning (ML), deep learning (DL), and more generally any artificial intelligence (AI)-based VNE solutions [56], [57],

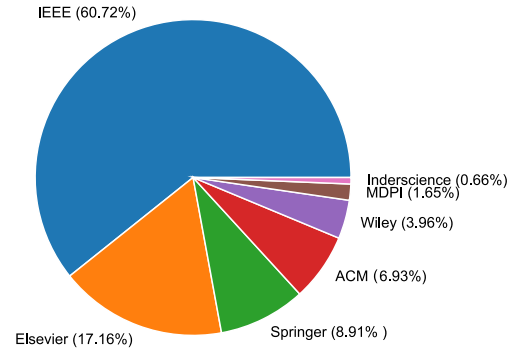


Fig. 3. Publisher-wise distribution of reviewed literature on VNE.

[58], [59]. This survey paper is highly novel and potentially impactful for several reasons: (1) it provides a comprehensive exploration of available simulation and emulation platforms, highlighting their advantages and disadvantages; (2) it discusses key enablers in both hardware and software; (3) it draws parallels between contemporary technologies like service chaining and network slicing with VNE; (4) it includes a generic mathematical formulation to aid readers' understanding and familiarize them with VNE; (5) it proposes a new taxonomy for classifying recent VNE solutions in the literature; (6) it offers detailed state-of-the-art analysis with explanations of exemplary solutions and sample working instances, emphasizing their pros and cons; (7) it presents a thorough comparative analysis of current VNE solutions; and (8) it highlights the most promising emerging directions for future research.

B. Survey Methodology

This article surveys the literature over the last decade, i.e., 2013-2024 on VNE. The articles discussed in this survey were identified using keywords such as *virtual network embedding*, *virtual data center embedding*, *virtual networks*, *virtual data centers*, and *embedding*. Most retrieved articles were directly relevant, and old articles, i.e., before 2013, were excluded as they were already reviewed in [15]. Additionally, these keywords were searched over various digital libraries such as *IEEE Xplore*, *ScienceDirect (Elsevier)*, *Wiley Online Library*, *ACM Digital Library*, *Taylor and Francis Online (T&F)*, and *MDPI*. Figure 3 captures the statistics of the overall distribution of the reviewed literature across the considered publishing platforms. These statistics state that *IEEE* is the predominant platform for publishing research works on VNE, followed by *Elsevier*, *Springer*, and *Wiley*.

C. Our Contributions

The key contributions of this survey can be outlined as follows.

Contribution 1: Comprehensive Review of the State-of-the-art VNE Strategies: We comprehensively analyze the full set of state-of-the-art VNE strategies over the last decade by considering the various related aspects and overcoming the shortcomings of existing surveys. For instance, the existing surveys by Belbekkouche et al. [14]

(2012) and Fischer et al. [15] (2013) are almost a decade old; the surveys by Cao et al. [18] (2018) and [16] (2016) have limited scope as they focus only on heuristic and exact solutions, respectively. Moreover, Yu et al. [17] (2016) and Jiang et al. [20] (2024) restricted to reviewing strategies for optical-connected and satellite networks. On the other hand, a recent survey by Wu et al. [21] (2024) focused solely on exploring AI-powered VNE solutions.

Contribution 2: Hardware and Software Enablers for VNE: To assist the readers in seamlessly understanding the background of NV, we provide detailed insights on the critical hardware and software technologies that make up the production-level NV environments, and touch upon their features, pros and cons.

Contribution 3: A Primer into VNE Modeling with Illustrative Examples: To acquaint the readers with virtualization, we include a primer on different and relevant virtualization technologies with visually appealing diagrams for comprehensible understanding. We also provide general-purpose modeling of the VNE problem by clearly highlighting the sub-issues related to VME and VLE and the constraints imposed. An extensive employment of examples should assist the readers in an easy and complete understanding of the overall problem.

Contribution 4: Detailed Insights on Evaluation Frameworks: Unlike existing surveys, this paper conducts a thorough investigation of the various simulation and emulation frameworks proposed in the literature for implementing, testing, and evaluating VNE algorithms. We provide detailed insights into individual platforms and include a comparative analysis highlighting essential features and identifying the gaps in each framework.

Contribution 5: Unique Classification, Thorough Review, and Comparative Study: Unlike existing surveys, we introduce a novel and detailed taxonomy of the reviewed literature, organized by solution approach, with several key advantages. It enhances the understanding of available solutions for researchers and practitioners, simplifies comparison, and guides future research by identifying gaps in existing approaches. It also guides future research by pinpointing gaps in existing strategies, revealing trends in VNE research, promoting standardization, fostering interdisciplinary connections, and serving as a foundational framework for education and training while encouraging innovation through hybrid approaches. Furthermore, the reviewed solutions are thoroughly analyzed and compared to highlight various design options' advantages, disadvantages, and suitability. We have included detailed explanations of exemplary solutions and illustrative scenarios to improve the survey's readability, making the content more fluid and precise.

Contribution 6: Detailed Review of the ML and/or DL-based Embedding Solutions: We also provide an in-depth review of ML, DL, and AI-powered intelligent VNE solutions. These techniques have recently added significant impetus to the VNE research and are included in this survey.

Contribution 7: Future Directions, Challenges, and Open Opportunities for Research: This survey also outlines numerous avenues for exploration and innovation that will shape the future of NV. Integrating VNE with modern technologies such as digital twins, blockchain, and NFV/VNF can unlock its potential and create a more secure, sustainable, and interconnected digital world.

D. Structure of the Survey

The rest of the survey is structured as follows. Section II presents a brief overview of hardware and software enablers that facilitate the adoption of NV. In Section III, we introduce the problem of VNE, formulate it, and present an illustrative example supported by visually appealing diagrams to give readers a clear understanding of the problem. Section IV classifies the literature based on the solution strategies adopted, namely exact and approximate methods. This section delves into the details of each approach with illustrative examples, highlighting the workings of some exemplary solutions. State-of-the-art embedding solutions based on ML and DL have been reviewed in Section V, while Section VI discusses the emerging research directions and the partially addressed issues on VNE. Finally, Section VII concludes the work. The overall organization of the survey is structured as per Figure 4.

II. ENABLERS OF VIRTUAL NETWORK EMBEDDING: HARDWARE, SOFTWARE, AND ABSTRACTION TECHNOLOGIES

NV is driven by advanced hardware, such as high-performance processors, optimized network interfaces, and sophisticated software like virtualization platforms and management tools. These innovations work together to enhance flexibility, functionality, and resource utilization. This synergy enables VNE's seamless integration and robust performance across diverse network environments.

A. Hardware Enablers

Network infrastructure transformation through NV heavily relies on cutting-edge hardware components such as programmable network devices, high-speed interconnects, and specialized virtualization accelerators to enhance the performance and scalability of hosting VNRs. The SN components encompass a range of hardware, including routers, switches, servers, network interface cards (NICs), cables, and connectors [60].

Routers and switches are pivotal in managing traffic flow within and outside the network, directing data packets to their destinations. Additionally, routers are responsible for traffic management and security enforcement, ensuring efficient and reliable network communication. Production-level SNs typically use routers that support high-capacity routing, IPv6 addressing, SDN integration, enhanced security, and cloud connectivity. The state-of-the-art routers such as Cisco CRS-X [61], Juniper Networks PTX Series [62], Arista 7500R Series [63], and Huawei NE Series [64] offer high capacity, advanced security

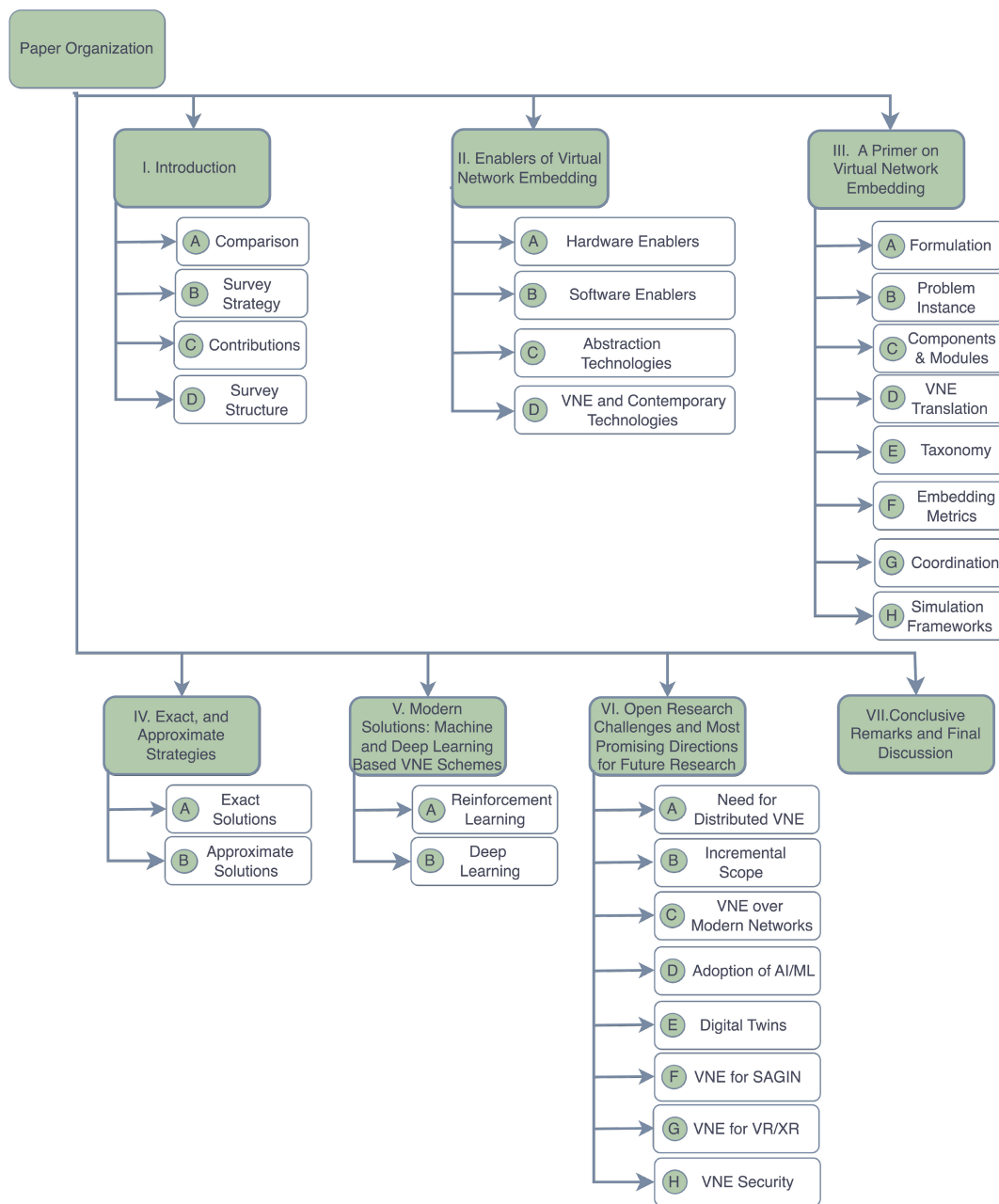


Fig. 4. Structure of this paper.

features, SDN capabilities, and AI integration for intelligent traffic management, making them ideal for SNs.

Servers are the computing and storage units responsible for running applications and managing data within a network. Modern servers are equipped with multi-core processors and high-density memory configurations. The widespread use of virtualization tools such as VMs, containers, and serverless functions [65], [66] allows for better resource utilization and support for diverse workloads. High-performance, energy-efficient, and secure servers like the Dell EMC PowerEdge [67] and HPE ProLiant series [68] are examples of modern technology supporting a broad range of enterprise applications.

NICs act as adapters, translating data between a device's internal language and the network's communication protocols. They enable high-speed connections, support virtualization, facilitate secure communication, and ultimately allow devices to talk to each other. NICs come in wired and wireless forms, offering advanced features like multi-gigabit Ethernet support, Wi-Fi 6/6E [69], and even upcoming Wi-Fi 7 standards [70].

Optical fiber cables assist in high-speed communication by transmitting data as light signals over long distances. Current state-of-the-art includes dense wavelength division multiplexing (DWDM), which increases the fiber capacity by allowing multiplexing over multiple wavelengths of light.

Some instances of the latest fiber optic technologies in production SNs include Corning's SMF-28 Ultra [71] and OFS's TrueWave RS [72], which offer higher data rates, greater flexibility, and improved resilience, meeting the QoS of high-speed networks.

B. Software Enablers

Software components in SNs play an indispensable role in enabling effective and scalable delivery of networking services. Though a plethora of software is pivotal in running a production SN encompassing multiple domains, we only discuss a few indispensable ones focused on assignment, management, and monitoring. To manage the NV setup, some popular Network Operating Systems (NOS) such as Cisco IOS [73], Juniper Junos [74], and Arista EOS [75] are often used in production level SNs and are capable of managing networked devices and performing crucial operations like routing and switching. On the other hand, for SDN-powered SNs, controllers such as OpenDaylight [76] and ONOS [77] offer centralized control over the SN, enabling programmability and dynamic configuration.

Specific to NV management technologies, multiple technologies such as VMware NSX [78], and open-source tools such as OpenStack [79] have gained significant popularity and can adequately abstract SN resources into VNRs, facilitating flexible service provisioning and management. Some automation tools, such as Cisco DNA Center [80], are also used to automate network provisioning and service orchestration across heterogeneous SNs. For large-scale SNs, resource monitoring is pivotal to provide real-time insights on the performance of the network, and tools such as Prometheus [81] and Grafana [82] assist in such operations. Together, these components enhance SNs' agility and efficiency in meeting diverse operational demands and supporting advanced network services.

C. Abstraction Technologies

This section provides an overview of various prevalent virtualization technologies often adopted by individuals and enterprises to support different information technology (IT)-based services.

1) *Server Virtualization*: Server virtualization involves partitioning server resources into distinct and isolated components known as VMs. This process is enabled by hypervisor or virtual machine monitor (VMM) software. Each VM is an independent system equipped with an operating system (OS) capable of executing various operations [84], [85]. Typically, a virtualized server can host multiple VMs, as depicted in Figure 5, where three VMs are shown, each running different applications with its dedicated OS. The VMM facilitates the creation, operation, and management of these VMs on the underlying hardware. Server virtualization yields several benefits, such as improved resource utilization, reduced maintenance costs, faster provisioning, enhanced flexibility, and expedited recovery. However, it also presents several research challenges, including resource provisioning and management [83].

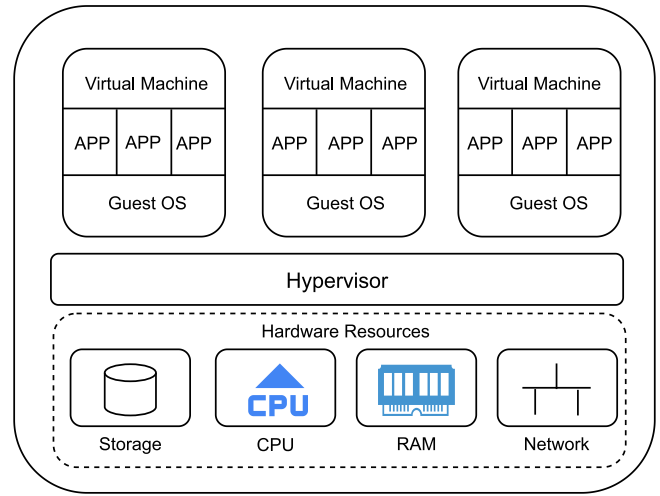


Fig. 5. An illustration of server virtualization [83].

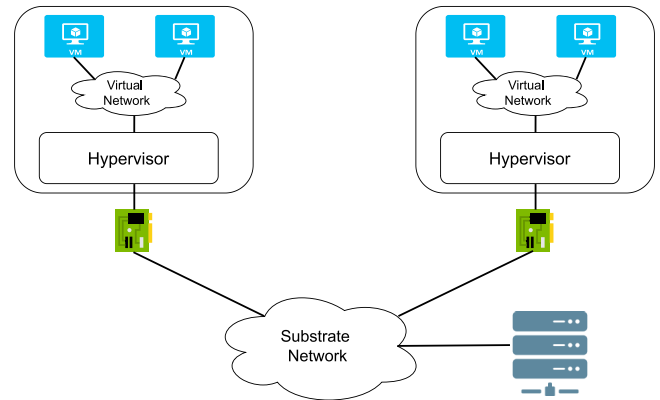


Fig. 6. Visualizing a network virtualized setup.

2) *Network Virtualization*: NV, as illustrated in Figure 6, logically partitions the physical or substrate network resources operating as a single independent unit often referred to as VNRs [86], [87], [88]. NV frees the underlying hardware from dependent software controlling individual components and exercises centralized control over all networking resources using a single software. A typical VN comprises multiple communicating VMs grouped capable of replicating an enterprise system. It encompasses virtualizing hardware components such as connections, switches, routers, etc., assisting the network administrator in efficiently exercising a single-point control without touching the underlying hardware.

Virtualizing networking resources has multiple benefits already presented in Section I. Additionally, NV has multiple applications such as (1) acting as a test bed for application deployment as it mimics real-world hardware and software, (2) facilitating the integration of multiple virtual networks on a shared substrate infrastructure, (3) is a perfect foil for application performance engineering and testing, and (4) assists in seamless and faster deployment of applications. However, it also introduces multiple research challenges, of which the most prominent and well-researched problem is that of virtual network embedding [15].

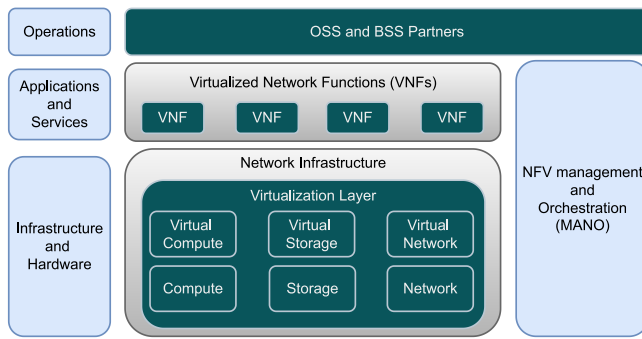


Fig. 7. The VNF stack.

3) *Virtual Network Functions (VNFs)*: In contrast to NFV, which aims to virtualize the entire network, VNFs focus on abstracting network functions as virtualized software-based entities [89], [90]. Specifically, VNFs are networking functionalities implemented as software and executed on virtual platforms, that is, VM in an NFV environment [91]. Traditionally, VNFs have been proposed to replace or supplement classical hardware-based functionalities such as firewalls, routers, and load balancers. The VNF stack with its components is illustrated in Figure 7. The Infrastructure and Hardware Layer consists of physical servers, storage, and networking equipment, hosting VMs or containers that run VNFs while networking devices facilitate communication. The Virtualization Layer abstracts these physical resources into virtual components, enabling flexible and scalable VNF operations using hypervisors or container runtimes. The VNF stack, which contains virtualized services like firewalls and routers, leverages these resources to deliver efficient, scalable services through service chaining. The NFV management and orchestration (MANO) module, which includes the NFV orchestrator (NFVO), virtualized infrastructure manager (VIM), and VNF manager (VNFM), is responsible for managing VNF deployment, resource allocation, and lifecycle operations. This ensures that the system remains adaptive and efficient throughout the execution of processes. OSS supports network functions through monitoring and fault detection, while BSS manages business processes such as billing and service agreements. The Applications and Services leverage VNFs to deliver Internet, VoIP, video streaming, and other services, dynamically scaling to meet customer demands while maintaining high performance and quality.

Moreover, VNFs offer multiple benefits to the infrastructure providers as discussed subsequently [91].

- *Flexible, Elastic, and Scalable Infrastructure*: As the VNFs are software-based, it is easier for the network admins to upscale and downscale on-demand by the variability in the workload without making extensive changes to the hardware.
- *Cost Benefits*: The dependencies on proprietary hardware are greatly reduced. These services can be deployed on off-the-shelf servers in the cloud, further reducing operational and management costs.
- *Fast Deployment*: VNFs are much faster to deploy in real-time as the delay in provisioning the hardware is eliminated. The modifications and updates can be rolled out in real-time, benefiting users.

- *Vendor Neutrality and Resource Optimization*: VNFs are hardware independent, thereby evading vendor lock-ins. Additionally, via virtualization, VNFs optimally utilize the network, compute, and storage resources more effectively.
- *Reduce Resource Footprints*: VNFs allow for multiple networking functionalities to be hosted on the same infrastructure, leading to reduced hardware and power saving for the SPs.
- *Automation, Orchestration, and Resilience*: Provisioning, configuration, and management can be easily automated and orchestrated using management and orchestration (MANO) systems, drastically reducing human dependencies. Moreover, VNFs are portable and can easily be migrated/replicated across locations, boosting the system's real-time disaster recovery and resilience.
- *Service Innovation*: VNFs enable the network providers to innovate quickly by introducing new features without being limited by the hardware.

We can safely deduce that VNFs enable setting up an agile, cost-effective, and adaptable network infrastructure that is perfectly capable of responding to the ever-changing needs of the users. After outlining the development of relevant virtualization technologies, our focus shifts to exploring the issue of VNE.

4) *Containerization*: In server virtualization, hypervisors manage the infrastructure, enabling various guest OSes to run different applications. The VM-based resource partitioning system suffered many pitfalls, such as (1) Significantly increased resource consumption. (2) Compatibility issues when migrating VMs between different OSes. (3) Improper isolation poses security risks and necessitates redundancy to safeguard against failures. (4) Communication among VMs is slower, and (5) VMs have significantly higher boot-up and creation times. Unlike VMs that virtualize the entire physical system, including hardware (CPU, memory, storage) and the OS, containerization, as illustrated in Figure 8, offers lightweight OS-level virtualization encapsulating applications and their dependencies into isolated units, facilitating execution across disparate environments [92], [93]. Managed by a container engine, such as Docker, containers can be easily deployed, scaled, and managed, facilitating agile development and continuous integration/continuous deployment processes [94], [95]. In practice, VMs and containers can complement each other. Combining these technologies provides strong isolation and increased flexibility and portability, enabling the efficient hosting of various applications.

Finally, Table III highlights the features, limitations, and use cases corresponding to different production-level abstraction techniques. In the following section, we introduce readers to a simple model of the problem and examine its research challenges. Additionally, to facilitate understanding, we provide illustrative scenarios highlighting the underlying issue of VNE.

D. VNE and Its Contemporary Technologies

VNE is a fundamental concept in NV that facilitates the efficient mapping of VNRs onto SN resources. However, it is essential to contextualize VNE alongside related concepts,

TABLE III
FEATURES, LIMITATIONS, AND USE-CASES OF DIFFERENT ABSTRACTION TECHNOLOGIES

Software Enablers	Abstraction Level	Isolation	Resource Overhead	Performance	Deployment Speed	Management Complexity	Security	Use Cases
Server Virtualization	Physical server hardware	Strong	High	Low	Slow	Moderate to high	High	Enterprise IT, SNs
Network Virtualization	Network resources	Moderate	Low to moderate	High	Moderate	High	Moderate	Large Scale SNs
Virtual Network Functions	Network functions	Deployment Specific (VMs or containers)	Moderate	Moderate to high	Fast to moderate	Moderate	Moderate	Cloud and Telecom networks
Containerization	Application and dependencies	Moderate	Low	High	Fast	Low to moderate	Moderate to low	Microservices and cloud-native apps

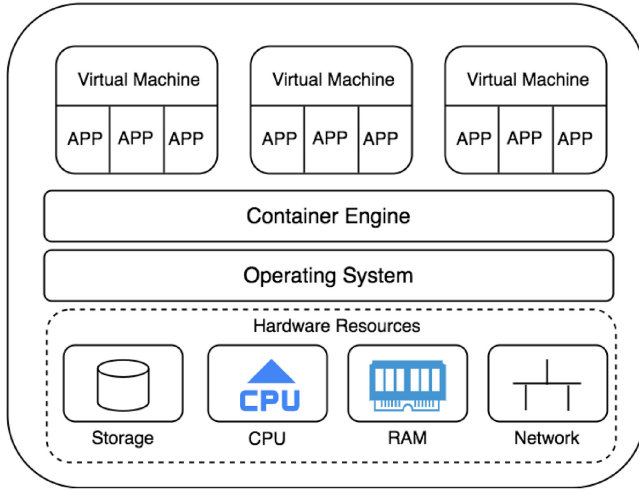


Fig. 8. An illustration of containerization.

such as service chaining and network slicing, as each serves a unique purpose within the broader NV framework. We subsequently explore these contemporary technologies and emphasize their relationship with VNE.

1) *Service Chaining*: It involves linking a series of network services, such as firewalls, load balancers, and intrusion detection systems (IDS), in a specified order to form a service path to redirect network traffic [96]. In this configuration, data packets are routed through a sequence of VNFs as shown in Figure 9, ensuring that these services process them before reaching their destination. This method enables efficient management and orchestration of network functionalities while supporting flexible, programmable, and scalable architectures that allow easy addition or modification of services without significant hardware changes [97], [98]. Service chaining can be categorized into various types according to the structure of the service chain, the types of network functions involved, and the environments in which it is implemented. We elaborate on the kinds of chaining structures subsequently with real-world use cases.

1) *Linear Service Chaining*: A sequential arrangement of network functions where traffic flows through each function in a pre-defined order, such as passing through a firewall, followed by an IDS, and then a load balancer. This approach is frequently utilized in security-centric configurations and traffic management applications where the sequence of services is critical.

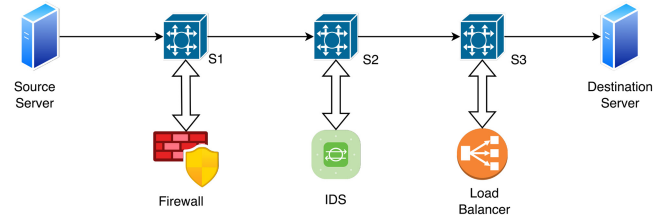


Fig. 9. An instance of a linear service chain for analyzing data packets between a source and a destination server [99].

- 2) *Branched Service Chaining*: It encompasses a non-linear service chain that branches to different network functions based on conditions or traffic types, enabling tailored processing. For example, general web traffic may pass through a basic firewall, while sensitive data traffic is routed through advanced security functions. This is particularly effective in environments with diverse types of traffic that require specific treatment.
- 3) *Parallel Service Chaining*: It allows functions to be used in tandem, enabling traffic to be split for concurrent processing before rejoining. For instance, traffic may be distributed across multiple firewalls or data-scrubbing devices to enhance redundancy and throughput. This method helps achieve high availability and load balance.
- 4) *Adaptive Service Chaining*: These chains are adjusted in real-time based on network conditions, traffic patterns, or user requirements, optimizing performance according to changing states. For instance, it can redirect traffic along different paths in response to network congestion or specific user profiles. This approach is especially advantageous in cloud environments and SDNs, where flexibility and rapid adaptation are uncompromisable.
- 5) *Hierarchical Service Chaining*: It consists of multiple levels of chains, where higher-level functions invoke or trigger lower-level chains based on specific criteria. For example, a high-level service chain may include a load balancer that, depending on traffic characteristics, activates various sub-chains with specialized security functions. This approach is particularly suitable for multi-tier architectures with varying levels of service and security.
- 2) *Network Slicing*: It refers to partitioning a physical network into multiple “slices,” each catering to specific application requirements [100]. Each slice operates independently

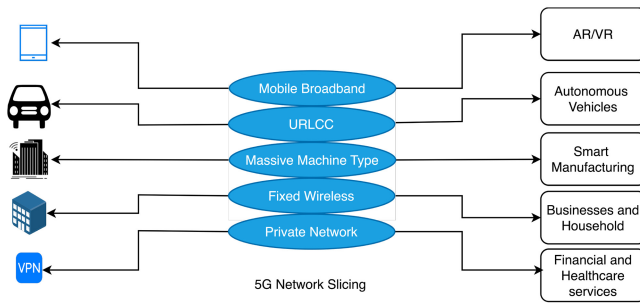


Fig. 10. Different slices in 5G networks and their use cases.

with dedicated resources and enforces a pre-defined QoS, supporting disparate applications, including consumer and enterprise-driven solutions. It leverages SDN, which separates the control plane from the data plane for centralized resource management, and NFV to execute networking operations on standard hardware. In conjunction with the techniques above, slicing facilitates the dynamic creation, allocation, and management of slices to accommodate various services. The slices are mostly created specifically for the services they cater to, as depicted in Figure 10.

- 1) *Enhanced Mobile Broadband*: It is designed for applications that require high data throughput and low latency, providing seamless connectivity for broadband services [101]. It is apt for stable yet bandwidth-consuming applications like video streaming and AR/VR.
 - 2) *Ultra-Reliable Low Latency Communication*: It prioritizes ultra-reliability and extremely low latency real-time communications [102]. The use cases include autonomous vehicles, industrial automation, remote surgery, and mission-critical communication systems.
 - 3) *Massive Machine-Type Communication*: Supports simultaneous communication with multiple connected devices with low data rates. Use cases include IoT applications such as smart cities, smart metering, and environmental monitoring [103].
 - 4) *Fixed Wireless Access*: Provides broadband connectivity through wireless connections instead of wired ones [104]. It delivers services to households and businesses with fiber or cable installations.
 - 5) *Enterprise or Private Network*: Tailored for enterprises or private users, offering customizable options and features. It is specifically designed for organizations needing dedicated resources, including secure communication channels for financial services or isolated slices for healthcare facilities.
- 3) *Virtual Machine Placement (VMP)*: With server virtualization support, multiple VMs can be co-hosted on the same physical server. A significant research challenge, as shown in Figure 11 and addressed by the community, is the optimal assignment of VMs to physical servers, known as the Virtual Machine Placement (VMP) [105]. The literature provides diverse solutions for the VMP and tailors them to specific deployment environments and objectives. Based

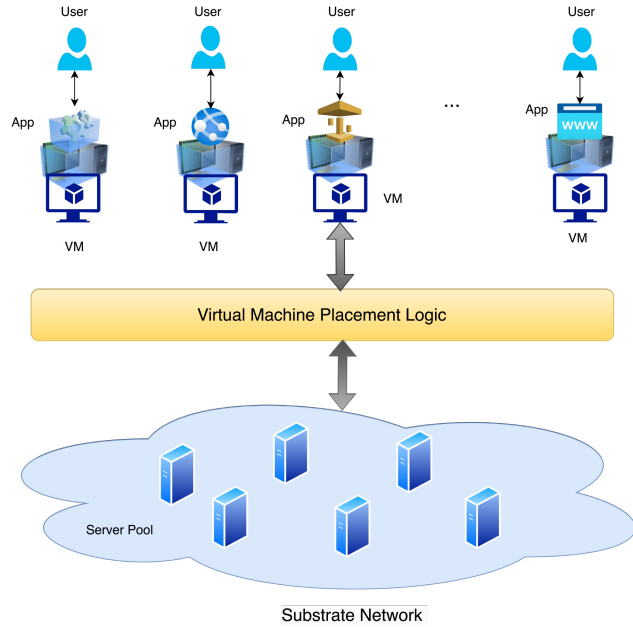


Fig. 11. An illustration of virtual machine placement problem.

on the types of applications running on the VMs, VMP schemes can be broadly categorized into static and dynamic approaches [106]. Static schemes are suited for applications with stable workloads. In contrast, dynamic schemes allow VMs to migrate between servers based on resource requirements, facilitating load balancing but introducing complexity and migration overhead [107]. A key focus has been on developing consolidation policies to reduce the number of active hosts by grouping VMs on energy-efficient servers, potentially enhancing system performance [108]. Various placement strategies include load-balancing [109], which evenly distributes VMs to improve reliability; cost-aware placement [110], which minimizes expenses at the expense of latency; network-aware placement [111], which reduces latency by co-locating frequently communicating VMs; and energy-aware placement [112], which emphasizes energy savings while maintaining performance. The choice of VMP strategy depends on the specific objectives of the data center, such as cost, performance, reliability, or energy efficiency. Table IV compares contemporary technologies and establishes their relationship with VNE.

III. A PRIMER ON VIRTUAL NETWORK EMBEDDING

Before we delve into the intricacies of the embedding problem, we first present a straightforward mathematical model that formally represents the critical components involved, namely VNRs and SN. Subsequently, we break down the embedding process into two sub-problems: virtual machine embedding (VME) and virtual link embedding (VLE). While various papers in the literature have provided disparate representations of VNE [113], [114], we believe that a simplified modeling approach can aid first-time readers and those unfamiliar with the domain in quickly grasping the

TABLE IV
COMPARATIVE STUDY OF VNE WITH CONTEMPORARY TECHNOLOGIES

Aspects	Network Slicing	Virtual Machine Placement	Service Chaining	Virtual Network Embedding
<i>Problem Definition</i>	Creation of virtual segments on shared networks	Strategic allocation of VMs to servers	Sequencing network functions for networking services	Mapping VNRs, i.e., VMs and VLs onto SNs
<i>Optimization Goals</i>	QoS delivery	Effective resource utilization	End-to-end traffic management	End-to-end service delivery
<i>Scope</i>	Entire network	VM Allocation in inter/intra DCs	Logical sequencing of VNFs	VM and VL assignment over SNs
<i>Use Cases</i>	5G services	Cloud DCs	Enterprise security and traffic management	Interconnected multi-tenant SNs
<i>Research Challenges</i>	Dynamic resource assignment and resource slicing	Objective-specific optimal server assignment	Ensuring seamless traffic flow with tolerable latency	Objective-specific optimal VM and VL assignment
<i>Interrelationship</i>	Allocating slices is similar to VMP in the context of networking resources	Supports slicing of the server resources for resource efficiency	Works within the network slices, utilizing VMP to optimally place VNFs	Supports concurrent slicing of the servers and networking resources for end-to-end services
<i>Advantages</i>	Customizable services with QoS guarantees	Optimized usage of DC resources	Streamlined service delivery policy adherence	Efficient usage of SN resources for multi-tenant models

TABLE V
COMPETITIVE STUDY OF DIFFERENT VIRTUAL NETWORK REPRESENTATION

Aspect	Directed Graphs	Undirected Graphs	Directed Graphs with Bidirectional Edges
<i>Use case</i>	CDNs and streaming services	Client-server	Sequence-sensitive operations
<i>Data Flow</i>	One-way data transmission	Flexible two-way traffic	Directional flow
<i>Resource Constraints</i>	Specific to a direction	Generic both direction	Distinct in each direction
<i>Traffic Rerouting</i>	Complex	Easier	Moderately complex
<i>Management Complexity</i>	Moderately complex	Less complex	Highly complex
<i>Resource Consumption</i>	Moderate	Low	Higher
<i>Traffic Management Flexibility</i>	Less	More	Moderate
<i>Implementation Challenges</i>	Moderately complex	Easier	More complex

complexities and comprehending the subsequent information presented.

A. Formulating the VNE Problem

Before formulating the VNE problem, we introduce the components involved in the embedding process. Before this, we discuss the various approaches for modeling a VNR. In this context, a directed graph representation may offer a more accurate model than the weighted undirected graph. Furthermore, the choice between directed and undirected graphs for modeling VNRs depends on the application's communication patterns and the specific goals of the embedding process. We will then offer insights on which model could be adopted based on the use case.

Directed vs. Undirected vs. Directed Graphs with Bidirectional Edges for VNR Modeling: Directed graphs are optimal representations for VNRs where data flow direction is essential, as seen in content delivery networks (CDNs) and streaming services that require one-way data transmission from servers to users [115]. This approach captures asymmetric resource constraints, such as distinct bandwidth and latency needs, and enables prioritization for critical paths, enhancing security through directional restrictions [116]. In contrast, undirected graphs suit bidirectional communication applications, like traditional client-server models, simplifying embedding by allowing two-way traffic and improving fault tolerance through rerouting [88], [117], [118]. This setup also facilitates flexible resource allocation in general-use networks, like VPNs, and handles dynamic traffic changes in MEC and IoT networks without re-embedding [118]. While directed graphs with bidirectional edges offer precision for sequence-sensitive tasks by highlighting dependencies among VMs, they add complexity, increasing computational demands and resource consumption, especially in more

extensive networks. Managing such models requires sophisticated algorithms and may necessitate protocol updates, complicating scalability and implementation. Table V captures a comparative study of different VNE models. Note that the appropriate model selection should align with the application's communication patterns, operational dependencies, and network requirements, ensuring the most effective representation for each scenario. As most of the literature reviewed models the VNR as an undirected graph, we have maintained this representation to ensure readers' consistency and ease of understanding. However, the undirected graph model can be adapted to a directed graph with minimal adjustments.

Definition 1 [Virtual Network Request (VNR)]: A VNR, as illustrated in Figure 1, comprises multiple interacting VMs with communication dependencies captured as VLs.

Modeling a VNR: Formally, a VNR is represented as an undirected weighted graph $\mathcal{G}^{Vr} = (\mathcal{N}^{Vr}, \mathcal{L}^{Vr})$. The set of vertices in the graph, i.e., $\mathcal{N}^{Vr} = \{v_1, v_2, \dots, v_i, \dots\}$ capture the VMs and the i^{th} VM is uniquely identified as v_i . The set of edges, i.e., $\mathcal{L}^{Vr} = \{(1, 2), (2, 4), \dots, (i, i'), \dots\}$ capture the VLs interconnecting any two VMs in $\mathcal{N}^{Vr}$. Note that an edge (i, i') implies that the VMs v_i and $v_{i'}$ exhibit communication dependencies. Every VM $v_i \in \mathcal{N}^{Vr}$ requests server resources for seamless operation and this demand is captured as $d(v_i)$ and the requisite physical bandwidth resource demand of a VL (i, i') interconnecting v_i and $v_{i'}$ is captured as $d(i, i')$.

Definition 2 [Substrate Network (SN)]: A substrate network refers to the physical organization of servers, switches, and physical links providing user services.

Modeling a SN: Formally, a SN is also modeled as an undirected weighted graph $\mathcal{G}^{Sn} = (\mathcal{N}^{Sn}, \mathcal{L}^{Sn})$. Here, $\mathcal{N}^{Sn} = \{s_1, s_2, s_k, \dots\}$ be the set of physical servers and $\mathcal{L}^{Sn} = \{(1, 2), (3, 4), \dots, (l, l'), \dots\}$ is the physical link set. The available resources of server $s_k \in \mathcal{N}^{Sn}$ is represented as

$a(s_k)$, and that of a substrate link $(l, l') \in \mathcal{L}_{Sn}$ is captured in $a(l, l')$.

Modeling the Computational Resources: The SN and the VNR resources are often multidimensional, and different authors have adopted disparate representations. Therefore, we would like to elucidate the readers on the various representations adopted in the literature before delving deeper into the modeling of VNE. Ideally, the VM and server resources are multi-dimensional, comprising CPU and memory demands. Most works express the demands individually for each resource type or as resource vectors [119]. However, such a model adds to the notational complexity and is often difficult for the readers to comprehend. Therefore, some works in [88], [117], [120], [121], [122] have adopted a more simplified approach to express the multi-dimensional resources. The authors use computational resource blocks (CRBs) as initially reported in [123], [124], wherein one CRB resource corresponds to 1 CPU core and 512 MB of memory. A two CRB demand implies 2 CPU cores and 1 GB of memory. Moreover, large-scale SPs widely adopt such sizing policies as they reduce resource management complexity. We adopt this resource modeling in the paper due to its obvious advantages. Therefore, the CRB demands of VM $v_i \in \mathcal{N}^{Vr}$ is given by $d(v_i)$. Similarly, the resources at the server $s_k \in \mathcal{N}^{Sn}$ denoted by $a(s_k)$ are also expressed in CRBs. Note that some works also consider the storage resources of the VMs and servers in addition to CPU and memory [125]. We would also like to mention that the above-discussed model is easily adaptable to multiple multi-dimensional resources. On the other hand, the bandwidth demand of a VL (i, i') and a physical link (l, l') is uni-dimensional and is captured as $d(i, i')$ and $a(l, l')$ (in Gbps).

Note that the VNE comprises two interconnected sub-problems: firstly, the allocation of VMs of VNRs to substrate servers, termed as virtual machine embedding (VME), and secondly, the assignment of interconnecting VLs of VNRs to physical/substrate paths after embedding the VMs termed as virtual link embedding (VLE). Both sub-problems are challenging, computationally intractable, and are proven to be $\mathcal{NP}$ -Hard [44], [126]. Next, we formally define each sub-problem and present high-level example scenarios to assist the readers in understanding the sub-problems.

Definition 3 [Virtual Machine Embedding (VME)]: Virtual machine embedding is a function $f_{vm} : \mathcal{N}^{Vr} \rightarrow \mathcal{N}^{Sn}$, wherein $\forall v_i \in \mathcal{N}^{Vr}, \exists s_k \in \mathcal{N}^{Sn}$, such that $d(v_i) \leq a(s_k)$.

The VME sub-problem is captured in Definition 3. It essentially states that at least one host machine $s_k \in \mathcal{N}^{Sn}$ should have enough resources (in CRBs) to accommodate the VM v_i . Ideally, a VM is assigned to only one server, but in cases of parallel workflows, a VM may be mapped onto multiple servers [127]. However, this model is explainable for applications with splittable jobs. On the other hand, a server can host multiple VMs of the same/other VNRs. However, some works in [88], [117], [120], [128], [129] have disallowed such assignments to facilitate distributed services and avoid single-point failures. Additionally, for a VNR, VME is said to be complete if all its VMs (set $\mathcal{N}^{Vr}$) obtain a successful assignment.

Definition 4 [Virtual Link Embedding (VLE)]: Virtual link embedding is a function $f_{vl} : \mathcal{L}^{Vr} \rightarrow \mathcal{L}^{Sn}$ between a VL and the substrate/physical link. The VL (i, i') between VMs v_i and $v_{i'}$ is assigned to a substrate paths (l, l') iff $d(i, i') \leq a(l, l')$.

In contrast to the VME, where a VM is assigned to a substrate server, VLE assigns a VL to a substrate path, constituting multiple physical links, each satisfying the VL's bandwidth requirement.

Definition 5 [Virtual Network Embedding (VNE)]: A VNR is said to be successfully embedded if all the VMs and VLs are respectively assigned substrate servers and paths satisfying their resource demands [88], [117].

Optimization Goals for VNE: VNE, which includes both VME and VLE, has been explored with various goals and objectives in the research community. Consequently, there is no standardized method or singular approach to defining these objectives. For instance, VME goals may include improving resource utilization [88], consolidating virtual machines on energy-efficient servers [130], increasing the acceptance ratio [122], and maximizing service provider revenue [131]. In contrast, the objectives of VLE typically focus on meeting various quality of service measures, such as bandwidth, communication, and delay requirements [15]. Additionally, both sub-problems entail a set of constraints essential for successful VNE. While it is challenging to mathematically encompass the various objectives and constraints found in the literature, we have concentrated on modeling a select few of the most critical metrics, providing clarity while recognizing the complexity of the topic.

Before formulating various objectives, we introduce two binary indicator variables, i.e., VM indicator (Equation (1)) and VL indicator (Equation (2)), which subsequently assist in defining the objectives and constraints of the VNE. The former is set to 1 if v_i is hosted on s_k , and the latter is set to 1 if the VL (i, i') is mapped onto the physical link (l, l') .

$$\chi_{i,k} = \begin{cases} 1 & v_i \text{ is assigned to server } s_k \\ 0 & \text{Otherwise} \end{cases} \quad (1)$$

$$\chi_{l,l'}^{i,i'} = \begin{cases} 1 & \text{If VL } (i, i') \in \mathcal{L}^{Vr} \text{ is mapped to the} \\ & \text{physical link } (l, l') \in \mathcal{L}^{Sn} \\ 0 & \text{Otherwise} \end{cases} \quad (2)$$

Objectives of VNE Schemes: We subsequently formulate some optimization goals of popular VNE schemes.

- 1) **Minimize Embedding Cost:** One of the key metrics that has gained significant attention in the literature on VNE is the reduction of embedding costs, which refers to the cost of SN resources utilized during embedding a VNR (Details of this metric can be found in Section III-F). The overall objective function is captured in Equation (3), where $c_{k,i}$ and $c_{l,l'}^{i,i'}$, respectively capture the cost of embedding v_i on s_k and (i, i') on the physical link (l, l') .

$$\begin{aligned} \min \quad & \sum_{s_k \in \mathcal{N}^{Sn}} \sum_{v_i \in \mathcal{N}^{Vr}} c_{k,i} \cdot \chi_{i,k} \\ & + \sum_{(i,i') \in \mathcal{L}^{Vr}} \sum_{(l,l') \in \mathcal{L}^{Sn}} c_{l,l'}^{i,i'} \cdot \chi_{l,l'}^{i,i'} \end{aligned} \quad (3)$$

- 2) *Maximize Revenue-to-cost ratio*: Another key metric that dictates the performance of VNE schemes is the revenue-to-cost ratio, which captures the ratio of the number of resources requested to the actual amount of SN resources used to embed a VNR (Details of this metric can be found in Section III-F). Equation (4) captures the overall objective of maximizing the ratio, which assists in compact embedding boosting the SPs revenue.
- 3) *Maximize Acceptance ratio*: It captures the number of VNRs successfully embedded (V^{pr}) to the number of VNRs in the system (V^{sys}) as depicted in Equation (5). A higher acceptance implies more profit for the SPs.

$$\max \frac{V^{pr}}{V^{sys}} \quad (5)$$

Constraints on VNE Schemes: Any VNE scheme must adhere to the following constraints.

- 1) *VM Embedding Constraint*: A VM v_i can be assigned to at most one server and is reflected in Equation (6).

$$\sum_{s_k \in \mathcal{N}^{Sn}} \chi_{i,k} = 1, \forall v_i \in \mathcal{N}^{Vr} \quad (6)$$

- 2) *Server Capacity Constraint*: A server s_k cannot host VMs beyond its capacity captured by $c(s_k)$ as shown in Equation (7).

$$\sum_{v_i \in \mathcal{N}^{Vr}} d(v_i) \cdot \chi_{i,k} \leq c(s_k), \forall s_k \in \mathcal{N}^{Sn} \quad (7)$$

- 3) *VL Embedding Constraint*: A VL $(i, i') \in \mathcal{L}^{Vr}$ must be mapped to at least one physical link as per Equation (8).

$$\sum_{(l, l') \in \mathcal{L}^{Sn}} \chi_{l, l'}^{i, i'} \geq 1, \forall (i, i') \in \mathcal{L}^{Vr} \quad (8)$$

- 4) *Link Capacity Constraint*: The number of VLs hosted by a physical link is bounded by its maximum capacity $c(l, l')$ as captured in Equation (9).

$$\sum_{(i, i') \in \mathcal{L}^{Vr}} \chi_{l, l'}^{i, i'} \cdot d(i, i') \leq c(l, l'), (l, l') \in \mathcal{L}^{Sn} \quad (9)$$

- 5) *Flow Constraint*: Given the mapping of v_i and $v_{i'}$ mapped onto servers s_k and $s_{k'}$, the flow conservation constraint for the VL (i, i') is expressed in Equation (10). The first case captures the flow constraint at intermediate points, whereas the others capture the start and end points dictated by the hosting servers.

$$\begin{cases} \sum_{(l, l') \in \mathcal{L}^{Sn}} \chi_{l, l'}^{i, i'} \cdot d(i, i') = \sum_{(l'', l''') \in \mathcal{L}^{Sn}} \chi_{l'', l'''}^{i, i'} \cdot d(i, i') \\ \sum_{(k, l) \in \mathcal{L}^{Sn}} \chi_{k, l}^{i, i'} \cdot d(i, i') = d(i, i') \\ \sum_{(l'', k') \in \mathcal{L}^{Sn}} \chi_{l'', k'}^{i, i'} \cdot d(i, i') = d(i, i') \end{cases} \quad (10)$$

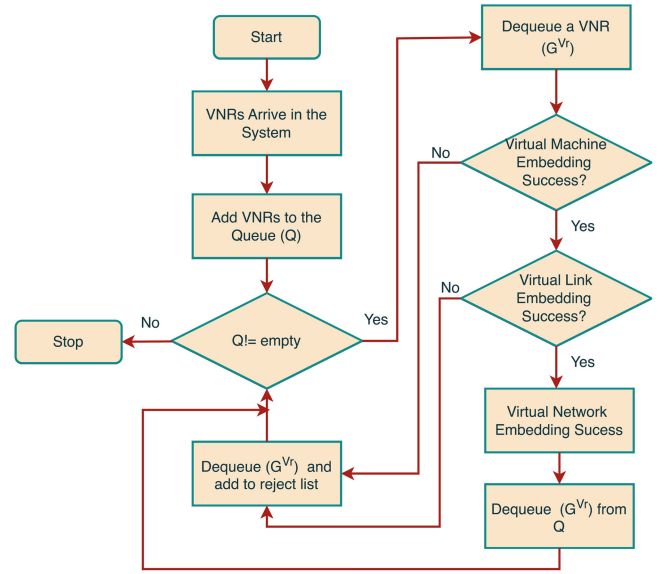


Fig. 12. Generic flowchart depicting stepwise working of any VNE scheme.

NP-Hardness: VNE is well-established to be $\mathcal{NP}$ – *Hard* and shares similarities with the multi-way separator problem [44]. Even after the VMs are embedded, allocating VLs to substrate paths reduces the unsplittable flow problem, which is also proven to be $\mathcal{NP}$ – *Hard* [132], [133], [134]. Thus, achieving an optimal solution for VNE essentially involves tackling two computationally complex problems: VME (Definition 3) and VLE (Definition 4). Additionally, due to the intractable nature of these problems, obtaining an optimal solution for larger problem instances poses significant challenges. To provide readers with a better understanding of the problem statement, a problem instance of VNE is subsequently discussed.

B. VNE: A High-Level Illustration of a VNE Problem Instance

Typically, the VNE encompasses the steps shown in Figure 12. To illustrate the problem of VNE more effectively, we provide an example scenario depicted in Figures 13 and 14. Referring to Figure 13, we have a SN consisting of servers $\{S_1, S_2, S_3, S_4\}$. A VNR with two VMs v_1, v_2 needs to be embedded into the SN. The resource demands of the VMs and the servers' capacities are expressed in CRBs, which are placed adjacent to the respective node in a rectangular box. The capacities of the hosts, expressed in CRBs, are denoted as numerals in rectangular boxes adjacent to the servers in Figure 13. For example, the capacity of server S_1 and VM v_2 is 2 and 1 CRB, respectively. Alternatively, the link demands are assumed to be gigabits per second (Gbps). For instance, the VL interconnecting v_1 and v_2 has a bandwidth demand of

$$\max \frac{\sum_{v_i \in \mathcal{N}^{Vr}} d(v_i) + \sum_{(i, i') \in \mathcal{L}^{Vr}} d(i, i')}{\sum_{s_k \in \mathcal{N}^{Sn}} \sum_{v_i \in \mathcal{N}^{Vr}} c_{k, i} \cdot \chi_{i, k} + \sum_{(i, i') \in \mathcal{L}^{Vr}} \sum_{(l, l') \in \mathcal{L}^{Sn}} c_{l, l'}^{i, i'} \cdot \chi_{l, l'}^{i, i'}} \quad (4)$$

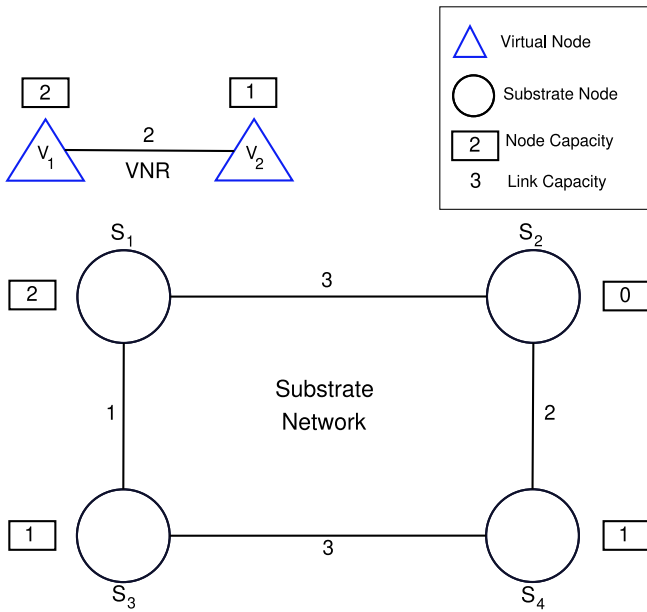


Fig. 13. An embedding instance. It consists of (i.) a VN with two VMs and a VL with their resource demands and (ii.) a substrate network with four servers and four physical links and their available capacities.

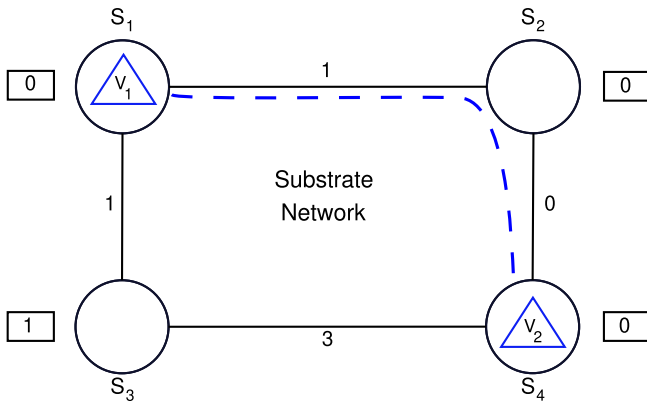


Fig. 14. The state of the substrate network after VM and VL embedding of the VNR in Figure 13.

2 Gbps. Similarly, the physical link between servers S_1 and S_2 has a capacity of 3 Gbps.

In the provided setup, an embedding solution is depicted in Figure 14. The VMs v_1 and v_2 are mapped onto servers S_1 and S_4 , respectively. The VL interconnecting the VMs is assigned onto the substrate path $S_1 - S_2 - S_4$. The updated capacities of the substrate resources after embedding are also indicated in Figure 14.

C. VNE: Components & Modules

In any VNE system, as shown in Figure 15, there are generally two main components: (1) the VNR and (2) the SN. The control modules are crucial for any embedding solution and are broadly categorized into (1) the Resource Manager and (2) the Embedding & Monitoring module. The Resource Manager primarily logs the status of resources, tracking both used and unused resources within the network. Meanwhile, the Embedding & Monitoring module is responsible for

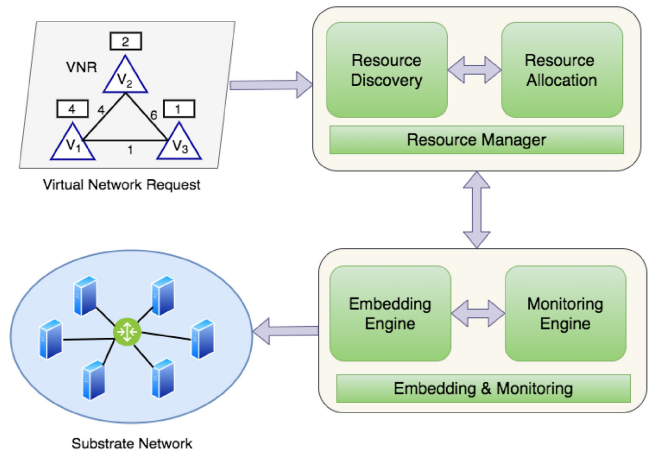


Fig. 15. Signal flow diagram for VNE.

identifying the optimal embedding on the SN and continuously monitoring the VNRs and SN resources. This module also collaborates with the Resource Manager to re-embed or re-allocate resources in response to dynamic network changes, such as fluctuating VM or VL demands of a VNR or in cases of hardware or software failures in the SN. Furthermore, these two modules interact regularly to execute essential management tasks, such as resource consolidation during off-peak times, disaster recovery, and failure management.

D. Actual Translation of VNE in Production SNs

The translation process encompasses the following steps: (1) Processing VNRs is initiated when a VNR with its detailed topology, including VMs and VLs, and their resource demands, like CPU, memory, and bandwidth, is received. (2) Resource Discovery: It encompasses a thorough discovery of available physical resources within the SN, encompassing servers, network devices, and storage units, to determine their current usage and capacities. (3) Virtual Node Mapping: The system identifies suitable physical nodes (servers) capable of hosting each VM based on resource availability and embedding-specific considerations, such as location, QoS, security levels, etc. (4) Virtual Link Mapping: For VLs connecting the VMs, the system identifies potential substrate paths meeting specified bandwidth and latency requirements. (5) Resource Allocation: Once mappings are determined, the system allocates resources (CPU, memory, bandwidth) on substrate servers and links to support VNRs operational needs. (6) Configuration and Deployment: VMs/containers are instantiated on the server using existing virtualization technologies. For communication via VLs, network configurations such as virtual switches and overlays (e.g., VLANs, VXLANs) are established. (7) Monitoring and Management: Continuous VNR performance and resource utilization monitoring enable proactive management and dynamic real-time adjustments to mappings and configurations. (8) Integration with Orchestration and Management Tools: Seamless integration with orchestration platforms and SDN controllers facilitates automated lifecycle management of the VNRs.

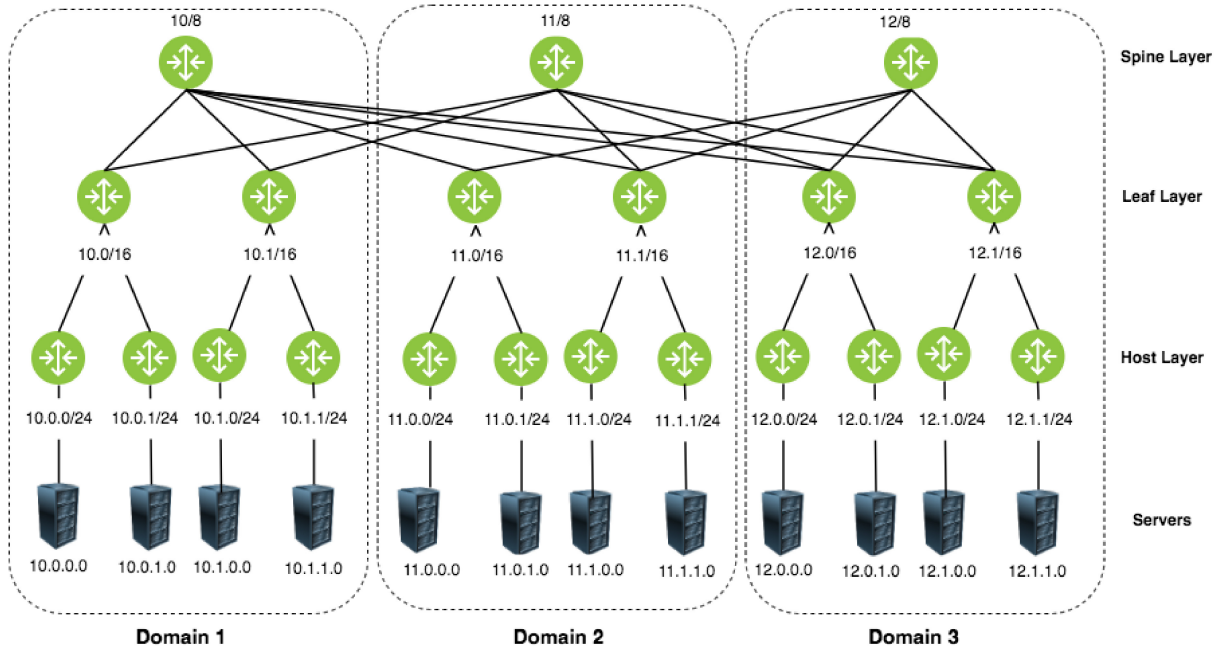


Fig. 16. Spine-leaf topology based SN.

To demonstrate the VNE problem in real-world production SNs, consider the example in Figure 16, depicting an SN topology with three layers: spine, leaf, and host, where servers connect directly to host-layer switches. Although the detailed spine-leaf architecture is discussed in [135], [136], this instance highlights how a spine-leaf topology supports a VNR in practical settings. Each spine layer signifies a separate domain, with three such domains or subnets shown in the figure. IP addresses for each device (switches and servers) are illustrated by varying octets based on layer positioning. In this context, embedding a VNR involves identifying paths for VLs and selecting servers for VMs. For example, Figure 17 shows the embedding of the VNR from Figure 1 on the SN, with highlighted servers and IPs hosting the VMs, and a dashed red line representing the VL between VMs v_1 and v_3 .

E. Taxonomy for VNE

Depending upon the nature of deployment, the VNE strategies are either (1) *static* or (2) *dynamic*. The infrastructure remains consistent, whereas the latter encompasses changing VNRs and SN. Moreover, the VNRs may be spread over multiple SNs owned by different SPs. In such cases, the VNE may be performed using resources of different SNs in a *distributed* manner instead of a *centralized* setup with a single SP. Considering the number of resources consumed to provision resources, the VNE can be classified as *precise*, where the VNRs are provisioned with minimum possible substrate resources, or *redundant*, where multiple substrate resources are provisioned together to realize a virtual component. On the other hand, depending on the stakeholders to which the embedding caters, the strategies can be broadly classified into *SP-centric* or *user-centric*. Based on the above-discussed classifiers, the overall literature on VNE can be categorized

as *static* or *dynamic*, *centralized* or *distributed*, *precise* or *redundant*, *SP-centric* or *user-centric*. Next, we discuss each property in detail and classify the studied literature.

Static or Dynamic (S/D): In a static setup, the embedding algorithm is exposed to a fixed set of VNRs executing over a SN for a pre-determined time with no change in the demanded resources. On the other hand, modern applications are dynamic because they arrive and leave the system dynamically. Additionally, during their stay in the system, the resource requirements of the components, i.e., VMs and VLs, fluctuate dynamically [137], [138]. In theory, the static embedding policies can operate in a dynamic setup but are not designed to handle the relocations of VNRs in their entirety or parts. Relocation or rearrangement of the VNRs' can also be performed to handle the fragmentation of resources caused by the dynamic arrival and departure of VNRs. Over a while, the SN may be upgraded to deal with scalability, calling for the re-organization of VNRs.

Centralized or Distributed (C/D): A single central entity generates the embedding for centralized setups. Such an approach has a significant advantage over non-centralized approaches due to the availability of global knowledge about the state of the SN and other VNRs, significantly assisting the embedding protocol in making optimal decisions. However, like any other centralized setup, these approaches suffer from single-point failures and congestion owing to overwhelming requests during peak load. On the other hand, in a distributed setup, multiple stakeholders generate the embedding, which improves the scalability compared to centralized approaches but requires proper synchronization to achieve optimal results [139]. Moreover, the lack of global information in decision-making mandates synchronization between decision-makers, resulting in an apparent trade-off between performance and communication overhead.

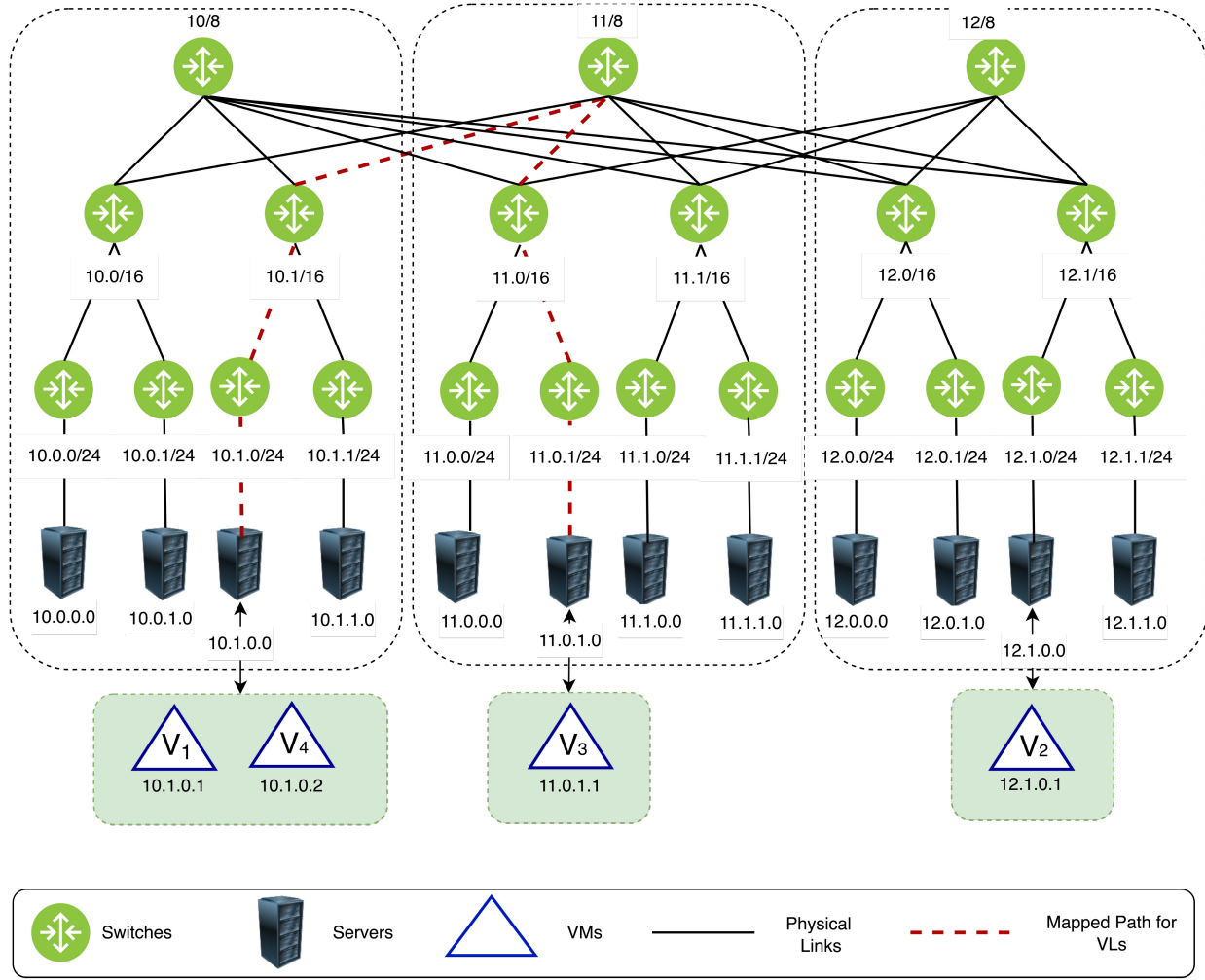


Fig. 17. An instance depicting the embedding of the VNR in Figure 1.

Precise or Redundant (P/R): Ideally, the VNR components, i.e., a VM and a VLs, are assigned to a substrate server and path having requisite resources. Such embedding policies are precise as they map the VNRs with minimum possible substrate resources without foreseeing the possibility of failure. However, reserving redundant resources for failure-sensitive applications is critical to recovering from unforeseen failures [140], [141]. An instance of such mapping is reserving multiple paths for a VL and splitting the requisite bandwidth over the selected paths. Unlike single-path mapping, the multi-path reservation policy keeps the communication channel intact when faced with failures in any of the paths. An obvious consequence of reserving additional resources is the embedding cost, and the users are often required to strike a trade-off between cost and reliability.

SP-Centric or User-Centric (S/U): Depending on the objective of the embedding strategies, the VNE schemes can be either SP-centric or user-centric. Note that if the VNE approaches focus on boosting the SP-specific metrics such as the embedding cost, revenue, revenue-to-cost ratio, and acceptance ratio, they are categorized as SP-centric. However, if the goal is to optimize parameters such as delay, throughput,

and path length directly impacting the user service, it is referred to as user-centric.

For ease of representation, we pick up the first character for each type. The character [S/D] denotes if the strategy caters to static or dynamic setups. Likewise, [C/D], [P/R], and [S/U] respectively denote if the VNE strategies are centralized or distributed, precise or redundant, SP-centric or user-centric. Moreover, an embedding belonging to S/C/R/U essentially denotes the static, centralized, redundant, and user-centric strategy. Such a representation policy eases the representation and increases the readability.

F. Embedding Metrics

To assess the performance of different embedding strategies, we consider multiple embedding metrics that can broadly be classified as *quality-of-service (QoS) based*, *cost-based*, *resilience-based*, and *other metrics* [15], [142], [143]. A visual representation/classification of different embedding metrics in the literature is provided in Table VI. A brief insight into each category and type is subsequently presented.

TABLE VI
THE RELEVANT METRICS USED IN THE LITERATURE TO EVALUATE DIFFERENT VNE SCHEMES

Embedding Metric	Category	Description
<i>Path Length</i>	<i>QoS</i>	It captures the number of substrate links reserved to provision a VL. A longer path length indicates a higher reservation of resources and escalated packet forwarding delays.
<i>Stress Levels</i>	<i>QoS</i>	It indicates the number of virtual entities assigned to a substrate entity. A higher stress level can negatively impact applications' QoS.
<i>Utilization</i>	<i>QoS</i>	It is the ratio between aggregating the resources provisioned/spent due to embedding to the total capacity of a substrate entity, i.e., a substrate server or a substrate link.
<i>Delay</i>	<i>QoS</i>	It is the maximum tolerable delay for communication between any two communicating VMs mapped onto the substrate servers.
<i>Embedding Cost</i>	<i>Resource Spending</i>	It refers to the aggregate amount of substrate resources expended in provisioning the VNRs, which includes the server resources for provisioning the VMs and the bandwidth resources for provisioning the VLs.
<i>Embedding Revenue</i>	<i>Resource Spending</i>	It is the total amount of resources a VNR requests considering both the VMs and VLs constituting it.
<i>Cost/Revenue</i>	<i>Resource Spending</i>	It is the ratio of resources provisioned to the demanded resource for a VNR.
<i>Acceptance Ratio</i>	<i>Resource Spending</i>	It captures the ratio between the number of VNRs successfully embedded and the number of VNRs received.
<i>Backups</i>	<i>Resilience and robustness</i>	It captures the number of backups reserved for a virtual component, i.e., VMs and VLs.
<i>Path redundancy</i>	<i>Resilience and robustness</i>	It is the ratio between the number of backup and direct paths.
<i>Node redundancy</i>	<i>Resilience and robustness</i>	It captures the number of running and backup nodes.
<i>Network criticality</i>	<i>Resilience and robustness</i>	It is a measure of robustness that dictates the ability of a network to be stable irrespective of environmental changes such as (1) changes in traffic request, (2) changes in topology and connections, and (3) changes in the state of transmitters and receivers.
<i>Algorithm runtime</i>	<i>Other</i>	It refers to the total execution overhead generating the embedding.
<i>Communication overheads</i>	<i>Others</i>	This captures the overhead for the exchange of messages between DCs or, in some cases, SPs. A higher degree of exchange can result in network congestion and delayed processing.
<i>Migration Overheads</i>	<i>Others</i>	It captures the relocation overheads of VMs and/or VLs. Such operations often incur overheads in relocation, including additional resources consumed/released, service disruption time, and time expended in generating the relocation.
<i>Security Features</i>	<i>Others</i>	This includes security features that are often added to improve embedding reliability.
<i>Energy Concerns</i>	<i>Others</i>	This parameter encapsulates the energy consumed by the nodes and/or links provisioned by the VNRs.

1) *Quality-of-Service (QoS)-Based*: These metrics measure service quality post-embedding of VNRs.

- 1) *Path Length*: It reflects the number of substrate links reserved to provision a VL. A longer path length indicates a higher reservation of resources and escalated packet forwarding delays, thereby deleteriously impacting the latency requirements of real-world applications such as video conferencing.
- 2) *Stress Levels*: It indicates the number of virtual entities assigned to a substrate entity. The more entities mapped, the higher the stress levels. A higher stress level can negatively impact applications' QoS. For instance, a substrate server loaded with multiple VMs keeps the operating system busy, resulting in degraded responses. On the other hand, a highly stressed physical link can result in congestion and packet transmission delay due to resource sharing.
- 3) *Utilization*: It is the ratio between aggregating the resources provisioned/spent due to embedding to the total capacity of a substrate entity, i.e., a substrate server or a substrate link.
- 4) *Delay*: Many real-world applications such as online gaming, video conferencing, and distributed computing often impose stringent restrictions on the maximum tolerable delay for communication between any two communicating VMs mapped onto the substrate servers. The VNE strategies must ensure that the delay requirements of the users are met [144].

2) *Resource Spending-Based*: These metrics discuss the number of resources expended in VNRs provisioning.

- 1) *Embedding Cost*: Embedding cost refers to the aggregate amount of substrate resources expended in provisioning the VNRs, which includes the server resources for provisioning the VMs and the bandwidth resources for provisioning the VLs. The dominant factor is the VL

embedding cost, as they are mapped onto substrate paths comprising multiple physical links. Other resources, such as storage, can also be considered while computing the embedding cost.

- 2) *Embedding Revenue*: It is the total amount of resources a VNR requests considering both the VMs and VLs constituting it.
- 3) *Cost/Revenue*: It is the ratio of resources provisioned to the demanded resource for a VNR. This metric helps compare different embedding protocols often tested over different topologies that make the embedding cost or revenue-based comparisons unfair.
- 4) *Acceptance Ratio*: It captures the ratio between the number of VNRs successfully embedded and the number of VNR requests received.
- 3) *Resilience and Robustness-Based*: These metrics focus on computing the resilience and reliability of an embedding.
 - 1) *Backups*: It captures the number of backups reserved for a virtual component, i.e., VMs and VLs. These backups can be used when the primary entity fails.
 - 2) *Path redundancy*: It is the ratio between the number of backup and direct paths. These redundant paths can be used as backups in case of total/partial failures of the primary path.
 - 3) *Node redundancy*: It captures the number of running and backup nodes. In contrast to path redundancy, this metric captures the demanded resources only and not the connectivity.
 - 4) *Network criticality*: It is a measure of robustness that dictates the ability of a network to be stable irrespective of environmental changes such as (i.) changes in traffic request, (ii.) changes in topology and connections, and (iii.) changes in the state of transmitters and receivers. The detailed computation of network criticality can be found in [145].

TABLE VII
COMPARATIVE STUDY OF DIFFERENT VNE SCHEMES BASED ON THE COORDINATION BETWEEN VME AND VLE

Coordination Strategy	Selective References	Description	Pros	Cons
Uncoordinated	[118], [120], [121] [88], [117]	VME and VLE are solved independently, with VME followed by VLE.	1. Simpler Implementation.	1. Sub-optimal resource Assignment. 2. Higher Embedding Costs. 3. Resource Fragmentation. 4. Prone to embedding failures. 5. Poor QoS guaranties.
Single Stage Coordinated	[146]–[148] [149], [150]	VME and VLE are solved simultaneously, mapping VMs and VLs.	1. Joint Optimization of VME and VLE. 2. Improved resource Utilization. 3. Lower embedding failures.	1. Higher Implementation complexity.
Two Stage Coordinated	[151]–[153]	VME is solved first with VLE requirements in mind, followed by a second stage for optimized VLE.	1. Balanced resource usage and scalability. 2. Suited for larger networks.	1. Less optimized than a single stage. 2. Higher risks of sub-optimal embedding.

4) *Other Metrics*: These focus on different aspects of embedding, such as security, energy, migration, and communication overheads during embedding/re-embedding.

- 1) *Algorithm runtime*: It refers to the total execution overhead generating the embedding. It is often a crucial metric for comparative studies of different embedding, especially for real-time applications.
- 2) *Communication overheads*: For distributed setups, synchronization is quintessential and involves the exchange of messages between DCs or, in some cases, SPs. A higher degree of exchange can result in network congestion and delayed processing.
- 3) *Migration Overheads*: For dynamic setup, there are scenarios where the VNRs have to be reorganized, implying the relocation of VMs and/or VLs. Such operations often incur overheads in terms of relocation, including additional resources consumed/released, service disruption time, and time expended in generating the relocation. Altogether, we categorize such costs as migration overheads.
- 4) *Security Features*: To avoid malicious attacks on VMs and mapped VL paths, security features are often added to improve embedding reliability.
- 5) *Energy Concerns*: As the VNR uses substrate resources such as CPU, memory, and bandwidth, energy is expended to use such resources. Therefore, this parameter encapsulates the energy consumed by the nodes and/or links provisioned by the VNRs.

G. Coordination Between VME and VLE

As discussed in Section III-A, VNE consists of two interdependent sub-problems: VME and VLE. Solving VME independently, without considering VLE, often results in suboptimal solutions due to a limited search space. Such methods, known as uncoordinated approaches, solve VME

first, using its output as input for VLE. Recent uncoordinated solutions, as explored in [88], [117], [118], [120], [121], aimed to improve the revenue-to-cost ratio and resource utilization in substrate networks. Although promising, these approaches face several challenges: (1) restricted VLE search space leading to suboptimal resource utilization, (2) higher embedding costs due to inefficient VME, increasing bandwidth usage, (3) increased likelihood of embedding failures when VLE lacks sufficient bandwidth resources after VME, (4) poor network performance with higher latency and inefficient paths, (5) difficulty in meeting diverse QoS requirements without joint optimization, and (6) added computational overhead when reprocessing is required due to embedding failures.

In contrast, VME and VLE can be coordinated using one of two approaches. The first is a single-stage approach, where VME and VLE are solved simultaneously, mapping the corresponding VLs immediately after the VMs are mapped. As discussed in [146], [147], [148], [149], [150], this approach jointly optimizes both VME and VLE, avoiding the inefficiencies of uncoordinated methods where sub-optimal VM placement can limit the VLE search space. Considering VMs and VLs holistically reduces latency, improves path efficiency, and minimizes the risk of embedding failures, especially in applications requiring strict performance guarantees, like low latency or high bandwidth. The second method is a two-stage approach, as illustrated in [151], [152], [153], where VME is first performed with future VLE requirements in mind. In the second stage, VLE is solved based on VM placement, ensuring optimized link embedding. Although the one-stage approach offers tighter resource optimization, it can be computationally expensive. In contrast, the two-stage method provides better scalability and flexibility, making it more suitable for more extensive networks or scenarios where computational efficiency is a priority without sacrificing significant performance. Table VII compares different VNE

schemes based on the coordination between the VNE and VLE.

H. Existing Simulation Frameworks

The research community has developed numerous simulation frameworks to design, develop, test, and validate embedding policies. This is especially advantageous for researchers due to the elevated expenses of conducting tests using commercial providers. Developed in C, GLPK (GNU Linear Programming Kit) [154] is an open-source solution tailored for tackling linear programming (LP) and mixed-integer programming (MIP) challenges. While it boasts a high-level modeling language for problem description, users must independently integrate SN and VNR inputs in a compatible format. However, notable limitations include its failure to address real-world scenarios comprehensively and its lack of support for multi-cloud environments and SDN. Many VNE schemes utilize randomly generated VNRs and SNs. However, tools like Georgia Tech Internetwork Topology Models (GT-ITM) [155], initially developed in C++ and later enhanced by various works, offer advanced features. GT-ITM facilitates the creation of diverse network topologies with customizable parameters such as node degree distribution, link bandwidth, and network size. It encompasses various network models, including random, power-law, and small-world graphs, enabling researchers to simulate a broad spectrum of network scenarios. Despite these capabilities, GT-ITM has limitations, notably its challenge in accurately capturing the complexities of real-world networks. While they can generate synthetic topologies with specified characteristics, they may not fully represent the intricate structures and behaviors in actual networks. Additionally, GT-ITM lacks inherent support for modeling dynamic network changes, potentially limiting its suitability for simulating evolving network environments.

Alternatively, a Python-based package called NetworkX Tool [156] has also been adopted for creating and studying complex networks in [157], [158]. It allows users to capture and analyze different graphs (directed, undirected, weighted, etc.). It is compatible with other Python libraries, such as NumPy and Matplotlib, for data analysis and visualization. Nevertheless, NetworkX has its limitations. It struggles with the performance of handling extensive graphs, particularly compared to specialized graph libraries implemented in lower-level languages such as C or C++. Although NetworkX is adequate for small to medium-sized graphs, users might face challenges when dealing with exceptionally large-sized VNRs and SNs. Additionally, the absence of built-in support for parallel computing can impact scalability in large-scale network analysis.

On the other hand, the authors in [142] introduced the ALEVIN toolkit, which integrates functionalities from GLPK [154]. Developed in JAVA, ALEVIN is a Graphical User Interface (GUI)-based tool that empowers users to customize SN and VNR topologies, incorporating characteristics such as node degree, link capacity, and traffic demands. It also provides a multi-layer visualization platform for developing VNE algorithms and includes various assessment measures

like revenue cost and hidden hop [159], [160], [161], [162], [163], [164]. However, ALEVIN is confined to single cloud implementations and lacks support for SDN. Additionally, ALEVIN entails longer computational times and presents complexity in terms of customization [142]. Alternatively, some research [88], [117], [120], [165] have also used another JAVA-based simulator called CloudSim [166] with BRITE topology generator [167]. This framework integrates real-time workload generators, time-sensitive scheduling algorithms, and event-driven techniques to simulate and evaluate real-time scenarios comprehensively. It offers a comprehensive modeling framework covering various aspects of cloud computing, including VMs, cloud services, DCs, and workloads. However, it lacks support for modern technologies such as SDN. Another framework, VNE-SIM [168], is a discrete event simulator developed in C++. It uses the rapid network simulation setup to construct the SN, whereas the VNRs are generated using the Boston University Representative Topology Generator [169], [170], [171]. However, VNE-SIM lacks an accurate representation of the intricate nature of real-world networks and multi-cloud architecture. Consequently, its utility in analyzing VNE strategies is limited.

Authors in [172] introduced Alib Utility, a Python-based simulator featuring built-in real-world SN topologies sourced from the *Internet Topology Zoo* [173], [174], [175]. Each substrate network's topology information is stored in a *yml* file containing data provided by network operators from publicly available sources. Since its inception in 2020, several works, including [122], [131], [175], [176], have made modifications and enhancements. For machine learning (ML)-based implementations, works like [177], [178] utilized Python versions 3.6/3.7 and PyTorch versions 1.2/1.4 [179]. Despite its numerous advantages, integrating VNE strategies into Alib Utility proves challenging due to its intricate structure. Additionally, it primarily caters to traditional VNE problem-solving and lacks support for SDN [175].

Potential Research Gaps: While numerous simulators exist in the literature, they suffer from several limitations, failing to offer comprehensive platforms for researchers in both industry and academia. These shortcomings include inadequate support for (1) handling evolving networks (both SNs and VNRs), (2) accurately emulating enterprise networks, (3) lacking SDN features, and (4) providing precise and accurate lab-scale testing and validations. Drawing from the preceding discussion, it is evident that existing tools suffer from *limited customizability*, *scalability constraints*, and *lack of flexibility*, thereby impeding their widespread adoption and usage. Moreover, traditional network management tools are tailored for static workloads with manual management and configuration of devices, unlike SDN, which relies on programmability and centralized control. Furthermore, these tools lack essential APIs or interfaces for programmatically controlling and configuring network elements, and they struggle to seamlessly integrate with SDN controllers, thereby missing opportunities to leverage the controller's capabilities for *network monitoring*, *traffic engineering*, and *policy enforcement*.

TABLE VIII
COMPARING THE FEATURES OF EXISTING SIMULATION PLATFORMS FOR VNE

Tool	Launch Year	Features									
		Language Support				Deployment Type		Open Source Availability	SDN Support	Scalability Tests	Integration With Real-World Data
		C	C++	Java	Python	Single Cloud	Multi Cloud				
GT-ITM [155]	1996	X	✓	X	X	X	X	✓	X	X	✓
GLPK [154]	2000	✓	X	X	X	X	X	✓	X	✓	X
Networkx Tool [156]	2008	X	X	X	✓	X	X	✓	X	X	X
CloudSim [166]	2009	X	X	✓	X	✓	✓	✓	X	✓	✓
ALEVIN [142]	2011	X	X	✓	X	✓	X	✓	X	✓	X
VNE-SIM [168]	2019	X	✓	X	X	✓	X	✓	X	X	X
Alib Utility [172]	2020	X	X	X	✓	✓	✓	✓	X	✓	✓

TABLE IX
LIMITATIONS OF THE EXISTING SIMULATION TOOLS

Tool	Limitation(s)
GT-ITM [155]	- Real-world networks complexities ignored. - No support for modeling evolving networks.
GLPK [154]	- Users were responsible for integrating VNR and SN topologies in a format compatible with the ILP solver. - Additional plugin support to simulate real-world scenarios. - No support for multi-cloud SN and SDN.
NetworkX Tool [156]	- No support for exceptionally large-sized VNRs and SNs. - No support for scalability analysis.
ALEVIN [142]	- Limited to single cloud infrastructure. - Topology customization is time consuming.
CloudSim [166]	Lacks support for modern technologies such as SDN.
VNE-SIM [168]	- Lacks real-world integration. - No support for multi-cloud SN and SDN.
Alib Utility [172]	- Complex Integration. - No support for multi-cloud SN and SDN.

Table VIII presents a comparative analysis of features such as supported languages, deployment environment, open-source availability, and support for real-world technology, and data integration among various simulation tools documented in the literature. We also summarize each tool's pitfalls in Table IX.

IV. EXACT AND APPROXIMATE VNE STRATEGIES

The VNE problem has been demonstrated to be $\mathcal{NP}$ -Hard [44]. Similarly, the issues of VME and VLE are also computationally intractable. Consequently, as the size of the problem increases, the complexity of finding an optimal solution within an acceptable time frame rises exponentially. Given these challenges, researchers have mainly adopted two different solution approaches, namely (1) *exact*, and (2) *approximate*. The solution strategies have been further sub-classified as per Figure 18.

In this context, the *exact* methods produce optimal solutions for small-scale problem instances. However, they prove to be *compute-intensive*, *time-consuming*, and *non-scalable* when confronted with more prominent test cases. Despite these drawbacks, exact solutions serve as benchmarks for evaluating the optimality bounds of *approximate* approaches. To address the limitations inherent in exact methods, some researchers have adopted *approximation* techniques, aiming to generate

satisfactory solutions within reasonable time constraints [182]. Building upon the optimization strategies outlined, we delve into the literature, thoroughly exploring each solution space and examining the proposed methods.

A. Exact Solutions

As mentioned, exact solutions guarantee optimal outcomes by thoroughly exploring all feasible solutions within a problem's search space. Various exact algorithms employ different techniques tailored to achieve optimality across diverse problem domains. *Branch and bound* algorithms meticulously navigate the search space, pruning unpromising branches to efficiently find optimal solutions, particularly effective for $\mathcal{NP}$ -Hard problems. Conversely, *exact exponential* algorithms traverse the entire solution space exponentially, ensuring optimality but often becoming computationally prohibitive for more significant instances. Subsequently, we discuss some exact solutions based on the strategies discussed above.

1) *Exact Exponential Solutions*: To achieve the optimal solution for VNE, it is essential to formulate it as an Integer Linear Programming (ILP) problem and address associated sub-problems such as VME and VLE. ILP involves optimizing a linear objective function under linear constraints, where decision variables must be integers. Solving ILP poses computational challenges in practical applications [15], but optimization solvers like CPLEX [183] and GLPK [184] are widely used to handle ILP instances effectively.

2) *Branch and Bound Solutions*: In their respective studies, the researchers in [180], [181] devised streamlined path-based models for online VNE, employing a branch-and-bound approach enhanced with column generation. Specifically, [181] employed a branch-and-price framework within the branch-and-bound procedure to handle integrity constraints, amalgamating column generation to efficiently obtain lower bounds for branch pruning. This framework, capable of tracking the optimal solution's lower and upper bounds, can produce a near-optimal solution with reduced computational time, ensuring per-instance closeness to optimality. Conversely, [180] introduced a path-based model named P-VNE, which utilizes dual formulations and a column generation process within a branch-and-bound framework to achieve efficient and optimal solutions in practical contexts, distinguishing it from the previous approach.

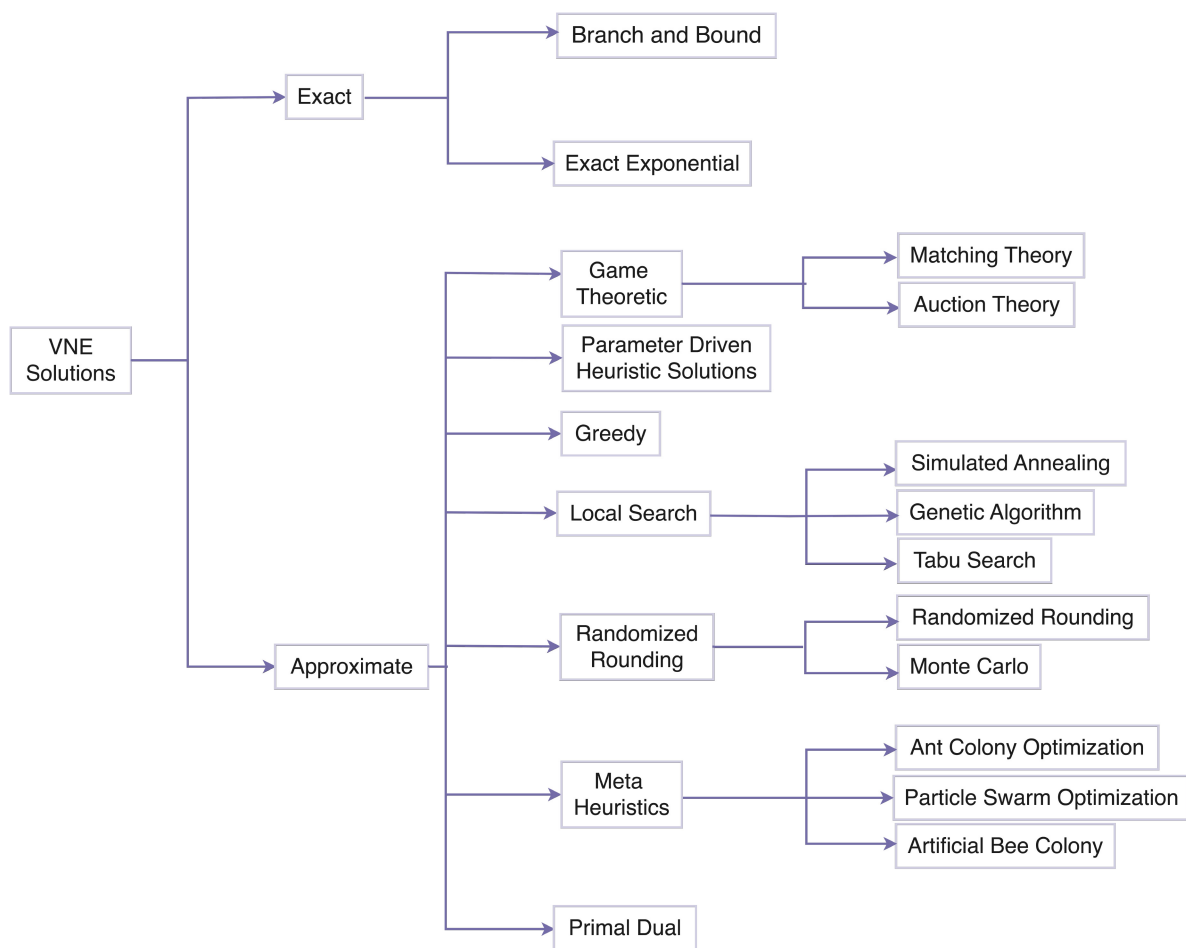


Fig. 18. Classification of exact and approximate VNE solutions.

TABLE X
INSIGHT ON THE EXACT VNE SCHEMES

Work	Optimization Mechanism	Use Cases	Pros	Cons
Hu <i>et al.</i> (2013) [180]	Column Generation	Smaller and medium-sized networks (VNRs and SN)	1. Optimal solution. 2. Works for medium-sized networks.	1. Compute-intensive 2. Time-consuming
Wang <i>et al.</i> (2016) [181]	Branch & Price	Smaller networks (VNRs and SN)	1. Almost-optimal solution with lower and upper bounds. 2. Per-instance closeness to optimality.	1. Compute-intensive 2. Time-consuming

TABLE XI
COMPARATIVE STUDY OF EXACT VNE SCHEMES

Reference	Quality-of-Service				Resource Spending				Resilience and Robustness				Others				
	q_1	q_2	q_3	q_4	s_1	s_2	s_3	s_4	r_1	r_2	r_3	r_4	o_1	o_2	o_3	o_4	o_5
Hu <i>et al.</i> (2013) [180]	X	X	✓	✓	X	X	✓	X	X	X	X	X	X	X	X	X	X
Wang <i>et al.</i> (2016) [181]	X	X	X	X	X	X	X	X	X	X	X	X	✓	X	X	X	X

Tables X and XI present a comprehensive comparative analysis of the exact VNE strategies. The first table outlines each scheme's optimization techniques, use cases, advantages, and limitations. In contrast, the latter analyzes the schemes based on the embedding metrics discussed in Section III-F.

B. Approximate Solutions

Approximate solutions encompass methods that offer near or almost-optimal solutions to complex optimization problems, vital when exact solutions are computationally impractical or unfeasible. Techniques such as primal-dual, randomized

TABLE XII
INSIGHT ON THE PRIMAL DUAL AND RANDOMIZED VNE SCHEMES

Work	Optimization Mechanism	Use Cases	Pros	Cons
Melo <i>et al.</i> (2013) [185]	Primal-dual and column generation	Small and medium-sized Networks (VNRs and SN)	1. Optimality guarantee. 2. Flexibility in constraint handling. 3. Polynomial-time approximation.	1. Computational complexity. 2. Sub-optimal solutions. 3. Resource fragmentation.
Hu <i>et al.</i> (2014) [186]	Primal-dual with MILP formulation	Medium and large-sized networks (VNRs and SN)	1. Almost optimal solutions. 2. Precise problem formulation. 3. Structured solution approach.	1. Computational complexity. 2. Tight feasibility margins. 3. Memory intensive. 4. Input sensitive.
Mijumbi <i>et al.</i> (2015) [187]	Primal-dual and column generation	Large-sized networks (VNRs and SN)	1. Almost optimal solutions. 2. High acceptance rate. 3. Higher utilization of SN resources. 4. Lower time complexity than optimal.	1. Limited to single domain SN. 2. Non-polynomial convergence.
Zhang <i>et al.</i> (2017) [188]	Primal-dual and dynamic programming	Small and medium-sized networks (VNRs and SN)	1. Polynomial time solution. 2. Higher link utilization. 3. Bounded VLE.	1. Compute-intensive. 2. Not VNE-optimal.

algorithms, local search, heuristic algorithms, meta-heuristics, and game-theoretic approaches aim to swiftly converge to nearly optimal solutions, prioritizing efficiency over absolute optimality. These approaches are widely adopted, particularly in solving challenges like VNE, where we explore numerous solutions proposed in the literature utilizing such methodologies.

1) *Primal-Dual Solutions*: The primal-dual solution approach solves linear programming (LP) problems by simultaneously addressing primal and dual formulations. It works on the philosophy of iteratively updating primal and dual variables that optimize the objectives while adhering to different constraints. This approach is efficient in combinatorial optimization problems such as VNE, offering efficient and often polynomial-time solutions. In the context of VNE, solutions based on primal-dual analysis are explored in works like [185], [187]. These studies propose scalable and coordinated VNE approaches: the former uses primal-dual analysis with column generation in a two-stage process involving path-based VM and VL assignment followed by dual variable pricing. At the same time, the latter employs multi-commodity flow constraints to balance resource consumption across substrate resources. Alternatively, in [186], the VNE problem was modeled using mixed-integer linear programming (MILP), ensuring near-optimal or almost-optimal solutions for each instance. Hu et al. utilized a primal-dual approach with iterative feedback between VM and VL mapping stages. In contrast, Zhang and Tang in [188] developed a polynomial-time approximation algorithm based on dynamic programming and primal-dual analysis to address the VLE problem under minimum hop constraints.

2) *Randomized Solutions*: Randomized rounding is a technique used to convert fractional solutions of linear programs into integer solutions through probabilistic decisions. Each fractional variable is rounded to 1 with a probability equal to its value and 0; otherwise, the expected values match the original solution. This method optimizes problems where exact integer solutions are challenging, preserving the problem's structure while providing near-optimal solutions. Tables XII and XIII analyze and provide the comparative

studies of randomized and primal dual-based VNE solutions.

3) *Local Search-Based Solutions*: Local search iteratively improves an initial solution by examining nearby solutions within a restricted region of the solution space. While these solutions do not ensure global optimality, they efficiently find a locally optimal solution within a practical time frame guided by heuristic principles. This presents an appealing method for addressing the VNE problem, particularly considering its exponential sample space. Researchers have thus turned to techniques such as *Tabu search*, *Genetic Algorithms (GA)*, and *Greedy Algorithms* to obtain locally optimal solutions within manageable time constraints.

Tabu Search: It iteratively explores solution spaces by preventing revisits to recently explored solutions or components, known as the “tabu list.” This philosophy aids the procedure in intelligently navigating the solution space while avoiding cycles and local optima and encourages exploration of diverse solution regions [189], [190], [191]. Moreover, Tabu search also employs diversification and intensification strategies to broaden the search and enhance promising solutions. Some effective VNE solutions powered by Tabu-search can be found in [192], [193].

The work in [192] modeled the multi-cloud SN and the VNRs as undirected weighted graphs captured as $\mathcal{G}^{Sn} = (\mathcal{N}^{Sn}, \mathcal{L}^{Sn})$ and $\mathcal{G}^{Vr} = (\mathcal{N}^{Vr}, \mathcal{L}^{Vr})$, respectively. The authors developed QoS-aware cross-domain splitting and embedding strategies across multiple providers. The authors made the following key contributions: (1) Developed a VNR splitting strategy leveraging limited information disclosed by providers considering their resource dynamics. The splitting efficiency was enhanced by considering providers' non-confidential information, such as network topologies and points-of-presence (PoPs). (2) The splitting was modeled as an ILP with constraints, and provisioning costs were defined based on the availability of providers' advertised resources. Tabu search was adopted to obtain a polynomial time solution with reduced communication overhead between providers and higher acceptance of VNRs. On the other hand, Ayoubi et al. [193] emphasized embedding multicast VNRs

TABLE XIII
COMPARATIVE STUDY OF PRIMAL-DUAL AND RANDOMIZED VNE SCHEMES

Reference	Quality-of-Service				Resource Spending				Resilience and Robustness				Others				
	q_1	q_2	q_3	q_4	s_1	s_2	s_3	s_4	r_1	r_2	r_3	r_4	o_1	o_2	o_3	o_4	o_5
Melo <i>et al.</i> (2013) [185]	✗	✓	✓	✗	✗	✗	✓	✓	✗	✗	✗	✗	✓	✗	✗	✗	✗
Hu <i>et al.</i> (2014) [186]	✗	✗	✗	✗	✓	✗	✗	✗	✗	✗	✗	✗	✓	✗	✗	✗	✗
Mijumbi <i>et al.</i> (2015) [187]	✗	✗	✗	✗	✗	✓	✗	✓	✗	✗	✗	✗	✓	✗	✗	✗	✗
Zhang <i>et al.</i> (2017) [188]	✓	✗	✓	✗	✗	✓	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗

TABLE XIV
INSIGHT ON TABU-SEARCH-BASED VNE SCHEMES

Work	Optimization Mechanism	Use Cases	Pros	Cons
Ayoubi <i>et al.</i> (2015) [193]	Dynamic programming based Tabu-search	Multi-cast VNRs	1. Higher SP revenue. 2. Balanced resource utilization. 3. Lower blocking ratio.	1. Limited to multi-cast VNRs only. 2. Limited to static VNRs.
Diallo <i>et al.</i> (2020) [192]	MILP-based Tabu-search	Multi-cloud VNE	1. Reduced communication overheads between SPs. 2. Reduced computing time. 3. Scalable protocol.	1. Limited to small-sized problem instance. 2. Limited to single objective optimization.

while adhering to end-delay and delay-variation constraints of various applications. The authors made several novel contributions: (1) They formulated the multicast VNE and theoretically established its intractability. (2) They proposed a novel three-step solution to the multicast VNE, introducing terminal embedding over multicast trees. A restrictive solution based on dynamic programming was discussed for homogeneous resource demands over multicast trees with a constant nodal degree. (3) They also extended the prior solution for heterogeneous resource demands over arbitrary SN topologies using Tabu search. Tables XIV and XV highlight the comparative study of the above two discussed schemes.

Genetic Algorithm: GA draws inspiration from natural selection, mimicking its mechanisms to effectively solve optimization and search problems. It employs bio-inspired operations like *selection*, *crossover*, and *mutation* to produce superior solutions [194]. The *selection* operator distinguishes between inferior and superior solutions, retaining the latter to improve convergence and solution quality. Different selection operators like *tournament selection*, *roulette-wheel selection*, *proportionate selection*, and *rank selection* are discussed in detail in [194]. The *crossover* operator produces fresh solutions, often referred to as offspring, by leveraging existing solutions within the population post-selection. Commonly, this operation entails swapping genetic information, specifically genes, between two parent chromosomes that have been selected. While the execution of crossover operations remains discretionary, its inclusion aids in diversifying the population, steering clear of local optima, and accelerating convergence rates. Following the identification of parent chromosomes, offspring can be engendered via single-point, multi-point, or uniform crossover techniques. Alternatively, the *mutation* operations randomly adjust the genetic information within chromosomes, generating fresh and varied solutions. Like crossover, the primary aim of *mutation* is to enhance population diversity and comprehensively explore the search space. Various methods for *mutation* can be employed, such as

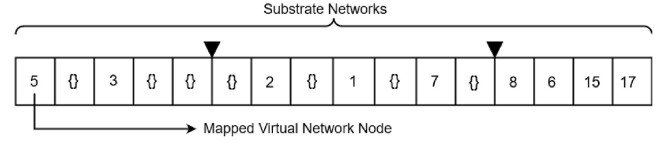


Fig. 19. Encoding of chromosomes in [195].

bit-flip, swap, and scramble operations. A critical factor in generating high-quality solutions with GA is achieving a balance between *exploration* and *exploitation* of the solution space. Exploration involves investigating new search areas to uncover diverse solutions, while exploitation intensively examines a restricted region of the current solution. Typically, *crossover* and *mutation* facilitate exploration, whereas the *selection* operator supports exploitation.

Different strategies for GA-based solutions typically differ in how they encode the population and configure the initial solution space, impacting the quality of the solutions they produce [196]. Some works solely focused on VME are discussed in [197], [198], [199], [200], [201]. On the other hand, more complex GA-based offline-VNE solutions, encompassing VME and VLE, have been proposed in the literature, such as those presented in [195], [197], [202], [203], [204], [205]. In [195], the authors focused on a set of SNs ($\mathcal{G}^{Sn}$) and VNRs ($\mathcal{G}^{Vr}$), implementing an embedding strategy based on classical GA. The encoding of a solution is depicted in Figure 19, where each allele represents a substrate node, and the corresponding numeral in the allele represents the hosted VM $v_i \in \mathcal{N}^{Vr}$. An empty allele denotes an unassigned substrate node. With this encoding, the authors employed a multi-point crossover, and only feasible solutions proceeded for bit-flip mutation. The objective was to enhance the SPs' long-term revenue and acceptance ratio. However, the solution suffered from delayed convergence.

To address these shortcomings, Zhang et al. [202] proposed a modified genetic algorithm, VNE-MGA. The population

TABLE XV
COMPARATIVE STUDY OF TABU SEARCH-BASED VNE SCHEMES

Reference	Quality-of-Service				Resource Spending				Resilience and Robustness				Others				
	q_1	q_2	q_3	q_4	s_1	s_2	s_3	s_4	r_1	r_2	r_3	r_4	o_1	o_2	o_3	o_4	o_5
Ayoubi <i>et al.</i> (2015) [193]	×	×	✓	✓	×	✓	×	✓	×	×	×	×	×	×	×	×	×
Diallo <i>et al.</i> (2020) [192]	×	×	×	✓	×	×	×	✓	×	×	×	×	✓	×	×	×	×

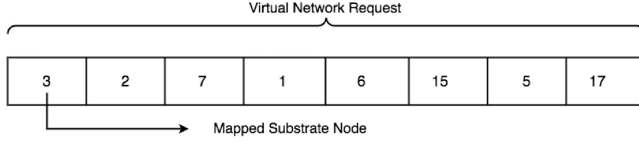


Fig. 20. Encoding of chromosomes in [202].

encoding is depicted in Figure 20, where the numerals represent the substrate node ($s_k \in \mathcal{N}^{Sn}$) onto which the VM ($v_i \in \mathcal{N}^{Vr}$) is mapped. Unlike the poor-quality, randomly generated initial solutions in [195], the authors implemented sophisticated mechanisms to develop the initial population. They adopted a partial mapping crossover, generating two random numbers r_1 and r_2 , such that $1 \leq r_1, r_2 \leq 6$, and exchanged genetic information between two selected parents. This was followed by an enhanced mutation operation where already mapped VMs were mutated with server indices in closer vicinity to reduce the VLE cost and boost the revenue-to-cost ratio.

Alternatively, Zhang et al. [203] adopted the same encoding as in Figure 19. However, they made some changes to improve the solution qualities compared to the work in [195]. The first change includes a probabilistic selection of the parents ranked in increasing order of the mapping cost instead of the biased selection of best-performing parents in [195]. Secondly, the crossover operation is upgraded and is only performed if the parents have dissimilar alleles, resulting in distinct children and assisting in exploring the search space. Thirdly, due to the restrictive crossover operation, the convergence of the crossover is compromised owing to similar optimal solutions. Therefore, the restriction on crossover for dissimilar parents is bounded by a threshold. This specific operation assists the solution in evading the local optima, boosting the execution time, and embedding cost reduction. Deviating from the classical GA-based embedding solutions, Martinez-Julia et al. [204] proposed a genetic-programming (GP)-based one-stage coordinated technique that discovers a sub-optimal solution by continuously seeking solution space within a bounded time and fixed iterations, thereby ensuring a minimum distance from the optimal solution. Compared to previous GA-based solutions, GP-based solutions provide faster convergence and up to $7\times$ better performance than the initial solutions.

VNRs frequently encounter fluctuating resource demands for VMs and VLs. This necessitates complete or selective reassignment to adapt to dynamic resource needs, balance

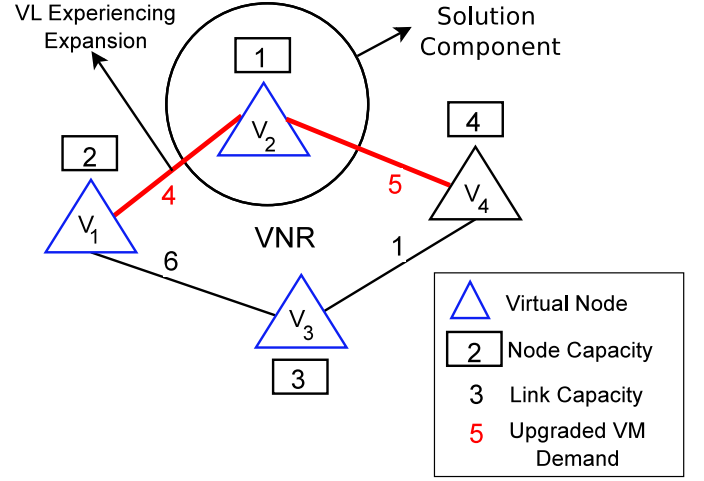


Fig. 21. An instance of solution component [129].

the load, reduce power consumption, and perform routine maintenance. In this regard, GA-based online embedding schemes catering to various goals have been explored in several studies, including [118], [121], [129], [206], [207], [208], [209], [210]. Dab et al. [129] specifically focused on selectively re-embedding solution components (SCs) that comprise a star topology with a central VM and its attached VLs with at least one of the VL facing bandwidth expansion. An instance of the SC centered at v_2 with VLs connecting VMs v_1 and v_4 , respectively, experiencing an expansion of 2 and 1 unit is illustrated in Figure 21. Note that relocating the SC also encompasses VM and VL embedding. The authors used a similar encoding scheme as shown in Figure 20, where gene entries capture the index of the migrated SC or 0 if none. The order of SCs in the chromosome reflects their migration sequence. Starting with a randomly generated initial population, the authors implemented an improved crossover method by computing the similarity index $SIM(C_1, C_2)$ of parents C_1 and C_2 , set to 1 if the genetic information matches for both parents and 0 otherwise. The hamming distance $H(C_1, C_2)$ indicating the number of 1's in $SIM(C_1, C_2)$ is then calculated. The crossover point is then identified depending upon $H(C_1, C_2)$, i.e., if $H(C_1, C_2) > 2$, a single point crossover operation is performed anywhere between $\alpha(C_1, C_2)$, where $\alpha(C_1, C_2)$ denotes the distance between the occurrence of the first 1 and the last 1, respectively. On the other hand, if $H(C_1, C_2) \leq 2$, the crossover is skipped, and the selected parents proceed directly to the bit-flip mutation. This method ensures the generation of distinct offspring and

the computation of fitness, aiming to reduce re-embedding costs and migration overheads.

On the other hand, works in [118], [121] discussed GA-based solutions focused on the complete re-embedding of online VDCs over a multi-domain substrate network. Motivated by the improved crossover and mutation operations in [129], the strategies focused on optimizing the re-embedding cost and migration overheads. From a computational complexity perspective, GA-based solutions are compute-intensive, and accurately estimating their complexity remains an open problem [202]. However, in [118], the authors attempted to compute the asymptotic complexity of their proposed VDC re-embedding scheme, identifying the most time-consuming operations: (1) selection, (2) crossover, (3) mutation, and (4) feasibility check. The selection operation consumes $O((P_{size})^2)$ time, where P_{size} is the population size. The crossover and mutation operations have complexities of $O(P_{size} * |\mathcal{N}^{Vr}|)$ and $O(|\mathcal{N}^{Vr}| * |\mathcal{N}^{Sn}|)$, where $\mathcal{N}^{Vr}$ and $\mathcal{N}^{Sn}$ denote the number of VMs in the VDC and substrate servers, respectively. Finally, the feasibility check, which verifies whether the VLs are mapped onto substrate paths with required resources, consumes $O(|\mathcal{L}^{Vr}| * |\mathcal{L}^{Sn}| * |\mathcal{L}^{Sn}|)$, where $\mathcal{L}^{Vr}$ and $\mathcal{L}^{Sn}$ represent the VLs and substrate links, respectively.

On the other hand, Nguyen et al. discussed two online VNE in [208], [209]. In [208], an independent two-stage procedure to assign resources to VNRs is discussed. In the first stage, VME is realized by considering the network topology attributes and resources that rank the virtual and substrate nodes before the embedding. The second stage adopts a distributed parallel GA-based strategy for VL mapping. The authors observed significant gains concerning resource utilization, average revenue-to-cost ratio, and operational overheads. To facilitate better coordination between VME and VLE, a joint resource allocation mechanism is proposed in [209]. To assist the GA-based procedure in achieving improved VLE, the authors consider the following modifications: First, in coordination with VM mapping, three path selection algorithms, (1) random path searching, (2) sequential single path searching, and (3) k-shortest path method is considered to boost the success of VLE. Secondly, to deal with infeasible VL mappings, the GA-based exploration procedure is augmented with a conciliation-based mechanism to minimize the relocations.

Bio-inspired operations such as crossover, mutation, and selection consume substantial time. To address this concern, Martinez-Julia et al. [206] employed GA that reduces the problem's upper bound to a lower order of complexity compared to previous solutions in the literature. The authors designed a specific heuristic for selecting configurations from the search space, evaluating the heuristic function during the construction of each configuration. These modifications significantly reduce execution time and drive the solution towards optimality. Alternatively, Lu et al. [207] restricted their focus to VLE and presented two distributed parallel GA-based solutions with different crossover and mutation operations. These algorithms utilize two variations of crossover and mutation schemes to achieve low latency and high efficiency.

On the other hand, Zhou et al. [211] addressed a more realistic scenario involving the batch arrival of multiple VNRs. Existing embedding solutions do not exploit the relationships between VNRs during the embedding process. This work proposes an optimal selection algorithm based on a GA to identify which VNRs should be hosted when the available SN resources are insufficient to accommodate all incoming VNRs. This approach aims to maintain a high acceptance ratio and maximize long-term revenue. To enhance robustness in the embedding process, the authors in [213] proposed a novel VNE strategy using a non-dominated sorting genetic algorithm (NSGA-II) to balance network centrality with the cost-to-revenue ratio. This approach distinguishes itself from existing literature in several ways. Firstly, the authors address the issue of robustness in the SN, a topic that has been largely neglected, by using classical and spectral graph robustness metrics to assess the network. Secondly, to avoid imbalanced mapping and improve resource efficiency, the strategy incorporates four-node attributes alongside dynamic capacity, employing TOPSIS to achieve an optimal solution. Thirdly, the proposed scheme conducts a single-stage VNE, in contrast to the two-stage methods commonly found in the literature, and considers resources for network nodes such as switches or routers.

For all the two-stage VNE schemes discussed above, the VMs are first mapped, followed by VL mapping between the mapped source and destination VM pairs. In contrast, Lu and Huang [205] proposed a coordinated one-stage VNE scheme wherein the VMs and VLs are mapped together, producing more optimal VNE owing to a higher degree of search space exploration. The encoding of the population identifies each gene entry as a path between two meta-nodes in the augmented graph. The authors adopt a random population generation strategy to obtain a feasible set of chromosomes, which then proceed to selection, crossover, and mutation operations. The authors adopt a single-point crossover operation with randomly selected parents and resolve the node mapping conflicts to avoid loops. However, the mutation operation is tricky, as the chosen gene may include a meta or substrate node. The latter replaces the path between intermediate substrate nodes with an alternative path randomly selected from path pools. However, in the former, the node mapping requires mutation followed by feasibility checks to ensure connectivity.

Researchers in [214] and [212] focused on generating the embedding of multi-cast VNRs in different environments. The former is dedicated to generating efficient embedding for tree topology-based VNRs algorithm to minimize the network resource consumption adhering to the end-to-end delay of applications. On the other hand, the latter provides a quick solution to the problem of VNE in a wireless environment, addressing interference and multicast properties, and demonstrates its effective performance and rapid execution compared to traditional optimization methods. On the other hand, [215] proposed a GA-based approach for various multicast protection schemes, including dedicated dual-tree, shared dual-tree, dedicated subgraph, and shared subgraph. For a given network topology and a group of optical multicast VNRs, the proposed scheme targeted a total reduction in the spectrum usage, number of blocked requests, and unprotected

TABLE XVI
COMPARATIVE STUDY OF GA-BASED VNE SCHEMES

Reference	Quality-of-Service				Resource Spending				Resilience and Robustness				Others				
	q_1	q_2	q_3	q_4	s_1	s_2	s_3	s_4	r_1	r_2	r_3	r_4	o_1	o_2	o_3	o_4	o_5
Dab <i>et al.</i> (2013) [129]	X	X	✓	X	✓	✓	X	✓	X	X	X	X	X	X	✓	X	X
Zhou <i>et al.</i> (2016) [211]	X	X	X	X	X	✓	X	✓	X	X	X	X	✓	X	X	X	X
Pathak <i>et al.</i> (2017) [195]	X	X	✓	X	✓	✓	X	✓	X	X	X	X	X	X	X	X	X
Zhang <i>et al.</i> (2019) [202]	X	X	✓	X	✓	✓	X	✓	X	X	X	X	✓	X	X	X	X
Afifi <i>et al.</i> (2019) [212]	X	X	X	X	X	X	X	X	X	X	X	X	✓	X	X	X	X
Shoosharian and Safaei (2019) [213]	X	X	X	X	X	X	✓	X	X	X	X	✓	X	X	X	X	X
Lu <i>et al.</i> (2020) [207]	X	X	✓	✓	X	X	X	X	X	X	X	X	✓	X	X	X	X
Satpathy <i>et al.</i> (2020) [121]	X	X	X	X	✓	X	X	X	X	X	X	X	X	X	✓	X	X
Nguyen <i>et al.</i> (2020) [208]	X	X	X	X	✓	✓	✓	✓	X	X	X	X	✓	X	X	X	X
Zhang <i>et al.</i> (2021) [203]	X	X	X	X	✓	X	X	✓	X	X	X	X	✓	X	X	X	X
Julia <i>et al.</i> (2021) [204]	X	X	✓	X	X	✓	✓	✓	X	X	X	X	✓	X	X	X	X
Julia <i>et al.</i> (2022) [206]	X	X	X	X	X	X	X	X	X	X	X	X	✓	X	X	X	X
Nguyen <i>et al.</i> (2022) [209]	X	X	X	X	✓	✓	✓	✓	X	X	X	X	✓	X	X	X	X
Guler <i>et al.</i> (2023) [214]	X	X	X	✓	✓	X	X	✓	X	X	X	X	X	X	X	X	X
Baez <i>et al.</i> (2023) [215]	X	X	✓	X	X	X	X	✓	X	✓	X	X	X	X	X	X	X
Satpathy <i>et al.</i> (2024) [118]	X	X	X	X	✓	X	✓	✓	X	X	X	X	X	X	X	X	X
Qiao and Huang (2024) [205]	✓	✓	X	X	X	X	X	✓	X	X	X	X	X	X	X	X	X

incrusted requests. Table XVI summarizes the literature on GA-based VNE solutions, highlighting the metrics considered.

Greedy Strategies: A greedy strategy is an approach to problem-solving that operates as follows: making a series of choices, each locally optimal, to obtain a global optimum. Some greedy-based online VNE strategies have been proposed in [128], [216], [217], [218], [219], [220]. The work in [216] focused on reducing the total operational cost and consolidating online VNRs onto the fewest servers across federated SNs. Specifically, the authors concentrate on (1) initial embedding, (2) decreased resource requirements, (3) increased resource requirements, (4) inter-data center migration, and (5) periodic consolidation. For the initial embedding, randomly selected a $v_i \in \mathcal{N}^{Vr}$ are assigned to a feasible server meeting the resource demands. The assignment procedure selects adjacent VMs in decreasing order of their communication bandwidth and prioritizes assignment over the same SN. Once embedded, the resource demand for VM and/or VLs may decrease, leading to (i) removal of VM, (ii) removal of VLs, or (3) release of VM/VL resources. On the other hand, increased resource requirements can lead to (i) increased resources for VMs, (ii) increased resources for VLs, or (iii) the addition of VMs. These additional resources sometimes require relocation/migration of VMs and/or VLs, and the authors discussed minimum cost (migration time/bandwidth) relocation techniques for the same. The authors, during embedding, ensure that the VLs are mapped with minimum inter-SN costs (in terms of backbone network bandwidth consumption). The dynamic updates in the VM and/or VL demands may render the initial assignment infeasible, especially in cases such as (i) removing expired VL, (ii) reducing the bandwidth allocation of existing VL, and (iii) increasing the bandwidth allocation of existing VL. These use cases require periodic consolidation to ensure the backbone network is not congested.

Focusing on topology-aware embedding schemes, Cao et al. [162], [163] discussed the VNE-NTANRC algorithm for offline environments, which considers network topology attributes and resources. In [162], they proposed two sub-approaches: one calculates all Node Values (NoVs) directly, while the other, inspired by the PageRank method, calculates NoVs in a stable state. Additionally, they introduced two VNE-NTANRC strategies, VNE-NTANRC-D and VNE-NTANRC-S, based on these sub-approaches. Unlike previous embedding schemes that only consider node capacity and link bandwidth, VNE-NTANRC accounts for four node-link constraints: (1) node location, (2) node capacity, (3) link bandwidth, and (4) link propagation delay. The consideration of the propagation delay ensures that mapped VNRs provide delay-guaranteed services. After calculating the NoVs, a two-stage approach is employed to solve the embedding problem: first, a greedy VM-to-server assignment, followed by shortest-path VLE to enhance substrate resource utilization. In [163], a generalized VNE energy cost model derived from various existing models was introduced. The authors adopted a novel node ranking approach to optimize embedding energy by quantifying multiple essential energy and revenue-related topological attributes and computing stable node ranking values iteratively using the Markov random model. The proposed ER-VNE algorithm efficiently embeds each VNR in polynomial time, considering multiple node-link constraints such as CPU, node storage, node location, link bandwidth, and link propagation delay. Alternatively, Hashmi and Gupta [221] also adopted the Google PageRank algorithm to assign ranks to the substrate servers based on the network's global and local topological features and resources. Similar to the works in [162] and [163], the proposed technique operates in two stages: in the first stage, a node ranking function is devised to identify the optimal physical node, and in the second

stage, both the hops of the paths and available bandwidth are analyzed to determine the optimal path for link mapping. The proposed scheme aims to boost the long-term acceptance and average revenue-to-cost ratios.

To add vigor to the ranking procedure, Ding et al. [222], T. G. et al. [122], and Guler et al. [223] devised novel node ranking schemes considering multiple correlated topological attributes. In this context, the work in [122] proposed a two-stage VNE procedure called NORD. To tackle the VME, NORD employs a hybrid entropy and TOPSIS-based ranking strategy for VMs and servers, considering systems and topological attributes. The former includes CRB and bandwidth strength, whereas the latter consists of metrics degree centrality, betweenness centrality, and Eigen centrality to capture dependencies effectively. Once the ranking is generated, a greedy VM embedding is followed by shortest path VL embedding to complete the assignment. The other works in [222], [223] respectively also considered centrality metrics, such as Eigen, and closeness in addition to the CPU, memory, and bandwidth constraints to rank the VMs. Once the ranking is obtained, a greedy VME, followed by the shortest-path VLE, completes the embedding procedure. To test the embedding over different topological attributes, Feng et al. [224] and Zhang et al. [225] discussed VNE methodologies that consider different combinations of topological attributes corresponding to the substrate servers and/or VMs for ranking the VMs, which is followed by VME and shortest-path VLE.

In [224], the authors focused on seven topological attributes: degree, strength, betweenness centrality, fairness, closeness centrality, Katz centrality, and Eigen centrality. Considering these topological metrics, the authors designed three unique topology-aware VNE algorithms by intelligently opting for a subset of attributes to better coordinate the VME and VLE. The paper [225] proposed a two-stage approach for VME. First, a new five-node ranking metric was considered to analyze network topologies to assess node importance. Second, the simplified ELECTRE method for MCDM is used for comprehensive evaluation to prevent unbalanced embedding. The proposed VNE algorithm greedily selects nodes with maximum potential for mapping and then utilizes a shortest path algorithm for link mapping.

Deviating from the previously discussed schemes, the authors in [227] identified common topological structures, such as paths and cycles, within VNRs and SN and performed VNE accordingly. For path embedding, they proposed transforming the problem into a Multiple Knapsack Problem (MKP) and a Multi-Dimensional Knapsack Problem (MDKP) by decomposing the SN into paths. They constructed a Weighted Directed Auxiliary Graph (WDAG) for cycle embedding to establish a one-to-one correspondence between each directed cycle in WDAG and each feasible embedding. Additionally, they presented a polynomial-time greedy solution that achieves the least resource-consuming VNE, thereby boosting the SP's revenue-to-cost ratio and overall revenue.

Alternatively, in [128], an online live migration strategy called VDC-M for re-embedding dynamic VNRs across geo-distributed SNs is presented. The authors consider the dependency among the VMs of the VNR in the re-embedding

process. The re-embedding procedure executes in two steps. The VM re-embedding plan is generated first, followed by the migration of VMs along the computed path with requisite bandwidth. This work aims to reduce re-embedding costs, blocking ratios, and migration overheads. The VM re-embedding plan is generated first, followed by the migration of VMs along the computed path with requisite bandwidth. This work aims to reduce re-embedding costs, blocking ratios, and migration overheads. To address VNR rejections caused by bottleneck substrate links, Fajjari et al. [217] proposed a reactive algorithm. The authors focused on re-embedding star components consisting of a central VM and its attached VLs (refer to Figure 21) rather than re-embedding the entire VNR. This approach reduces VN rejections and accommodates more VNRs at a minimal re-embedding cost. Similarly, Cao et al. [218] introduced two algorithms to enhance the resource utilization of the SN. The initial embedding procedure manages the initial deployment of virtual components (VMs and VLs), while the QoS-aware re-embedding is triggered when the virtual components' demands are unmet. Focusing solely on resource management, Giles et al. [220] developed a resource management scheme that utilizes VM migrations to reduce fragmented resources caused by the dynamic arrival and departure of online VNRs. Depending on the state of the SN and the demanded VNR resources, authors either adopt a greedy heuristic with zero migrations when there is no fragmentation and sufficient resources or use a Subgraph Mixed Integer Program with possible migrations to embed a request online when resources are fragmented and inadequate.

Divergent from the above-discussed schemes, Duan et al. [219] looked at the VNE problem from a novel perspective. Instead of merely devising a mapping strategy for VNRs, the authors aimed at identifying the minimum amount of SN resources, such as switches and links, required in a fat-tree SN to host multi-cast VNRs under resource constraints. They proposed two VNE schemes for multi-cast VNRs. In the first scheme, the VMs are placed in close vicinity to reduce the embedding cost without interfering with the pre-existing traffic. The second scheme re-distributes a minimal number of VMs using VM migrations to achieve an even more compact placement. This study offers practical insights into provisioning the SN infrastructure to alleviate bottlenecks and optimize computing resource utilization in SNs.

Though the above-discussed schemes were effective, they were limited to offline setups with static VNRs. In contrast, the work in [228] developed a dynamic VNE scheme considering storage, adjacent links and node resources, computing, and topological parameters for ranking to enhance SN resource utilization; however, it suffered multiple rejections. Lu et al. [226] proposed a collaborative embedding and re-embedding policy to remedy this. This policy was based on a uniquely defined resource importance metric for nodes and links, considering multiple topological attributes. Finally, Table XVII provides a comparative study of greedy VNE schemes.

4) *Parameter Driven Heuristic Solutions*: Heuristic solutions are strategies for tackling complex optimization problems like VNE efficiently within a reasonable time frame. These

TABLE XVII
COMPARATIVE STUDY OF GREEDY VNE SCHEMES

Reference	Quality-of-Service				Resource Spending				Resilience and Robustness				Others				
	q_1	q_2	q_3	q_4	s_1	s_2	s_3	s_4	r_1	r_2	r_3	r_4	o_1	o_2	o_3	o_4	o_5
Feng <i>et al.</i> (2014) [224]	X	X	X	X	✓	✓	✓	✓	X	X	X	X	X	X	X	X	X
Duan <i>et al.</i> (2015) [219]	X	X	X	X	✓	X	X	X	X	X	X	X	X	X	✓	X	X
Ding <i>et al.</i> (2015) [222]	✓	X	X	X	X	X	✓	✓	X	X	X	X	✓	X	X	X	X
Guler <i>et al.</i> (2016) [223]	✓	X	✓	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Sun <i>et al.</i> (2017) [216]	X	X	X	X	✓	X	X	X	X	X	X	X	X	X	X	X	✓
Gilesh <i>et al.</i> (2017) [220]	X	X	X	X	X	X	X	✓	X	X	X	X	X	X	✓	X	X
Sun <i>et al.</i> (2018) [128]	X	X	X	X	✓	X	X	✓	X	X	X	X	X	X	✓	X	X
Cao <i>et al.</i> (2018) [163]	X	X	✓	X	X	X	✓	✓	X	X	X	X	X	X	X	X	X
Cao <i>et al.</i> (2018) [162]	X	X	✓	X	X	X	✓	✓	X	X	X	X	X	X	X	X	X
Zhang <i>et al.</i> (2018) [225]	X	X	X	X	✓	X	✓	✓	X	X	X	X	✓	X	X	X	X
Lu <i>et al.</i> (2018) [226]	X	✓	✓	X	✓	✓	✓	✓	X	X	X	X	X	X	X	X	X
Cao <i>et al.</i> (2019) [218]	X	X	✓	✓	X	X	X	✓	X	X	X	X	X	X	X	X	X
Hashmi and Gupta (2020) [221]	✓	X	X	X	X	✓	✓	✓	X	X	X	X	✓	X	X	X	X
Wu <i>et al.</i> (2020) [227]	X	X	✓	X	X	✓	X	✓	X	X	X	X	X	X	X	X	X
Xia <i>et al.</i> (2022) [228]	X	✓	✓	X	X	✓	X	✓	X	X	X	X	X	X	X	X	X
Tg <i>et al.</i> (2023) [122]	X	X	X	X	X	X	✓	✓	X	X	X	X	X	X	X	X	X

solutions are practical and efficient, often yielding approximate results rather than exact ones. The search for solutions relies on intuition or educated guesses, making them ideal for $\mathcal{NP}$ -Hard problems where exhaustive searches are impractical, and they usually utilize domain-specific knowledge to improve performance. These schemes can be classified based on the specific metrics or objectives they aim to optimize, helping to determine their primary focus and select the appropriate algorithm for the problem.

In this context, some energy-aware VNE schemes are discussed in the literature [158], [160], [229], [232], [233], [234], [236], [239], [241], [244], [246], [247], [250]. The work in [247] adopted node and link capabilities to develop a location-constrained energy-aware VNE strategy for enterprise networks. On the other hand, Amokrane et al. [229] and Triki et al. [233] proposed green resource management strategies to minimize the carbon footprint in provisioning VNR resources through a two-phase process: partitioning VNRs and assigning sub-VNRs to substrate resources based on electricity prices, power usage, renewable resource availability, request priorities, and carbon footprint. Pham et al. [160] aimed to balance traffic load and energy consumption in embedding multiple VNRs over an interconnected SN using a multi-objective heuristic for VME and SDN-assisted VLE, achieving energy minimization by tracking and deactivating inactive servers. Gong et al. [250] examined energy consumption variations across servers and proposed a joint ranking strategy using TOPSIS, considering the energy quotient and revenue for SPs in embedding decisions. Alternatively, the authors in [236] and [246] developed energy-saving mechanisms to optimize server and physical link energy by reorganizing embedded VNRs during off-peak periods. Extending the works to multi-domain SNs, the proposals in [244] and [241] presented energy-efficient models operating in multiple stages,

providing reasonable performance improvements compared to single-domain protocols. For VNRs with fluctuating resource demands, the research by Chen and Li [232] discussed two VLE strategies based on undirected network minimum cost flow to consolidate link resources quickly. The work in [234] outlined two embedding algorithms that leverage the dynamic characteristics of VNRs to reduce energy consumption without compromising the SP's revenue. Alternatively, Nam et al. [239] presented server consolidation strategies for embedding and re-embedding dynamic VDCs, focusing on resource and energy efficiency as VDCs continuously join and leave the system. On the other hand, Guan et al. [158] proposed an energy-efficient solution that considers expected energy consumption at different times, virtual network operation, future migration costs, and data center architecture for online VNE.

To address failures, several reliable VNE schemes have been discussed in the literature [231], [240], [242], [248], [249]. Zhang et al. [242] improved embedding reliability by using a Markovian random walk model to determine topology-aware reliability rankings for substrate nodes and proposed two embedding strategies, RRRW-MaxMatch and RDCC-VNE, aimed at enhancing QoS and QoE of embedded services. Extending reliable embeddings to wireless networks, Li et al. [249] focused on QoS improvement by considering the mean link quality of all outgoing links, guiding the protocol to map virtual links to paths with lower hop counts. On the other hand, works in [240] and [245] designed robust backup topologies to enhance link survivability and balance the load. Liu et al. used a reserved quota shared via multiplexing to achieve these objectives, while Zheng et al. proposed dividing networking resources into primary and secondary flows, used during normal operations and failures, respectively. To adhere to SLAs during physical infrastructure failures, the authors

TABLE XVIII
COMPARATIVE STUDY OF PARAMETER-DRIVEN HEURISTIC VNE SCHEMES

Reference	Quality-of-Service				Resource Spending				Resilience and Robustness				Others				
	q_1	q_2	q_3	q_4	s_1	s_2	s_3	s_4	r_1	r_2	r_3	r_4	o_1	o_2	o_3	o_4	o_5
Amokrane <i>et al.</i> (2013) [229]	✓	✗	✗	✗	✗	✓	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗
Gong <i>et al.</i> (2014) [230]	✗	✗	✗	✗	✗	✓	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗
Liao <i>et al.</i> (2014) [231]	✗	✗	✗	✗	✓	✗	✗	✗	✗	✗	✓	✗	✗	✗	✗	✗	✗
Chen <i>et al.</i> (2014) [232]	✗	✗	✗	✗	✓	✓	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✓
Triki <i>et al.</i> (2015) [233]	✗	✗	✓	✓	✓	✓	✗	✗	✗	✗	✗	✗	✓	✗	✗	✗	✓
Zhang <i>et al.</i> (2015) [234]	✗	✗	✗	✗	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓
Shanbhag <i>et al.</i> (2015) [235]	✓	✗	✗	✗	✓	✓	✓	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗
Guan <i>et al.</i> (2015) [158]	✗	✗	✗	✗	✗	✗	✗	✓	✗	✗	✗	✗	✓	✗	✓	✗	✓
Ghazisaeedi and Huang (2015) [236]	✗	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓
Jiang <i>et al.</i> (2016) [237]	✓	✓	✓	✗	✗	✗	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗
Chowdhury <i>et al.</i> (2016) [238]	✓	✗	✗	✗	✗	✗	✓	✓	✗	✓	✗	✗	✓	✗	✗	✗	✗
Nam <i>et al.</i> (2017) [239]	✗	✗	✓	✗	✗	✗	✗	✓	✗	✗	✗	✗	✗	✗	✓	✗	✓
Liu <i>et al.</i> (2017) [240]	✗	✗	✗	✗	✓	✓	✗	✓	✗	✓	✗	✗	✗	✗	✗	✗	✗
Dahir <i>et al.</i> (2017) [241]	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓
Zhang <i>et al.</i> (2018) [242]	✗	✗	✗	✗	✗	✓	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗
Gao <i>et al.</i> (2018) [243]	✗	✗	✓	✗	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗
Cao <i>et al.</i> (2018) [244]	✗	✗	✗	✗	✓	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓
Zheng <i>et al.</i> (2019) [245]	✓	✓	✓	✗	✗	✓	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗
Liu <i>et al.</i> (2019) [246]	✗	✗	✗	✗	✗	✗	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✓
Cong <i>et al.</i> (2020) [247]	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	✗	✗	✓
Pham <i>et al.</i> (2020) [160]	✓	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	✗	✗	✓
Shahriar <i>et al.</i> (2020) [248]	✗	✗	✗	✗	✓	✗	✗	✗	✗	✓	✗	✗	✓	✗	✗	✗	✗
Li <i>et al.</i> (2021) [249]	✓	✗	✗	✗	✗	✗	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗

in [238] allow SPs to provision redundant backup resources for VNRs. One approach is to reserve a dedicated backup for each VNR component, enabling fast VNR recovery and high uptime guarantees for SPs. This paper proposes 1 + 1-Protected VNE (1 + 1-ProViNE) and implemented DRONE, formulates integer linear programming (ILP) encompassing two solutions (1) optimal (OPT-DRONE) and a (2) heuristic (FAST-DRONE) to manage assignment computational complexity.

To boost the resiliency of VNRs, Sun et al. [251] discussed a survivable embedding strategy represented as a MILP. They presented two solution strategies: (1) Lagrangian relaxation-based and (2) heuristic-based algorithms to balance time efficiency and embedding cost. Unlike these models, Shahriar et al. [248] proposed a weakly survivable approach with lower resource overheads. To address scalability, they decomposed the embedding into three sub-problems: (1) VNR graph augmentation, (2) identifying VLs for disjoint mapping, and (3) computing VNE with disjoint constraints, and proposed heuristics for each. While these techniques are limited to unicast VNRs, the work in [231] discussed a survivable VNE using an improved set cover algorithm for multicast VNRs.

Some load-balancing heuristics have been discussed in [230], [235], [237], [243], [252]. Shanbhag et al. [235] presented VHub, a framework that models the embedding problem as a p-hub median problem, performing VM and VL embedding in a single stage to achieve balanced VNR embedding. Gong et al. [230] defined a global resource

capacity (GRC) metric to capture the embedding potential of substrate nodes, developing a load-balanced embedding scheme that enhances revenue and the revenue-to-cost ratio for SPs. On the other hand, authors in [252] introduced a re-optimization mechanism distinguishing between node and link resources based on their impact on connectivity, providing a re-embedding strategy to accept previously rejected VNRs by fixing bottleneck requirements, thus improving the acceptance ratio and load balance across the SN. Alternatively, Gao and Rouskas [243] proposed a VNR reconfiguration strategy that dynamically reconfigures VMs and VLs to balance the load, decomposing the problem into VM selection and VM re-embedding, with the former solved as a linear programming problem and the latter using a Markovian random walk model. On the other hand, Jiang et al. [237] focused on improving transmission quality and load balancing, developing a reconfiguration algorithm to address these twin objectives and achieve optimization benefits. Finally, Table XVIII captures the comparative behavior of different heuristic schemes.

5) *Game Theoretic Approaches*: Game theory is the study of strategic decision-making where outcomes depend on the actions of multiple agents, each with their own preferences.

Matching Theory: A particular branch of game theory, matching theory, has garnered significant attention for addressing the VNE problem. Matching theory focuses on pairing agents from two (or sometimes three) sets to ensure none of the matched agents have incentives to deviate from their current assignments unilaterally, thus maintaining stability.

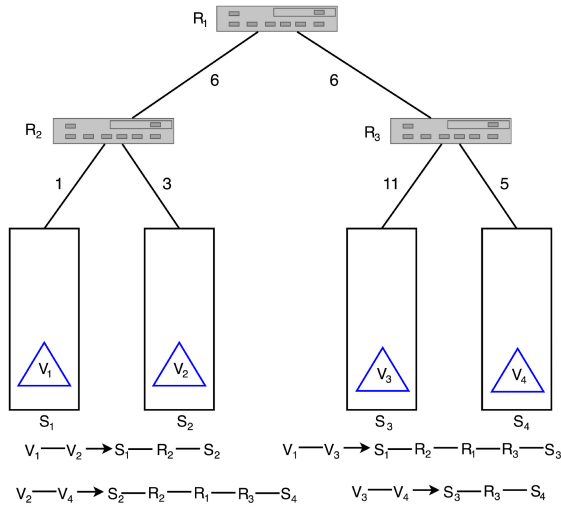


Fig. 22. A possible initial assignment of the VNR (VMs and VLs) over a two-tier SN for the VNR depicted in Figure 1. The numerals along the physical links represent their available resources.

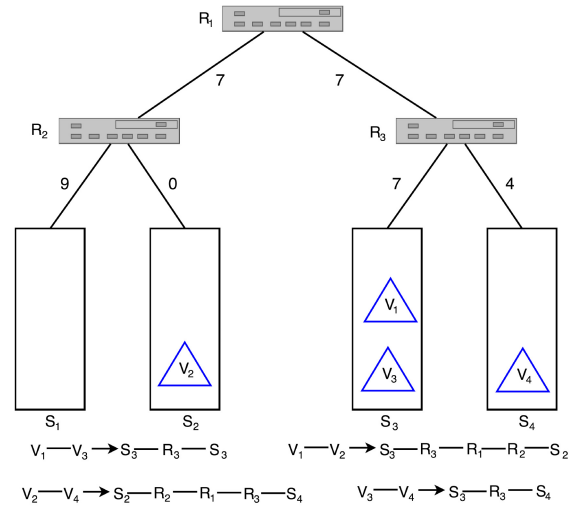


Fig. 23. A possible reassignment of the VNR (VMs and VLs) over a two-tier SN for the VNR depicted in Figure 21. The numerals along the physical links represent their available resources.

Some matching theoretic VNE solutions are presented in [88], [117], [120], [131]. In this regard, the authors in [131] modeled the VME as a one-to-many matching game between the VMs and the servers and solved it using the classical deferred acceptance algorithm (DAA) [253]. The preferences of the VMs and servers were generated using the hybridization of entropy and TOPSIS [254], [255]. The former was used to derive the criteria weights, whereas the latter ranked the alternatives. After the VME, the VLE is performed using the shortest path algorithm.

The overall objective of the work was to maximize the revenue-to-cost ratio of VNRs by considering a blend of system and topological attributes during ranking to accurately capture the dependencies among a VNR's VMs. Though the above protocol was effective, it was limited to offline VNRs. To remedy this lacuna, Satpathy et al. discussed three correlated dynamic resource reconfiguration policies in [88], [117], [120] and focused on reduced re-embedding cost and improved utilization of resources. These works specifically focused on the complex use case wherein the resource expansion of VMs and/or VLs required relocations. For instance, considering the assignment of a VNR $\mathcal{G}^{Vr}$ with VMs $\mathcal{N}^{Vr} = \{v_1, v_2, v_3, v_4\}$ and VLs $\mathcal{L}^{Vr} = \{e_{1,2}, e_{1,3}, e_{2,4}, e_{3,4}\}$. Additionally, the SN $\mathcal{G}^{Sn}$ comprises $\mathcal{N}^{Sn} = \{s_1, s_2, s_3, s_4\}$ servers and $\mathcal{L}^{Sn} = \{e_1, e_2, e_3, e_4, e_5\}$ physical links. The SN is also assumed to have three routers $\{R_1, R_2, R_3\}$. As per the above setting, a possible embedding of the VNR (refer to Figure 1) is captured in Figure 22. Note that the numerals on the physical links of the SN reflect its available resources $a(e_l)$. The VNR experiences resource expansion for the VLs interconnecting VMs v_1-v_2 and v_2-v_4 of 2 and 1 unit as captured in Figure 21.

Though the existing mapped path for VL v_2-v_4 has enough available resources to provide for the additional VL demands of 1 unit, it is impossible for the mapped path for the VL v_1-v_2 to meet the extra demand of 2 units due to the bottleneck link S_1-R_2 . To successfully host the VNR, the

additional demand for all the VLs must be met. Therefore, the provider here has no option but to relocate the VM v_1 , inevitably leading to the relocation of the VLs v_1-v_2 and v_1-v_3 . After the relocation, the VL demand of all the VLs can be easily met, and the updated embedding with the available resources can be found in Figure 23.

For such use cases, the work in [88] adopted an analytical hierarchy process (AHP)-based ranking of servers. DAA-based VME is performed in [120]; however, it fails to capture the heterogeneous demands of the VMs. Moreover, the DAA does not provide a stable allocation in such cases [123]. To overcome these issues, the work in [88] discusses a modified, revised deferred acceptance algorithm (MRDA) that generates a partially stable allocation with heterogeneous VM demands. As reflected in the example, selective relocation of SCs [256] is performed considering multiple agendas of SPs. Though [88] generated acceptable solutions, the agents' preferences were static and were computed before initiating the re-embedding procedure. The detailed working of the protocol is elucidated in the flowchart presented in Figure 24. However, such preferences can be inaccurate, especially when multiple dependent SCs of VNRs are re-embedded. This is inescapable as a dynamic snapshot of the adjacent VMs should be considered when generating the preferences of VMs that cannot be obtained apriori. To overcome this, [117] proposes a protocol that maintains an additional data structure in the form of a priority list to dynamically capture the preferences considering the current state of the substrate network and allocations of dependent VMs. From the perspective of computation complexity, the work in [88] incurs a time complexity of $O(|\mathcal{N}^{Vr}| * |C|^2)$ in generating the preferences, where C is the set of all criteria. On the other hand, the time taken by the matching is $O(|\mathcal{N}^{Vr}|^2 * |\mathcal{N}^{Sn}| + |\mathcal{N}^{Vr}|^2)$, with the assumption of $|\mathcal{N}^{Vr}| \gg |\mathcal{N}^{Sn}|$, hence the worst case complexity of VME is $O(|\mathcal{N}^{Vr}|^2)$. The VLE following the VME consumes $O(|\mathcal{N}^{Vr}|^2)$ time. Therefore the asymptotic complexity can be derived as $O(|\mathcal{N}^{Vr}| * |C|^2) + O(|\mathcal{N}^{Vr}|^2)$.

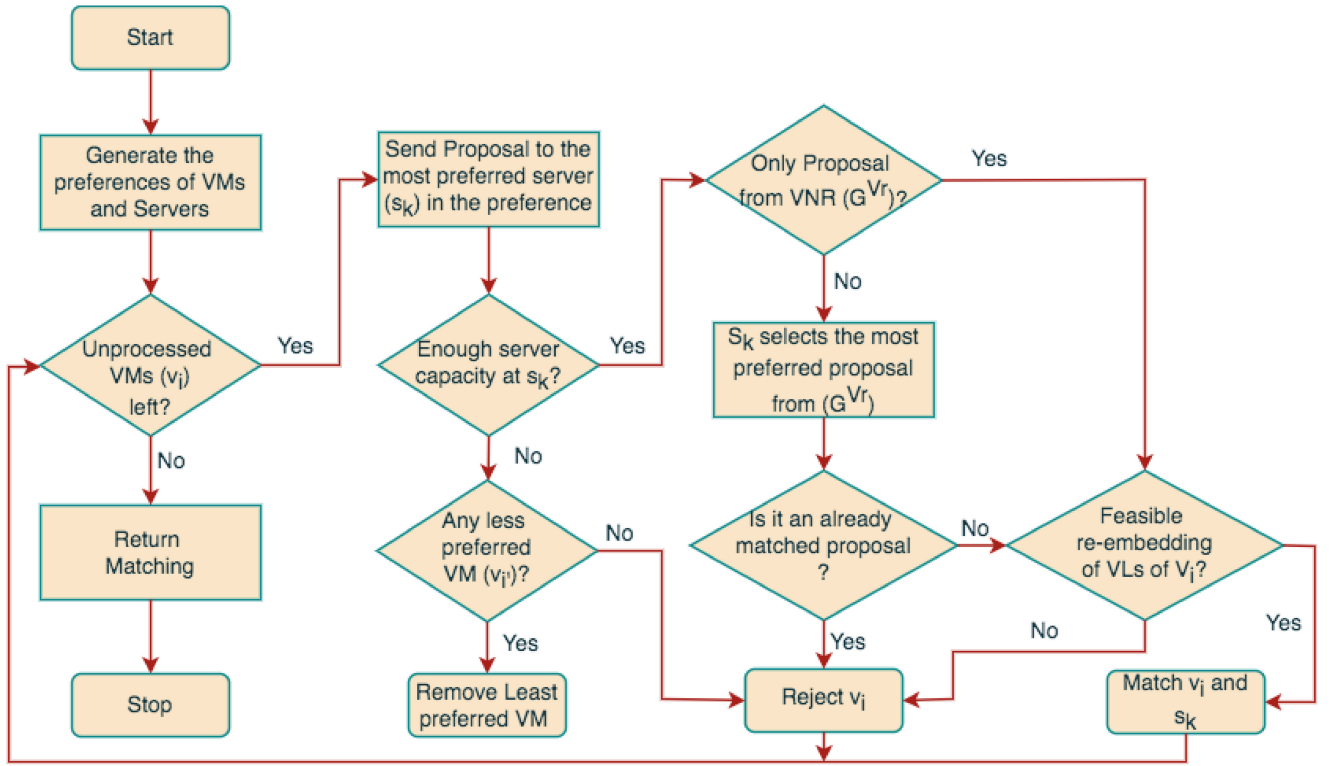


Fig. 24. The working of ReMatch [88].

Alternatively, in [117], which is a similar solution incurs a time complexity of $O(|\mathcal{N}^{Vr}|) + O(|\mathcal{N}^{Vr}| * |\mathcal{N}^{Sn}| * |\mathcal{C}|^2)$, wherein the additional $O(|\mathcal{N}^{Vr}|)$ is consumed due to normalization list (NL) construction. Note that this list captures the processing order of unprocessed VMs.

Auction Theory: In contrast to the centralized matching-based solutions discussed above, the research in [257], [258] proposed distributed resource provisioning approaches for VNRs across geo-distributed DCs based on auction theory. Deviating from the fixed pricing of substrate resources, the former adopted a dynamic pricing model to induce competition among the users in an open auction cloud market, thereby boosting the overall revenue of the SPs. The latter developed iterative bidding and consensus to embed the VMs, followed by VLE. Table XIX effectively captures the comparative study of different game-theoretic VNE schemes.

6) Meta-Heuristic Solutions: VNE can be viewed as a combinatorial problem within a discrete search space. As the problem size increases, the solution space expands, making it more challenging to find an optimal solution. In these cases, meta-heuristic approaches like ant colony optimization (ACO) [259], particle swarm optimization (PSO) [260], and various others have been employed to find sub-optimal solutions efficiently. Next, we will discuss different meta-heuristic-based solutions for VNE in greater detail.

Ant Colony Optimization: The ant colony optimization (ACO) algorithm is a population-based algorithm inspired by the foraging behavior of ants [259], [261]. Typically, these virtual agents construct solutions by traversing a graph (reflecting the solution space) guided by pheromone trails and heuristic information—the likelihood of selecting a path

in the graph is influenced by the pheromone levels and the heuristic value. After solutions are formed, pheromone levels are updated through evaporation to prevent premature convergence and deposition based on the quality of the solutions. This process iterates until a stopping criterion is reached. Some ACO-based VNE solutions are presented in [149], [210], [262], [263]. In this regard, Fajjari et al. solved the VNE problem in [149] and [262] using two different variants of Max-Min Ant system-based-ACO. The authors modeled the SN an undirected weighted graph $\mathcal{G}^{Sn} = (\mathcal{N}^{Sn}, \mathcal{L}^{Sn})$. Additionally, each $s_k \in \mathcal{N}^{Sn}$ is characterized by (i) residual processing power, (ii) residual memory, (iii) type, and (iv) geographical location. Alternatively, the physical links $e_l \in \mathcal{L}^{Sn}$ available resources are reflected in $a(e_l)$. For the VNRs, it is also modeled as an undirected weighted graph $\mathcal{G}^{Sn} = (\mathcal{N}^{Sn}, \mathcal{L}^{Sn})$. Given this setting, in [149] adopted the Max-Min Ant system, which operated as follows. (1) The SCs (refer to Figure 21) were identified and sorted. Note that the SCs are sorted based on the number of hanging links that do not have one of the virtual nodes (VMs) mapped and are processed in that order. (2) A set of ants was allowed to explore the search space, i.e., the SN graph, in parallel for a predefined number of iterations. While constructing a solution (mapping the SCs in the predefined order), an ant tries to embed it by identifying potential candidates for each SC in the search space and selecting the one using a stochastic local decision process such that the number of hops and the distance from other mapped nodes is minimized. The decisions are guided by heuristic information and pheromone trails, reflecting the transition moves' desirability. The mapping decision for the SC is taken based on a probabilistic function, considering both

TABLE XIX
COMPARATIVE STUDY OF GAME-THEORETIC VNE SCHEMES

Reference	Quality-of-Service				Resource Spending				Resilience and Robustness				Others				
	q_1	q_2	q_3	q_4	s_1	s_2	s_3	s_4	r_1	r_2	r_3	r_4	o_1	o_2	o_3	o_4	o_5
Esposito <i>et al.</i> (2016) [258]	✓	✗	✓	✗	✗	✗	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗
Satpathy <i>et al.</i> (2020) [120]	✗	✗	✓	✗	✓	✗	✓	✓	✗	✗	✗	✗	✓	✗	✗	✗	✗
Metwally <i>et al.</i> (2020) [257]	✗	✗	✓	✗	✗	✓	✗	✓	✗	✗	✗	✗	✓	✗	✗	✗	✗
Satpathy <i>et al.</i> (2022) [117]	✗	✗	✓	✗	✓	✗	✓	✓	✗	✗	✗	✗	✓	✗	✗	✗	✗
Satpathy <i>et al.</i> (2023) [88]	✗	✗	✓	✗	✓	✗	✓	✓	✗	✗	✗	✗	✓	✗	✗	✗	✗
Tg <i>et al.</i> (2023) [131]	✗	✗	✗	✗	✗	✗	✓	✓	✗	✗	✗	✗	✓	✗	✗	✗	✗

the heuristic information and pheromone deposits as per their relative importance. (3) The best solution (based on the least cost) is selected once the artificial ant constructs the solution. (4) The pheromone trails are updated by local evaporation (exploration) and reinforcement (exploitation) as per the best solution obtained. The above steps are repeated for a designated number of iterations, and the one with the best fitness is considered the final solution. On the other hand, the work in [262] adopted a similar model as discussed but focused on developing a scalable embedding policy for online VNRs over a private substrate backbone network that utilizes minimum physical resources while meeting the resource demands (QoS) of the VMs and VLs.

Deviating from the classical VNE solutions, which suffered from (1) generic embedding schemes without considering VNR topologies, thereby ignoring critical relationships between the VNR components (2) Existing ACO-based VNE schemes using graph decomposition are based on simple graph structures such as stars. (3) The existing schemes disregard reliability by allowing hosting multiple VLs on a physical path. To overcome these pitfalls, authors in [263] proposed a graph decomposition-based embedding scheme. The authors first analyzed the VN, decomposed it into ring and tree structures, and devised separate embedding policies for each. This was achieved by a pre-computation based on ant random walking based on pheromones, which also assists the protocol in identifying the ring structures. Once the structures are identified, a random node for each virtual ring (say v_0) is mapped (say s_0), and other nodes (say $v_1, v_2, \dots$) are also subsequently mapped clockwise or counterclockwise. Other nodes are mapped near s_0 to minimize VL mapping overhead. This procedure is repeated for all such ring structures and finally leads to a load-balanced assignment with higher long-term SP revenue.

In contrast to two-stage VNE solutions with node and link embedding, Zheng et al. [265] solely targeted optimizing the latter. For VME, the authors adopted three different ranking strategies considering (1) CPU demand, (2) total bandwidth of outgoing links, and (3) product of CPU and total bandwidth of outgoing links. The VMs were processed in a non-increasing order of their ranks. To optimize the VLE, the authors develop a link heuristic function (say $\eta_{i,k}$) that captures the aggregate bandwidth consumed by embedding v_i on s_k . Therefore, a smaller value of $\eta_{i,k}$ is desirable for VLE. For instance, consider a VNR with 3 VMs and 3 VLs, as depicted in

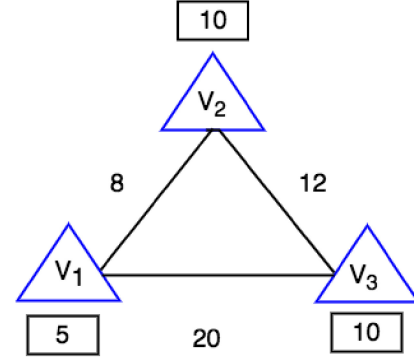


Fig. 25. A sample instance of a VNR with three VNRs and three VLs.

Figure 25. The state of the SN post embedding the VNR is captured in Figure 26. Note that the assignments partition the SN due to the exhaustion of the link resources at $S_1 - S_7$ and $S_3 - S_6$. The VL connecting v_1 and v_3 is mapped onto $S_1 - S_5 - S_4 - S_3$ (indicated by $\dots$). Now, if v_2 is embedded onto S_2 , the resource demand of VL $v_2 - v_3$ cannot be met on the link $S_1 - S_2$, hence $\eta_{2,2}$ for this assignment can be set to 0 (indicated by $\times$). On the other hand, if v_2 is hosted on S_7 or S_8 , then $\eta_{2,7} = \eta_{2,8} = 0$. However, if v_2 is embedded onto servers S_5 or S_4 , then $\eta_{2,5} = 28$ and $\eta_{2,4} = 32$, the former being a better option. Therefore, these heuristic decisions assist the ants in assignments with decreased bandwidth consumption and boost the possibility of successful VLE. A similar solution focused only on VLE was discussed in [266] wherein the VLs routes were determined using ACO over a complex topology of Aeronautical Radio Incorporated (ARINC) 664 Avionics Full Duplex Switched Ethernet (AFDX) networks, considering multiple effective parameters such as bandwidth utilization, bandwidth flexibility limit for further extensions, port usage, total harness weight, latency, and redundancy.

On the other hand, Chang et al. [210] discussed an embedding strategy focused on an ant building a VME solution. Whenever a VM is embedded, the feasibility of the assignment is verified using the k-shortest-path algorithm. Note that the solution generated by an ant is only considered feasible if all the VLs incident on the embedded VM have a successful embedding. This work aims to boost the long-term SP revenue, acceptance ratio, and embedding costs. On the other hand, [267] discussed a scalable and fast ACO-based

TABLE XX
COMPARATIVE STUDY OF ACO-BASED VNE SCHEMES

Reference	Quality-of-Service				Resource Spending				Resilience and Robustness				Others				
	q_1	q_2	q_3	q_4	s_1	s_2	s_3	s_4	r_1	r_2	r_3	r_4	o_1	o_2	o_3	o_4	o_5
Chang <i>et al.</i> (2013) [210]	X	X	X	X	✓	✓	✓	✓	X	X	X	X	X	X	X	X	X
Fajjari <i>et al.</i> (2014) [262]	X	✓	✓	X	✓	✓	X	✓	X	X	X	X	X	X	X	X	X
Qiang <i>et al.</i> (2014) [264]	X	✓	X	X	X	X	X	✓	X	X	X	X	✓	X	X	X	X
Zhu <i>et al.</i> (2016) [263]	X	✓	✓	X	✓	✓	X	✓	X	X	X	X	X	X	X	X	X
Zheng <i>et al.</i> (2017) [265]	X	X	X	X	X	X	✓	✓	X	X	X	X	X	X	X	X	X
Bayram and Ozcan (2023) [266]	X	X	✓	✓	X	X	X	✓	X	✓	X	X	X	X	X	X	X
Ikoma <i>et al.</i> (2024) [267]	X	X	✓	X	X	X	X	✓	X	X	X	X	✓	✓	X	X	X

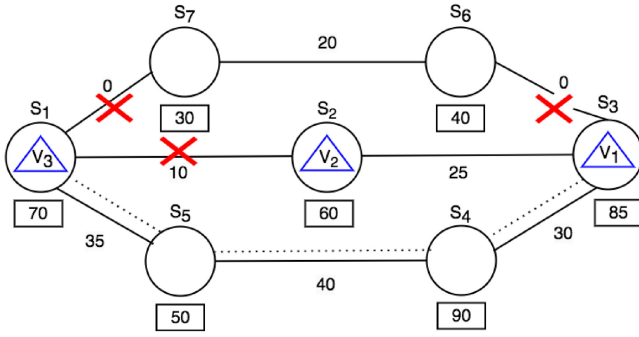


Fig. 26. A sample instance of the SN post embedding of VNR in Figure 25.

VNE scheme for disaggregated data centers considering the communication overheads. Finally, Table XX captures the characteristics of all the strategies discussed above.

Particle Swarm Optimization: PSO is an iterative optimization technique that utilizes a population of candidate solutions known as particles. Each particle adjusts its position in the solution space based on its own experience and the collective experience of the swarm, aiming for optimality by balancing exploration and exploitation [278], [279]. Like ACO, PSO operates through the following steps: (1) Initialization: Particles are initialized with random positions and velocities in the search space. (2) Evaluation: The fitness of each particle is computed. (3) Update Personal and Global Bests: Update the best-known positions based on current evaluations. (4) Velocity Update: Adjust each particle's velocity based on its own experience and that of neighboring particles. (5) Position Update: Update each particle's position based on its new velocity. (6) Termination: Repeat steps (2)-(5) until the termination condition, such as a maximum number of iterations, is met [280].

In this context, Zhang *et al.* [269] discussed a discrete PSO-based two-stage coordinated VNE protocol to address the twin objectives of revenue maximization and energy reduction. The authors modeled the SN and the VNR as weighted undirected graph $\mathcal{G}^{Sn} = (\mathcal{N}^{Sn}, \mathcal{L}^{Sn})$ and $\mathcal{G}^{Vr} = (\mathcal{N}^{Vr}, \mathcal{L}^{Vr})$. Each particle in PSO generates a solution, which is a D dimensional vector $\mathbb{X}_i = (x_i^1, x_i^2, x_i^3, \dots, x_i^D)$ and x_i^1 reflects the i^{th} particle position in the first iteration, which is later reevaluated using a fitness function. For the node mapping, the authors adopted the same D -dimensional vector, wherein x_i^1 captured

the assignment of the VM $v_1 \in \mathcal{N}^{Vr}$ corresponding to the first VNR, where D captures the cardinality of the VNR, i.e., $|\mathcal{N}^{Vr}|$. VLE follows this, making decisions based on revenue and energy parameters. Finally, the particle's velocity is updated based on addition, subtraction, and multiplication operations.

Alternatively, Song *et al.* discussed three unique one-stage coordinated embedding strategies in [271], [273], [274]. In [274], they proposed a distributed framework called HA-VNE-PSO, which operates in two steps. First, PSO is used to achieve a sub-optimal embedding for VNRs. Second, an archive-based strategy that records the latest information on frequently repeated tasks is introduced to improve performance further. In [271], a divide-and-conquer-based evolutionary approach is presented that works on the principle of cooperative co-evolution by partitioning a VNR into multiple smaller VNRs, thus aiming to improve the acceptance ratio. The research in [273] differs from the previous approaches in several ways. Firstly, the state of the particles in PSO is updated at every solution construction step, achieving better coordination between VME and VLE. Secondly, positions and velocities are encoded as discrete vectors and matrices, enhancing the search process. Thirdly, heuristic information related to the substrate path is introduced to guide the protocol toward compact placements, thereby reducing embedding costs and rejection rates.

Deviating from traditional DCNs, Chai *et al.* [275] studied the problem of VNE over SDN-enabled DCNs. The authors first discussed a hierarchical virtualization-enabled SDN architecture designed to optimize network load and embedding reliability jointly. They propose a locally optimal solution for VME and VLE using the ideal point method. After computing the local solution, deviation from optimal discrete particle swarm optimization (DPSO) is applied to achieve a globally optimal solution. Another aspect of VNE involving one-to-many (multicast) interactions, in the form of virtual multicast trees (VMTs), was explored in contrast to the traditional VNE problem, which focuses on one-to-one (unicast) communication, in [277]. The authors proposed an embedding framework inspired by VMTE-PSO to map VMTs onto a shared SN. Their approach aimed to minimize bandwidth consumption on the SN while meeting the computing demands of VMs and reducing redundant substrate link usage.

TABLE XXI
COMPARATIVE STUDY OF PSO-BASED VNE SCHEMES

Reference	Quality-of-Service				Resource Spending				Resilience and Robustness				Others				
	q_1	q_2	q_3	q_4	s_1	s_2	s_3	s_4	r_1	r_2	r_3	r_4	o_1	o_2	o_3	o_4	o_5
Guo <i>et al.</i> (2015) [268]	X	X	X	X	✓	X	X	X	X	X	X	X	X	X	X	X	X
Zhang <i>et al.</i> (2016) [269]	X	X	X	X	X	✓	X	X	X	X	X	X	X	X	X	X	✓
Wang <i>et al.</i> (2017) [270]	X	X	X	X	✓	X	X	X	X	X	X	X	X	X	X	X	X
Song <i>et al.</i> (2019) [271]	X	X	X	X	✓	✓	✓	✓	X	X	X	X	X	X	X	X	X
Boyang <i>et al.</i> (2019) [272]	X	X	X	X	X	✓	✓	✓	X	X	X	X	X	X	X	X	X
Song <i>et al.</i> (2019) [273]	X	X	X	X	✓	✓	✓	✓	X	X	X	X	X	X	X	X	X
Song <i>et al.</i> (2019) [274]	X	X	X	X	✓	✓	✓	✓	X	X	X	X	X	X	X	X	X
Chai <i>et al.</i> (2020) [275]	X	✓	X	X	X	X	X	✓	X	X	X	X	X	X	X	✓	X
Chiang <i>et al.</i> (2020) [276]	X	X	X	✓	✓	✓	✓	✓	X	X	X	X	X	X	X	X	X
Guler <i>et al.</i> (2023) [277]	X	X	✓	X	X	X	X	X	X	✓	X	X	X	X	X	X	X

Considering multi-domain SNs, some effective VNE strategies have been presented in [268], [270]. In [268], an efficient VNR partitioning strategy leverages the evolution process of particles to achieve a sub-optimal solution with reduced embedding costs. Conversely, [270] adopts a two-step embedding process: the first step involves inter-domain embedding, considering only node and inter-domain link resources due to limited information sharing between SPs. In contrast, the second step uses Kruskal's algorithm for fast intra-domain mapping. An intriguing study in [276] extends VNE to multi-access edge computing (MEC), optimizing end-to-end latency constraints by considering asymmetric processor technologies of the SNs. The authors also focus on activity on vertex (AOV) networks, which closely resemble a VNR in a MEC setup, to optimize latency through instant processing. The comparative study of different PSO-based VNE schemes is presented in Table XXI.

Other Meta-heuristics: Other meta-heuristic-based strategies, such as the artificial bee colony (ABC), membrane computing, group search algorithm, and memetic algorithms, have also been adopted to devise solutions for VNE across different environments. For instance, the works in [281], [282], [284] employed ABC to design solutions focused on increasing SPs' revenue and enhancing their acceptance ratio. In contrast, Qiang et al. [264] proposed a survivable VNE scheme without reserving any backup resources, utilizing an ABC-based approach that dynamically migrates VMs and reassigns VLs during failures. Additional meta-heuristic solutions, including membrane computing, group search, bat algorithm, spider monkey, dragonfly, memetic algorithms, and artificial algae algorithms aimed at improving the revenue-to-cost ratio, are presented in [196], [283], [285], [286], [288], [289], [290], [291], [292], [293].

Hybrid Approaches: Deviating from traditional single meta-heuristic-based approaches, some hybrid strategies that combine the strengths of multiple methods have been discussed in [272], [287], [294]. In [272], the authors presented a GA-based VNE strategy that uses an adaptive crossover operation and replaces the mutation operation with simulated annealing (SA) to enhance convergence speed. Alternatively,

in [287], SA was integrated into PSO to prevent premature convergence and increase the likelihood of finding an optimal solution. Both approaches aim to improve the acceptance ratio and reduce the fragmentation of SN resources. Finally, Table XXII offers a comprehensive comparative analysis of the aforementioned meta-heuristic-based VNE schemes, taking into account the different embedding metrics discussed in Section III-F.

Comparative Study of Different Meta-heuristic Solutions: Meta-heuristic solutions offer robust embedding by balancing exploration and exploitation within the search space. Hybrid approaches, which combine elements of various meta-heuristics, show significant improvements in performance metrics such as acceptance ratio, convergence speed, and resource utilization. The specific requirements for faster convergence, high solution quality, or scalability often guide the choice of technique. Considering these aspects, we provide a comparative study of the reviewed literature in Table XXIII.

V. MODERN SOLUTIONS: MACHINE- AND DEEP-LEARNING-BASED VNE SCHEMES

With the rise of artificial intelligence (AI) and machine learning (ML), these technologies have significantly advanced the VNE space, providing innovative and sophisticated solutions [314]. ML-based solutions are preferred over classical optimization schemes because they can handle high-dimensional data and learn complex patterns without explicit programming. They are highly scalable, leveraging distributed computing to manage vast amounts of data, and are adaptable across various domains with minimal modifications. These techniques also automate feature extraction from raw data, reducing the need for manual intervention and domain-specific knowledge. Additionally, they are effective in the way they model non-linear and complex relationships, offering better training for unseen data. Furthermore, ML-based solutions can execute over real-time systems for continuous learning and decision-making, making them ideal for dynamic environments such as VNE. However, despite their numerous advantages, challenges such as data acquisition, interpretation

TABLE XXII
COMPARATIVE STUDY OF OTHER META-HEURISTICS

Reference	Quality-of-Service				Resource Spending				Resilience and Robustness				Others				
	q_1	q_2	q_3	q_4	s_1	s_2	s_3	s_4	r_1	r_2	r_3	r_4	o_1	o_2	o_3	o_4	o_5
Pathak and Vidyarthi (2015) [281]	X	X	X	X	✓	X	X	✓	X	X	X	X	✓	X	X	X	X
Liu <i>et al.</i> (2016) [282]	X	X	X	X	✓	✓	✓	✓	X	X	X	X	✓	X	X	X	X
Infuhr and Raidl (2016) [283]	X	X	X	X	✓	X	X	✓	X	X	X	X	X	X	X	X	X
Pathak <i>et al.</i> (2018) [284]	X	X	X	X	✓	X	X	✓	X	X	X	X	✓	X	X	X	X
Yu <i>et al.</i> (2018) [285]	X	X	X	X	X	✓	✓	✓	X	X	X	X	X	X	X	X	X
Yi <i>et al.</i> (2019) [286]	X	X	X	X	✓	X	✓	X	X	X	X	X	X	X	X	X	X
Boyang <i>et al.</i> (2019) [272]	X	X	X	X	X	X	✓	✓	X	X	X	X	X	X	X	X	X
Zhang <i>et al.</i> (2020) [287]	X	X	X	X	✓	X	✓	✓	X	X	X	X	X	X	X	X	X
Balamurugan <i>et al.</i> (2020) [288]	X	X	X	X	X	X	X	✓	X	X	X	X	✓	X	X	X	X
Li <i>et al.</i> (2022) [289]	X	✓	X	X	✓	✓	✓	X	X	X	X	X	X	X	X	X	X
Xu <i>et al.</i> (2023) [290]	X	X	✓	X	X	X	✓	X	X	X	X	X	✓	X	X	X	✓
Balamurugan <i>et al.</i> (2024) [291]	X	X	✓	X	X	X	X	✓	X	X	X	X	✓	X	X	X	X

TABLE XXIII
COMPARATIVE STUDY OF DIFFERENT META-HEURISTICS FROM THE PERSPECTIVE OF VNE

Algorithm	Metrics		Scalability	Convergence Speed
	Acceptance Ratio	Revenue-to-cost Ratio		
PSO	Moderate	Moderate	High	Fast
ACO	High	High	Moderate	Slow
ABC	High	High	Moderate	Moderate
SA	Moderate	Moderate	High	Slow
Memetic	High	High	Moderate	Slow
Hybrid	High	High	High	Fast

of results, and the need for substantial computational resources can hinder their performance. Furthermore, biased training data and non-inclusive datasets can lead to errors.

Various ML-based solutions have been explored in the literature [157], [296], [297], [298], [299], [308], [310]. Nasiri *et al.* [157] proposed a distributed VNE strategy for SDN-enabled networks, utilizing multi-agent systems (MAS). These agents, distributed across the SN, communicate via messages and make embedding decisions collaboratively. In this context, [313] proposed a hierarchical cooperative multi-agent reinforcement learning (MARL), combining two-level hierarchical reinforcement learning (HRL) and multi-agent collaboration to optimize the VNE. The authors focused on maximizing revenue, minimizing cost, and improving the acceptance ratio. On the other hand, He *et al.* [308] and Jahani *et al.* [297] focused on energy-efficient VNE strategies. He *et al.*'s approach used field theory to establish SN features and identify energy-saving opportunities, while Jahani *et al.* employed a self-organizing map neural network to record successful embeddings for model training. Sun *et al.* [296] extended embedding to hybrid VNRs, proposing a partitioning strategy to differentiate uni-cast and multi-cast VNRs, followed by a spectral-clustering-based VNE. For SDN-enabled networks, Yan *et al.* [298] adopted K-Means clustering for VME and K-shortest path for VLE.

Alternatively, Zheng *et al.* explored two different VNE approaches in [310] and [299] for static and dynamic

embedding setups, respectively. For static VNRs, they modeled the embedding as a finite coordinated Markov decision process, where each VM's embedding corresponds to a state organized by importance. Conversely, in [299], they adopted a supervised radial basis function neural network to anticipate fluctuations in the resource demands of VNRs, enabling seamless reallocation of substrate resources.

A. Reinforcement Learning Based

Among various ML approaches, reinforcement learning (RL) based solutions have been widely adopted in [130], [178], [295], [300], [301], [302], [303], [304], [305], [307], [309], [311], [312] due to their ability to make decisions by receiving rewards or penalties for their actions. The works in [130], [312] discussed a multi-layer graph convolutional network (GCN) aided VNE strategy that extracts high-order spatial structure information from the SN and the VNRs. It is augmented with policy gradient algorithms and meta-reinforcement learning (meta-RL) to adjust tuning parameters to enhance the GCN's dynamic performance. However, real-time VNE faces two significant challenges: the lack of historical information and the inability to predict future demands. To address these issues, researchers in [301] proposed an effective multi-stage VNE scheme and demonstrated its efficacy through exhaustive simulations. Alternatively, the authors in [307] introduced a scalable and fast RL-based embedding scheme called neighborhood enhanced policy adaptation (NEPA), inspired by the Nested Rollout Policy Adaptation (NRPA), a variant of Monte Carlo tree search (MCTS).

Yao *et al.* discussed RL-based solutions for VNE in [295], [304], and [303]. In the first protocol, they utilized a well-trained policy gradient method based on historical data to make optimal VME decisions. In the second protocol, they proposed a sophisticated three-step VNE procedure: initially representing substrate network (SN) nodes and links in attribute and adjacency matrices, capturing dynamic updates using perturbation theory, and finally applying RL to establish

TABLE XXIV
COMPARATIVE STUDY OF AI AND REINFORCEMENT LEARNING-BASED VNE SCHEMES

Reference	Quality-of-Service				Resource Spending				Resilience and Robustness				Others				
	q_1	q_2	q_3	q_4	s_1	s_2	s_3	s_4	r_1	r_2	r_3	r_4	o_1	o_2	o_3	o_4	o_5
Yao <i>et al.</i> (2018) [295]	X	X	X	X	X	✓	✓	✓	X	X	X	X	✓	X	X	X	X
Sun <i>et al.</i> (2018) [296]	X	X	X	X	✓	X	X	X	X	X	X	X	✓	X	X	X	X
Jahani <i>et al.</i> (2019) [297]	X	X	✓	X	✓	✓	✓	✓	X	X	X	X	✓	X	X	X	✓
Yan <i>et al.</i> (2019) [298]	X	X	X	✓	X	X	X	✓	X	X	X	X	X	X	X	X	X
Zheng <i>et al.</i> (2019) [299]	X	X	X	X	✓	✓	X	✓	X	X	X	X	X	X	X	X	X
Andreoletti <i>et al.</i> (2020) [300]	X	X	X	X	✓	X	X	X	X	X	X	X	X	✓	X	X	X
Thakkar <i>et al.</i> (2020) [301]	X	X	✓	X	X	X	X	X	X	X	X	X	✓	✓	X	X	X
Wang <i>et al.</i> (2020) [178]	X	X	✓	X	X	X	✓	X	X	X	X	X	X	X	X	X	X
Lu <i>et al.</i> (2020) [302]	X	X	✓	X	X	X	✓	✓	X	X	X	X	X	X	X	X	X
Yao <i>et al.</i> (2020) [303]	X	X	X	X	X	✓	✓	✓	X	X	X	X	X	X	X	X	X
Yao <i>et al.</i> (2021) [304]	X	X	X	X	X	✓	✓	✓	X	X	X	X	X	X	X	X	X
Nasiri <i>et al.</i> (2021) [157]	✓	X	X	X	X	X	✓	X	X	X	X	X	✓	✓	X	X	X
Zhang <i>et al.</i> (2021) [305]	X	X	X	X	X	✓	X	✓	X	X	X	X	X	X	X	X	X
Cheng <i>et al.</i> (2021) [306]	X	X	X	X	X	✓	✓	✓	X	X	X	X	X	X	X	X	X
Elkamel <i>et al.</i> (2022) [307]	X	X	X	X	X	X	✓	✓	X	X	X	X	X	✓	X	X	X
He <i>et al.</i> (2022) [308]	X	X	✓	X	X	X	X	X	X	X	X	X	X	X	X	X	✓
Lie <i>et al.</i> (2022) [309]	X	X	✓	✓	X	X	X	✓	X	X	X	X	X	X	X	X	X
Zheng <i>et al.</i> (2022) [310]	X	X	✓	X	X	X	✓	✓	X	X	X	X	X	X	X	X	X
He <i>et al.</i> (2022) [130]	X	✓	✓	X	X	X	✓	✓	X	X	X	X	X	X	X	X	✓
Gao <i>et al.</i> (2022) [311]	X	X	✓	X	✓	X	X	✓	X	X	X	X	X	X	X	X	✓
Ma <i>et al.</i> (2023) [312]	X	X	X	X	X	✓	✓	✓	X	X	X	X	X	X	X	X	X
Lim <i>et al.</i> (2024) [313]	X	X	X	X	✓	✓	✓	✓	X	X	X	X	X	X	X	X	X

the relationship between the SN and the VNR, resulting in efficient VNE. In contrast, in [303], they modeled VNE as a time-series problem, treating the continuity of VME with a recurrent neural network (RNN) that leverages the history of the SN. A gradient-based policy updates the RNN parameters based on a unique average revenue-to-cost ratio. Alternatively, Wang *et al.* [178] introduced a pointer network to assist in VME using an attention mechanism and designed a reward function with a policy gradient to capture VM dependencies.

Several proposals have discussed hierarchical VNE strategies, including those in [302] and [306]. In [302], the focus was on partitioning the VNR into multiple smaller VNRs before performing VNE. Conversely, [306] assessed the long-term and short-term impacts of embedding a VNR using high-level and low-level agents, where the former identifies a feasible VNR, and the latter embeds the VNRs through different actions.

Research in [305] and [300] proposed secure VNE schemes utilizing RL. In [305], a policy network acts as an agent and creates a feature matrix comprising extracted attributes from the SN. Each substrate server is assigned a mapping probability, which assists the model in secure VME, followed by BFS-based VLE. Alternatively, [300] proposed a distributed VNE that securely processes user and SP embedding information using the Shamir Secret Sharing (SSS) scheme.

Additionally, Li *et al.* [309] and Gao *et al.* [311] extended VNE to elastic optical networks and wireless sensor networks

(WSNs), respectively. They proposed coordinated VNE strategies and designed reward functions catering to different objectives, such as acceptance ratio, revenue-to-cost ratio, and load balancing. A detailed study of the comparative behavior of varying AI and RL schemes can be found in Table XXIV.

B. Deep Learning Based

In addition to RL-based models, the research community has also embraced deep learning (DL)-based solutions, which model complex patterns in data using multi-layer neural networks and large datasets. Several DL-based VNE schemes are discussed in [171], [177], [315], [316], [317]. In this context, Yan *et al.* [171] introduced a DRL-based VNE scheme utilizing a novel neural network based on GCN to extract spatial features from the SN topology. They employed a parallel policy gradient method to train the model and designed a unique reward function to optimize acceptance, revenue-to-cost ratio, and load balance. Conversely, Park *et al.* [177] employed an encoder-decoder model for VNE, where the encoder handled graph embedding tasks while the decoder managed node-level mapping during embedding. Yuan *et al.* [316] utilized Q-learning with a specially crafted reward function incorporating VLE considerations and updated via unsupervised learning.

In contrast, Dolati *et al.* [315] transformed grid-like VNRs into images and employed a CNN within a DRL framework for

TABLE XXV
COMPARATIVE STUDY OF DEEP LEARNING-BASED VNE SCHEMES

Reference	Quality-of-Service				Resource Spending				Resilience and Robustness				Others				
	q_1	q_2	q_3	q_4	s_1	s_2	s_3	s_4	r_1	r_2	r_3	r_4	o_1	o_2	o_3	o_4	o_5
Dolati <i>et al.</i> (2019) [315]	X	X	✓	X	✓	✓	X	X	X	X	X	X	X	X	X	X	X
Yuan <i>et al.</i> (2020) [316]	X	X	X	X	X	✓	✓	✓	X	X	X	X	X	X	X	X	X
Yan <i>et al.</i> (2020) [171]	X	X	✓	X	X	✓	✓	✓	X	X	X	X	X	X	X	X	X
Park <i>et al.</i> (2021) [177]	X	X	X	X	X	X	X	✓	X	X	X	X	✓	X	X	X	X
Zhang <i>et al.</i> (2022) [317]	X	X	X	X	X	✓	✓	✓	X	X	X	X	X	X	X	X	X
Duan <i>et al.</i> (2024) [318]	X	X	X	X	X	✓	✓	✓	X	X	X	X	X	X	X	X	✓

TABLE XXVI
COMPARISON BETWEEN RL AND DL-BASED VNE SOLUTIONS:
FEATURES, METRICS, AND IMPLEMENTATION CHALLENGES

Aspects	Reinforcement Learning	Deep Learning
Approach	Interactive training with reward and penalty based decisions	Multi-layer neural networks to learn patterns from large datasets
Training Complexity	Iterative learning through trial-and-error	Deep neural networks trained with large datasets, relying on backpropagation for parameter adjustment
Application Area	Dynamic VNE with changing requirements	More applicable to static VNE
Evaluation Metrics	Convergence rate and average reward	Accuracy, precision, recall
Implementation Challenges	Exploration-exploitation trade-offs, scalability, and reward sensitivity	Data dependency, overfitting, and interpretability

embedding solutions. Zhang *et al.* [317] explored embedding solutions for space-air-ground integrated networks (SAGIN) using a five-layered policy network operating on a feature matrix extracted from network attributes. Post-training, their DRL method determined node and link embedding probabilities for VNE. On the other hand, [318] presented a DRL-based VNE scheme utilizing Proximal Policy Optimization (PPO). The authors employed a hybrid feature extraction approach combining handcrafted and GCN-derived features to effectively coordinate VME and VLE, aiming to optimize multiple objectives such as acceptance rate, revenue, and energy consumption.

Table XXV provides a comprehensive and comparative view of different DL-based VNE schemes considering different metrics discussed in Section III-F.

Comparison of ML and DL-based VNE Solutions: Table XXVI presents a comparative study highlighting various aspects of ML and DL-powered VNE solutions, including application areas, performance metrics, training complexity, and implementation challenges. However, the selection between these approaches hinges on application-specific needs such as network characteristics and available data resources. RL thrives in adaptive decision-making and dynamic re-assignments of VNRs, whereas DL excels in learning complex embedding patterns and making predictions through extensive data analysis. Integrating both approaches can synergistically enhance VNE performance by leveraging their respective strengths.

Finally, Table XXVII provides a comprehensive taxonomy of the VNE schemes reviewed in this work. This table systematically categorizes each protocol based on the criteria discussed in Section III-E. In addition to classifying the VNE schemes, we provide detailed insights into the

specific deployment environments where each technique is applied, such as SN, wireless networks, IoTs, and IoVs. Furthermore, we emphasize the unique contributions of each method and optimization technique, offering a deeper understanding of each VNE solution's strengths and practical applications.

VI. OPEN RESEARCH CHALLENGES AND MOST PROMISING DIRECTIONS FOR FUTURE RESEARCH

This section discusses the most promising future research directions and the scope for improvements regarding the current VNE state-of-the-art. We present the scope for future research from different aspects as illustrated visually in Figure 27, including distributed VNE strategies, some unaddressed/partially addressed issues requiring incremental improvements, implications, challenges, and the future scope of using NV over modern networks. We also discuss the research directions for incorporating advanced AI/ML techniques for VNE, its integration with digital twins, and augmented reality/virtual reality technologies. Finally, we wrap up the discussion by identifying potential issues vulnerable to security threats and possible solutions that can be looked into.

A. Need for Distributed VNE

As discussed in Section III-A, VNE is proven to be a computationally intractable problem. Therefore, the research community was focused on developing polynomial-time heuristics by relaxing multiple constraints. Although such solutions were effective, they were not remarkably scalable for large networks and were restricted to a single SN with a centralized controller taking embedding decisions. However, a centralized controller improves the embedding time but suffers from multiple issues, such as single-point failures and congestion during peak load. Moreover, with the fast-paced modern network, geo-separation is realizable, and the requirement of distributed services is crucial. This calls for distributed VNE strategies that cater to multiple different agendas.

In this regard, some distributed VNE schemes were discussed in [196], [207], [208], [213], [270], [274], [294], and [284]. These models were based on optimization solvers such as GA, PSO, Harmony Search Algorithm (HSA),

TABLE XXVII
A TAXONOMY OF VNE STRATEGIES

Category	Reference	Optimization Technique	Environment	Contribution
S/D/P/S	Amokrane <i>et al.</i> (2013) [229]	Heuristic	Multi-domain SN	Improve the SPs revenue with minimum carbon foot-prints.
D/C/P/S	Chang <i>et al.</i> (2013) [210]	Metaheuristic (ACO)	SN	Hybrid GA and ACO model for VNE.
D/C/P/S	Melo <i>et al.</i> (2013) [185]	Primal Dual	SN	Integer linear programming (ILP) formulation for online VNE.
S/C/P/S	Hu <i>et al.</i> (2013) [180]	Exact (Branch and Bound)	SN	Optimal VNE using column generation embedded into a branch-and-bound procedure.
D/C/P/S	Dab <i>et al.</i> (2013) [129]	Metaheuristic (GA)	SN	Selective re-embedding of dynamic VN components facing resource bottleneck using GA.
S/C/P/S	Gong <i>et al.</i> (2014) [230]	Heuristic	SN	Global resource capacity (GRC)-based VNE with selective admission policy.
S/C/P/S	Feng <i>et al.</i> (2014) [224]	Local Search (Greedy)	Single-domain SN	Topology-aware VNE with seven distinct topological attributes.
S/C/P/S	Hu <i>et al.</i> (2014) [186]	Primal Dual	Single Domain SN	Optimal resource consumption.
S/C/P/S	Fajjari <i>et al.</i> (2014) [262]	Metaheuristic (ACO)	Multi-Domain SN	ACO-based online VNE over the private substrate backbone network.
S/C/R/S	Liao <i>et al.</i> (2014) [231]	Heuristic	Multi-Domain SN	Survivable VNE of multicast VNRs.
D/C/P/S	Chen and Li (2014) [232]	Heuristic	SN	Novel VLE algorithms for path splitting with reduced energy consumption and real-time performance.
S/C/P/S	Qiang <i>et al.</i> (2014) [264]	Metaheuristic (ABC)	Single Domain SN	Artificial bee colony based survivable VNE with node and link remapping.
D/C/P/S	Mijumbi <i>et al.</i> (2015) [187]	Primal Dual	Single Domain SN	One-shot unsplittable flow VNE based on column generation.
S/C/P/S	Guo <i>et al.</i> (2015) [268]	Metaheuristic (PSO)	Multi-Domain SN	Particle Swarm Optimization (PSO) based VNR partitioning followed by embedding.
S/C/P/S	Ghazisaeedi and Huang (2015) [236]	Heuristic	SN	Energy-aware VNE by optimizing the VM and VL energy consumption during the off-peak period using re-configuration.
D/C/P/S	Shanbhag <i>et al.</i> (2015) [235]	Heuristic	SN	Balanced embedding with fewer substrate resources.
D/C/P/S	Zhang <i>et al.</i> (2015) [234]	Heuristic	SN	Energy-aware VNE with dynamic demands.
D/C/P/S	Guan <i>et al.</i> (2015) [158]	Heuristic	SN	Energy-aware VNE by considering the computing and networking energies.
D/D/P/S	Triki <i>et al.</i> (2015) [233]	Heuristic	SN	Energy and resource efficient VNE.
D/C/P/S	Pathak and Vidyarthi (2015) [281]	Metaheuristic (ABC)	SN	Interactive artificial bee colony based VNE for dynamic arrivals.
D/C/P/U	Ayoubi <i>et al.</i> (2015) [193]	Exact	SN	QoS aware multi-cast VNE using Tabu search.
S/C/P/S	Duan <i>et al.</i> (2015) [219]	Local Search (Greedy)	SN	Compact multicast VNE with VM migrations.
S/C/P/S	Ding <i>et al.</i> (2015) [222]	Local Search (Greedy)	SN	Topological attributes considered VNE.
S/C/P/S	Liu <i>et al.</i> (2016) [282]	Metaheuristic (ABC)	SN	Artificial bee colony-based VNE to boost the acceptance ratio.
D/C/P/S	Zhang <i>et al.</i> (2016) [269]	Metaheuristic (PSO)	SN	PSO-based VNE for maximizing embedding ratio and minimizing energy consumption.
D/C/P/S	Zhou <i>et al.</i> (2016) [211]	Metaheuristic (GA)	SN	GA-based resource-aware and selective VNE of VNRs in batches.
S/C/P/S	Zhu and Wang (2016) [263]	Metaheuristic (ACO)	SN	Graph decomposition-based VNE using ACO.
D/C/P/U	Jiang <i>et al.</i> (2016) [237]	Heuristic	SN	Focused on closely embedding/re-embedding policies for VNRs hosting similar services.
D/C/P/S	Wang <i>et al.</i> (2016) [181]	Exact (Branch and Price)	SN	Branch-and-price framework with column generation for efficient VNE.
S/C/P/S	Guler <i>et al.</i> (2016) [223]	Local Search (Greedy)	SN	Closeness-centrality based multicast VNE.
D/D/P/S	Esposito <i>et al.</i> (2016) [258]	Game Theory (Auction Theory)	SN	Distributed auction mechanism for VNE.
D/C/P/S	Inführ and Raidl (2016) [283]	Metaheuristic (Memetic)	SN	Memetic algorithm based VNE.

(Continued)

and ABC, which inherently suffered from non-scalability and consumed excess time in embedding, thereby limiting their adoption. Alternatively, two distributed protocols

based on auction theory were discussed in [257], [258] but suffered from issues of exaggeration and collusion. The work in [128] discussed a brute-force strategy to relocate

TABLE XXVIII
(Continued.) A TAXONOMY OF VNE STRATEGIES

S/C/P/S	Gong <i>et al.</i> (2016) [250]	Heuristic	SN	A energy aware heterogeneous VNE with power-varying diversity of physical nodes.
S/C/P/S	Zhang <i>et al.</i> (2016) [319]	Heuristic	SDN-enabled SN	VNE using virtual node combination.
S/C/P/S	Zhang and Tang (2017) [188]	Primal Dual	SN	VLE based on primal–dual method.
S/D/P/S	Wang <i>et al.</i> (2017) [270]	Metaheuristic (PSO)	5G-enabled SNs	Cross-domain VNE using DPSO.
D/C/P/S	Gilesh <i>et al.</i> (2017) [220]	Local Search (Greedy)	SN	Service disruption and resource fragmentation aware VDCE.
D/C/P/S	Pathak <i>et al.</i> (2017) [195]	Local Search (GA)	Multi-domain SN	Classical GA-based VNE with multi-point crossover.
D/D/P/S	Sun <i>et al.</i> (2017) [216]	Local Search (Greedy)	Multi-domain SN	Consolidation of dynamic VDCs.
D/C/P/S	Nam <i>et al.</i> (2017) [239]	Heuristic	SN	Dynamic VNE with server consolidation.
D/C/P/S	Zheng <i>et al.</i> (2017) [265]	Metaheuristic (ACO)	SN	Ant colony systems based VNE.
S/C/R/S	Liu <i>et al.</i> (2017) [240]	Heuristic	SN	Survivable VNE with robust backup topology for recovery.
S/C/P/S	Dahir <i>et al.</i> (2017) [241]	Heuristic	Multi-domain SDNs	Energy efficient VNE with strict adherence to the bandwidth requirements of network links.
S/C/P/S	Sun <i>et al.</i> (2018) [296]	Machine Learning	Multi-Domain SN	Hybrid embedding strategy for uni-cast and multi-cast VNRs.
S/C/P/S	Cao <i>et al.</i> (2018) [244]	Heuristic	Multi-Domain SN	Energy-aware and geo-distributed VNE.
S/C/P/S	Cao <i>et al.</i> (2018) [162]	Local Search (Greedy)	Single-domain SN	Google page rank-based node ranking for topology-aware VNE.
S/D/P/S	Metwally <i>et al.</i> (2018) [257]	Game Theoretic (Auction Theory)	Multi-domain SN	Auction theory-based competitive pricing model for VNE.
D/C/P/S	Gao <i>et al.</i> (2018) [243]	Heuristic	SN	Dynamic reconfiguration of VMs and VLs for balancing the load of the substrate network.
D/C/P/S	Sun <i>et al.</i> (2018) [128]	Local Search (Greedy)	Multi-domain SN	Optimizing the migration overheads for dynamic VDCs.
D/C/P/U	Lu <i>et al.</i> (2018) [226]	Local Search (Greedy)	SN	Collaborative and topology-aware VNE based on resource importance measures.
S/C/P/S	Cao <i>et al.</i> (2018) [163]	Local Search (Greedy)	SN	VNE with Google PageRank-based node ranking with multiple topological attributes.
S/D/P/S	Zhang <i>et al.</i> (2018) [225]	Local Search (Greedy)	SN	Five-node ranking metric based unified ranking using Electre for VNE.
S/C/P/S	Yao <i>et al.</i> (2018) [295]	Machine Learning	SN	Reinforcement learning based VNE.
D/C/P/S	Yu <i>et al.</i> (2018) [285]	Metaheuristic (Membrane Computing)	SN	Membrane computing based parallel VNE.
D/D/P/S	Pathak <i>et al.</i> (2018) [284]	Metaheuristic (ABC)	SN	Artificial bee colony based dynamic VNE.
D/C/P/S	Zhang <i>et al.</i> (2019) [202]	Metaheuristic (GA)	SN	Improved initial population and crossover for GA-based VNE.
S/C/P/S	Boyang <i>et al.</i> (2019) [272]	Metaheuristic (PSO)	SN	Robust GA-based VNE with improved crossover and simulated annealing based mutation.
D/C/P/U	Cao <i>et al.</i> (2019) [218]	Local Search (Greedy)	SN	Dynamic re-embedding for VNRs.
D/D/P/S	Song <i>et al.</i> (2019) [274]	Metaheuristic (PSO)	SN	Archive-assisted PSO-based VNE.
D/C/P/S	Song <i>et al.</i> (2019) [271]	Metaheuristic (PSO)	SN	Divide-and-conquer evolutionary algorithm with overlapping decomposition (ODEA) for large scale VNE.
D/C/P/S	Jahani <i>et al.</i> (2019) [297]	Machine Learning	SN	Self-organized green VNE that utilizes historical data of successful VM mapping for training a self-organizing map neural network.
D/C/P/S	Song <i>et al.</i> (2019) [273]	Metaheuristic (PSO)	SN	Constructive particle swarm optimization-based VNE.
S/C/P/S	Yan <i>et al.</i> (2019) [298]	Machine Learning	SDN	Latency-aware resource-efficient VNE, where VME and VLE are performed using K-Means clustering and K-shortest path, respectively.
S/C/P/S	Zheng <i>et al.</i> (2019) [245]	Heuristic	SN	Survivable and balanced VNE to deal with network failures.

(Continued)

components of VNRs dynamically over a geo-distributed infrastructure, but the approach was neither scalable nor computationally inexpensive. Note that a primary overhead for

distributed VNE is exchanging information between different SPs or geo-distributed SNs for embedding decisions. This requires sending/receiving many coordination/synchronization

TABLE XXIX
(Continued.) A TAXONOMY OF VNE STRATEGIES

D/C/P/S	Zheng <i>et al.</i> (2019) [299]	Neural network	SN	Supervised radial basis function based neural network to predict the changes in VNRs with dynamic reallocation of substrate resources.
D/C/P/S	Dolati <i>et al.</i> (2019) [315]	Machine Learning	SN	Automated selection of problem features from SN and VNR 2-D images.
S/C/P/S	Liu <i>et al.</i> (2019) [246]	Heuristic	WSNs	A reactive VNE scheme that allows SN components to be put to sleep/activate to optimize resource usage and energy consumption.
S/C/P/S	Yi <i>et al.</i> (2019) [286]	Metaheuristic (BA)	Multi domain SN	A bat algorithm based VNE.
S/C/P/S	Afifi <i>et al.</i> (2019) [212]	Metaheuristic (GA)	WSN	GA-based VNE over wireless networks with interference and multi-cast properties.
D/D/P/S	Shooshtarian and Safaei (2019) [213]	Metaheuristic (GA)	SN	Non-dominated sorting-based GA for robust, online, and balanced VDCE.
S/C/P/S	Balamurugan <i>et al.</i> (2020) [288]	Metaheuristic (DO)	SN	Multi-criteria Dragonfly based Graph Theory for VNE with reasonable resource utilization.
D/D/P/S	Nguyen <i>et al.</i> (2020) [208]	Metaheuristic (GA)	SN	Topology-aware node embedding followed by VL mapping using fast GA.
D/C/P/S	Satpathy <i>et al.</i> (2020) [120]	Game Theory (Matching Theory)	Multi-domain SN	Matching-based selective re-embedding of online VDCs.
D/D/P/U	Lu <i>et al.</i> (2020) [207]	Metaheuristic (GA)	SN	Fast and parallel VL embedding using GA.
D/C/P/S	Satpathy <i>et al.</i> (2020) [121]	Metaheuristic (GA)	Multi-domain SN	Improved crossover operator assisted GA-based complete re-embedding of VDCs.
S/C/P/S	Wu <i>et al.</i> (2020) [227]	Local Search (Greedy)	SN	Multiple Knapsack Problem and Multi-Dimensional Knapsack Problem-based VNE for path and cycle topology VNRs.
S/C/P/S	Chai <i>et al.</i> (2020) [275]	Metaheuristic (PSO)	SDN-enabled SN	Globally optimal VNE scheme based on PSO.
D/C/P/S	Lu <i>et al.</i> (2020) [302]	Machine Learning	SDN-enabled SN	A unified and flexible multi-layer VNE based on reinforcement learning.
D/C/P/S	Yao <i>et al.</i> (2020) [303]	Machine Learning	SN	Continuous-decision VNE scheme based on Reinforcement Learning.
S/C/P/S	Yan <i>et al.</i> (2020) [171]	Machine Learning	SN	Deep reinforcement learning with neural network for automatic VNE.
D/C/P/S	Cong <i>et al.</i> (2020) [247]	Heuristic	SN	Energy-optimized VNE with location constraints.
D/C/P/S	Pham <i>et al.</i> (2020) [160]	Heuristic	SDN-enabled SN	Congestion and energy-aware Pareto-optimal coordinated VNE scheme.
S/C/P/S	Zhang <i>et al.</i> (2020) [287]	Metaheuristic (SA+PSO)	Multi-domain SN	Simulated annealing and PSO-based hybrid VNE.
D/C/P/U	Chiang <i>et al.</i> (2020) [276]	Metaheuristic (PSO)	Multi-access Edge Computing Network	PSO-based VNE scheme for optimal end-to-end latency constraints.
S/D/P/S	Andreoletti <i>et al.</i> (2020) [300]	Machine learning	Multi-domain SN	A secure VNE scheme based on reinforcement learning to process data shared by the customer and SPs under the Shamir Secret Sharing.
S/C/P/S	Thakkar <i>et al.</i> (2020) [301]	Machine Learning	SN	Reinforcement Learning based prediction model for the efficient Multi-stage VNE.
D/C/P/S	Yang <i>et al.</i> (2020) [320]	Heuristic	Satellite Networks	Topology and QoS aware VNE.
S/C/R/U	Shahriar <i>et al.</i> (2020) [248]	Heuristic	SN	Weakly survivable and connectivity-aware VNE.
S/C/P/S	Wang <i>et al.</i> (2020) [178]	Machine Learning	SN	Reinforcement Learning-based VNE with a pointer network-based VME and a reward function to connect the VME and VLE.
S/C/P/S	Hashmi and Gupta (2020) [221]	Local Search (Greedy)	SN	Topology adaptable VNE with Google PageRank based VME and hops and available bandwidth based optimal VLE.
S/C/P/S	Yuan <i>et al.</i> (2020) [316]	Machine Learning	SN	Q-learning based VNE.
D/C/P/S	Fan <i>et al.</i> (2021) [321]	Heuristic	IoVs	Dynamic and hierarchical VNE based on efficient topology control scheme for mobile nodes.
D/C/P/S	Cheng <i>et al.</i> (2021) [306]	Machine Learning	SN	Hierarchical reinforcement learning-based resource-aware and proactive VNE.
S/C/P/S	Zhang <i>et al.</i> (2021) [203]	Metaheuristic (GA)	Multi-domain SN	Probabilistic selection and crossover for GA-based VNE.

(Continued)

messages over the network. Not a lot of attention has been given to this aspect by the research community, and it can be a good line of future research. Sometimes, these messages contain sensitive information, such as SP topology, pricing, and network details, traversing the public network and raising security concerns. Therefore, integrating modern security

TABLE XXX
(Continued.) A TAXONOMY OF VNE STRATEGIES

D/C/P/U	Li <i>et al.</i> (2021) [249]	Heuristic	WSNs	VNE with reliability and delay requirements for wireless networks
D/C/P/S	Julia <i>et al.</i> (2021) [204]	Metaheuristic (GA)	SN	Efficient VNE using constrained GA.
S/C/P/S	Yao <i>et al.</i> (2021) [304]	Machine Learning	SN	Reinforcement learning based dynamic attribute matrix representation (RDAM) for VNE.
D/C/P/U	Park <i>et al.</i> (2021) [177]	Machine Learning	SN	Reinforcement Learning-based graph embedding and mapping framework for VNRs.
S/D/P/S	Nasiri <i>et al.</i> (2021) [157]	Artificial Intelligence	SDN-enabled SN	Distributed VNE for SDNs using multi-agent systems.
S/C/P/S	Zhang <i>et al.</i> (2021) [305]	Machine Learning	SN	Security-aware VNE based on reinforcement learning.
S/C/P/S	Zheng <i>et al.</i> (2021) [310]	Machine Learning	SN	A finite Markov decision process based VNE.
S/C/P/U	Julia <i>et al.</i> (2022) [206]	Metaheuristic (GA)	SN	Optimal and fast embedding using a guided heuristic based on GA.
S/C/P/S	Zhang <i>et al.</i> (2022) [317]	Machine Learning	Space-air-ground integrated network	Deep reinforcement learning based heterogeneous resource orchestration for multi-domain VNE.
D/C/P/S	Satpathy <i>et al.</i> (2022) [117]	Game Theory (Matching Theory)	Multi-domain SN	Selective re-embedding using a coalitional matching-game with dynamic preferences of VMs and servers.
D/C/P/S	Nguyen <i>et al.</i> (2022) [209]	Metaheuristic (GA)	SN	GA-based coordinated VNE with conciliation to deal with infeasible VL mappings.
D/C/P/S	Elkael <i>et al.</i> (2022) [307]	Machine Learning	5G Networks	Reinforcement learning and neighborhood search assisted VNE.
D/C/P/S	He <i>et al.</i> (2022) [308]	Machine Learning	SN	Spectral clustering based on field theory, extracting SN features for energy efficient VNE.
D/C/P/S	He <i>et al.</i> (2022) [130]	Machine Learning	SN	Graph convolutional network based feature extraction of VNR and SN for effective VNE.
D/C/P/S	Xiao <i>et al.</i> (2022) [228]	Local Search (Greedy)	SN	Dynamic resource sharing based VNE considering compute storage and topological metrics.
D/C/P/S	Gao <i>et al.</i> (2022) [311]	Machine Learning	WSN	Realizing load balance with reward function considering the execution and residual ratio of substrate network and cost of VNR.
S/C/P/S	Yuan <i>et al.</i> (2022) [322]	Machine Learning	SN	Neural network assisted VNE.
D/C/P/S	Li <i>et al.</i> (2022) [309]	Machine Learning	Elastic Optical Networks	Deep reinforcement learning based VNE.
S/C/P/S	Li <i>et al.</i> (2022) [289]	Metaheuristic (GSA)	SN	Group search algorithm based balanced VNE considering multiple objectives.
D/C/P/S	Satpathy <i>et al.</i> (2023) [88]	Game Theory (Matching Theory)	Multi-domain SN	Selective re-embedding of VDCs using one-to-many matching with static preferences of VMs and servers.
S/C/P/S	T. G. <i>et al.</i> (2023) [131]	Game Theory (Matching Theory)	Single-domain SN	Offline multi-attribute based embedding of VNRs using the classical deferred acceptance algorithm (DAA).
S/C/P/S	T. G. <i>et al.</i> (2023) [122]	Local Search (Greedy)	Single-domain SN	Multi-criteria-based topology-aware greedy VNE.
D/C/P/S	Ma <i>et al.</i> (2023) [312]	Machine Learning	IoT	Graph convolutional network aided VNE.
D/D/P/S	Fan <i>et al.</i> (2023) [323]	Heuristic	SN	Node essentiality based evaluation model for dynamic, distributed, and collaborative VNE.
D/C/P/S	Guler <i>et al.</i> (2023) [214]	Metaheuristic (GA)	SN	Modified GA-based VNE for embedding multicast tree based VNs.
D/D/P/S	Qiao and Huang (2024) [205]	Meta-heuristic (GA)	SN	Distributed GA-based coordinated VM and VL mapping.
D/C/P/S	Duan <i>et al.</i> (2024) [318]	Deep Learning	SN	Hybrid feature extraction based GCN for VNE.
D/C/P/S	Satpathy <i>et al.</i> (2024) [118]	Meta-heuristic (GA)	Multi-domain SN	Improved crossover and mutation based VNE.
S/D/P/S	Balamurugan <i>et al.</i> (2024) [291]	Meta-heuristic (Whale Optimization)	SN	Whale Optimization-based graphical VNE.
D/D/P/S	Lim <i>et al.</i> (2024) [313]	Machine Learning	SN	Two-level hierarchical reinforcement learning (HRL) and multi-agent collaboration for VNE
S/C/P/S	Ikoma <i>et al.</i> (2024) [267]	Meta-heuristic (ACO)	Multi-domain SN	QoS aware VNE considering communication dependencies.

protocols, such as Blockchain [324] and [325], can cater to the security in such a distributed setup and be considered a future direction.

The existing distributed VNE solutions are unreliable and do not offer resilient solutions. Most existing literature on resilient embedding models focuses on providing backups and

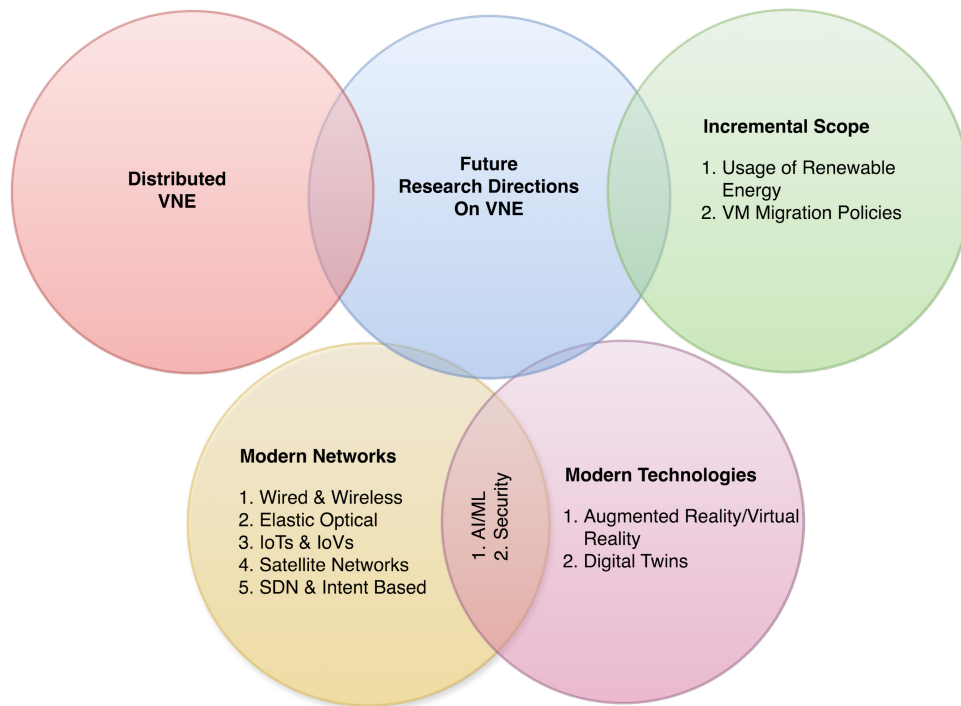


Fig. 27. A visual representation of future research directions for VNE.

operating centrally. However, with the outburst of distributed computing, the need for real-time and resilient embedding strategies is almost inescapable.

B. Incremental Scope

Lately, the VNE research community has piqued interest in the following, which has some partially addressed issues and scope for further advancements.

1) *Using Renewable Energy Sources:* Energy consumption presents a significant operational challenge for SPs, affecting efficiency, sustainability, and costs. According to the 2021 report from the International Energy Agency (IEA), large-scale DCs consume approximately 200-250 terawatt-hours (TWh) of electricity annually, accounting for around 1-2% of global electricity usage, comparable to the energy demands of some mid-sized countries. Projections indicate this could rise to as much as 8% of the worldwide market by 2030, driven by expanding digital services such as cloud computing, AI, and IoT. Major corporations like Google, Amazon, and Microsoft are leading the charge in renewable energy adoption, collectively making up over 50% of global corporate renewable purchases in 2020 to power their DCs. Integrating renewable energy into VNE poses significant research challenges, primarily due to the intermittency of sources like solar and wind, necessitating adaptive algorithms to manage fluctuating availability. This increases computational demands and may require expensive energy storage solutions, which, despite advancements, remain insufficient for long-term storage. Furthermore, geographic limitations restrict renewable energy usage to specific regions, complicating optimal VNE placement. Inconsistent regulations

and inadequate upfront infrastructure for harnessing renewable energy impede widespread adoption. The research community has proposed various approaches to address SPs' energy concerns. One approach is to embed VNRs on servers powered by renewable energy sources like solar or wind, substantially reducing SPs' carbon footprint. Additionally, locally sourced renewable energy can alleviate DC operational costs, especially in areas with high electricity prices, enhancing the economic sustainability of embedding VNE solutions. Research in [229], [326] and [233] have focused on optimizing the energy consumption of servers and networking resources. Alternative strategies include placing unused servers in sleep mode [160] or consolidating VNRs onto fewer servers during off-peak periods, as discussed in [236], [239], [246], and [232]. While these models were effective, they also had several shortcomings. For instance, putting servers in sleep mode results in significant energy consumption during the power-up phase, leading to spikes in power usage and delayed response times for users during peak loads, creating undesirable outcomes. Dynamic consolidation during off-peak periods involves relocating VNR components, such as VMs and VLs, which is cumbersome and cannot be performed frequently. Additionally, the energy profiles of servers and networking components fluctuate based on various factors, including hardware state, load, and application characteristics. These parameters change in real-time, necessitating complex profiling mechanisms and challenging, effective management.

A few aspects the research community can explore are consolidation mechanisms prioritizing embedding in regions with higher renewable energy availability. Additionally, strategies can be developed to leverage multi-time-scale electricity markets and the regional and temporal fluctuations in electricity

prices influenced by renewable energy sources [327], [328]. Aligning VNE systems with environmental objectives can help qualify them for incentives, promote green networking practices, and enhance network resilience. Researchers should also focus on developing adaptive and predictive algorithms that optimize VNE based on real-time renewable energy availability and forecasted energy patterns. This includes creating energy-efficient VNE schemes that dynamically respond to fluctuations in renewable energy supply without compromising the QoS. Utilizing ML models to predict renewable energy availability could significantly enhance the energy efficiency and resilience of VNE systems. Furthermore, exploring hybrid energy strategies that integrate renewable and non-renewable sources can ensure continuous service while maximizing green energy usage. Addressing these areas will support sustainable growth in network services and contribute to achieving global environmental goals. Moreover, focusing solely on energy reduction may prove inadequate; therefore, energy-aware policies should be integrated with load-balancing mechanisms that consider topology to enhance acceptance and improve the revenue-to-cost ratio.

2) *VM Migration Policies*: Dynamic consolidation of VNRs often requires the relocation of VMs and VLs. VM migration is a sophisticated tool that allows the SPs to seamlessly move a VM from a source server to a destination server [107]. It comprises the transfer of the memory and storage and re-establishment of the network connections, thereby introducing additional overheads in terms of *migration time*, *network traffic*, and *service degradation*. Multiple techniques in the literature exist to migrate a VM live, such as pre-copy, post-copy, and hybrid-copy [329]. However, in this context, a complex decision involves appropriately scheduling the migration of multiple VMs, especially in dynamic setups with frequent re-embedding. The only protocol that considers scheduling the VMs of VNRs is discussed in [330]. It focused on partitioning the VMs into resource-independent groups that do not share migration paths and destination/source servers—however, the protocol provided no upper bounds on the service downtime. As a future direction, researchers can develop a migration plan that considers VNR and SN topologies and prioritizes scheduling VMs corresponding to different VNRs.

3) *Need for a More Comprehensive Simulation/Emulation Platforms*: Although the research community is propelled to develop solutions for VNE, there is a lack of a comprehensive platform that can be used to evaluate all the embedding schemes. Though there are some simulation platforms, such as ALEVIN [142], CloudSim [166], VNE-SIM [168], and Alib Utility [172]. However, these tools were designed for specific use cases, and it is cumbersome to integrate all embedding algorithms catering to different metrics into a single framework. Additionally, the simulation frameworks are unrealistic and do not capture the real-world features of the VNRs and SNs. To realize the real-world potential of an embedding scheme, it is always advisable to test it over test-bed or emulated test-bed setups rather than simulation tools. An emulated setup allows the researchers and developers to emulate the setup at the lab scale without actually using the infrastructure of commercial CSPs. Further,

it assists researchers and developers in designing, evaluating, and prototyping VNE strategies at the lab scale. It also enables seamless benchmarking without separately implementing it for specific platforms. Developing an emulator for VNE is challenging and introduces multiple design challenges, including (i.) emulating a geo-distributed SN topology, (ii.) emulating real-world workloads, i.e., VNRs, (iii.) seamless portability to the cloud, and (iv.) concrete evaluation and implementation for validation. The research community can look into these challenges as a future direction and develop exciting solutions.

C. NV for Future Networks: Current State-of-the-Art and Future Scope

NV has matured as a technology over the last decade. With its advantages, it has found applications in different modern networks, such as wireless networks, optical networks, satellite networks, IoT networks, etc. These environments are quite different and introduce unique challenges that the researchers can look into.

1) *Wired vs. Wireless Networks*: Initially developed for the stable environment of wired networks, VNE has now extended into wireless domains, introducing unique challenges due to the distinct characteristics of wireless applications and SN infrastructure. VNE faces ongoing challenges in wired networks, such as high infrastructure costs requiring optimized resource utilization and managing resource fragmentation from fixed routing paths through precise allocation strategies [88]. Additionally, reducing latency is critical for real-time applications, requiring VNE to optimize routes and alleviate congestion delays [117]. Scalability in expanding networks poses another challenge, as more complex topologies need advanced algorithms. Conversely, VNE in wireless networks faces wireless environments' dynamic, broadcast-driven nature. Wireless applications are often location-sensitive and data-driven, allowing minimal error tolerance, which limits physical node deployment options. Furthermore, time-sensitive data relay requirements complicate VL allocation, while the broadcast nature of wireless links can cause interference on adjacent channels. Addressing these issues calls for advanced resource allocation methods, such as time-division and frequency-division multiplexing, as studied in [331] and [332].

Research efforts, such as those in [246], [311], [333], have advanced energy-efficient, load-balanced wireless VNE solutions with improved load-balancing techniques. However, essential factors unique to wireless networks, like mobility and distribution, remain under-explored. For instance, Zhang et al. [334] proposed a three-stage resource allocation method for 5G wireless VNE, focusing on VLE and incorporating algorithms to reduce delay and improve link utilization. Nevertheless, these approaches lack comprehensive integration of user mobility, a critical component in mobile networks where reconfiguration may be necessary as nodes move. Moreover, ad-hoc wireless networks, which lack centralized management, pose challenges in requiring distributed control. Addressing these gaps in future research could involve exploring space-division multiplexing for improved channel

sharing and developing VNE schemes that bridge the unique demands of wired and wireless networks.

2) *Elastic Optical Networks*: Traditional Gigabit Ethernet networks face bandwidth bottlenecks, single points of failure, and limitations in their tree-based structure, which restrict their ability to handle modern one-to-many and many-to-many communication demands [335]. To address these challenges, the research community has developed optical networks with wavelength division multiplexing (WDM), which enables better fiber utilization and provides high bandwidth with low latency [336]. Due to limited fiber resources, virtualization has become essential for optimal use, making VNE over optical networks a prominent research area. Solutions are generally classified as transparent or opaque mappings, where the former assigns the same set of wavelengths for VL while the latter does not. Although VNE over optical networks is relatively new, advanced strategies have emerged, such as in [165], [309], [337], aiming to enhance SP acceptance rates. Unresolved challenges include (1) high reconfiguration costs and complexity, complicating frequent re-embedding; (2) physical layer constraints such as fiber loss, dispersion, and non-linearities, which degrade signal quality over long distances and necessitate adaptation in VNE schemes; (3) wavelength-granular bandwidth limitations, restricting flexible resource allocation and complicating fractional bandwidth requirements; (4) resource fragmentation from scattered wavelengths, requiring detailed planning; and (5) high costs of amplification for long-haul transmission, necessitating route optimization in VNE to manage energy costs effectively. Future research could further address these challenges, paving the way for efficient solutions in optical network VNE.

3) *IoT and IoVs*: The rapid proliferation of IoTs and IoVs and their resource limitations have meant that the adoption of NV was inevitable [338]. A critical challenge in this regard is meeting the stringent and heterogeneous QoS requirements of real-time applications such as industrial IoT, high-definition video monitoring, AR, and VR. To address the strict QoS of real-world applications, authors in [312] discussed the GCN-based embedding to establish the relationship between substrate nodes. On the other hand, [339] focused on mitigating multidimensional resource fragmentation to maximize the SPs' profit. The authors formulated a VNE as a bi-level optimization (BiVNE), considering the interdependence between VMs and VLs. They introduced a nested bi-level optimization framework and adopted an ant colony system (ACS) for VME and a Dijkstra algorithm augmented with an exact-fit spectrum slot assignment method for VLE. However, little impetus is given to achieving a balanced assignment, budgetary constraints, and energy consumption. These metrics can be considered immediate future work to set up a comprehensive IoT network. On the other hand, IoVs represent a connected transportation system that continuously interacts and exchanges data for intelligent decisions [340]. NV fits appropriately in this setup and enables the users to utilize the resources optimally. For such environments, the adoption of NV is relatively in the beginning stage, and some preliminary research has been conducted by Fan et al. [321] and focused on maximizing the revenue-to-cost ratio and

utilization of the substrate resources by considering both topological and non-topological attributes. However, these environments are inherently dynamic and require effective re-embedding schemes to deal with the dynamic arrival and departure of VNRs. Another promising research direction explored in [341] focuses on supporting IoV services with diverse QoS requirements. The authors propose integrating NFV and SDN, two complementary technologies, to reduce function provisioning costs and efficiently coordinate heterogeneous physical resources in space-air-ground integrated networks (SAGIN) [342].

4) *Satellite Networks*: Satellite networks offer global Internet access, multimedia broadcasting, and emergency communication, but they pose unique VNE challenges: (1) High Mobility and Latency: Satellite movement causes high latency and intermittent connections, complicating VNE for latency-sensitive applications, which need adaptive algorithms to maintain stable links; (2) Limited Bandwidth and Resources: With less bandwidth and computing power than terrestrial networks, satellite VNE requires highly efficient resource management; (3) Power constraints: As satellites rely on solar and battery power, energy-efficient VNE methods are essential to reduce consumption; (4) Diverse QoS Requirements: Applications in satellite networks have varying QoS needs, requiring VNE solutions to balance these with limited onboard resources; (5) Frequent Topology Changes: Constant shifts in low earth orbit (LEO) and medium earth orbit (MEO) constellations necessitate frequent re-embedding of VNRs; (6) Environmental Vulnerabilities: Atmospheric conditions can increase the risk of data loss, requiring robust error correction in VNE; (7) Security Concerns: The vulnerability to jamming and interception requires advanced VNE security measures. While multi-constraint strategies like those in [320] address some challenges, further innovation is essential to meet satellite networks' unique demands fully.

D. Inclusive Adoption of AI/ML for VNE

The adoption of AI/ML has gained significant impetus in the recent past. In this context, multiple works, as discussed in [157], [297], [308] adopted multiple ML-inspired strategies such as MAS, field theory, and K-Means clustering to optimize the resource allocation for VNRs. Alternatively, RL and DL-based solutions [130], [171], [177], [301], [312], [315], [316], [317] have also attained significant popularity owing to their apparent abilities to handle high-dimensional data, ease of automation, flexibility, adaptability, end-to-end learning, and real-time decision making [343], [344], [345], [346]. However, with the ever-increasing adoption of AI and ML, it is expected to evolve and play a significant role, especially in resource allocation, boosting the performance of VNRs, and automating network management activities. A few aspects that can be looked into are network automation and orchestration, optimal resource allocation, predictive analytics, security and anomaly detection, dynamic scaling, QoS adherence, multi-domain VNE, and energy efficiency. The rapid adoption of federated learning (FL) [347], [348], [349], [350] facilitates training across decentralized sources with local data without

exercising central control. Such a model can be adopted to enforce privacy and data security, where the models can be distributively trained without sharing sensitive data.

E. Digital Twins

The digital twin is the virtual representation or digital counterpart of a physical object, system, or process [351], [352], [353]. It is created by integrating data from various sources, such as sensors, IoTs, and multiple data streams, enabling real-time processing, analysis, and decision-making [354]. It presents an exciting frontier for researchers to use digital data to optimize, manage, and exercise control of VNRs. Once the digital twin evolves and reaches standardization, it will accurately mimic the physical networks in real time, assisting in dynamic and precise decision-making. Another line of research that can be explored is the integration of AI and ML techniques with digital twins in the development of self-healing networks. This can be a game-changer in embedding decisions, especially by predicting network failures, anticipating traffic patterns, and restoring network functionality from the digital data. Other research challenges on digital twin-powered VNE are security and privacy, energy efficiency, multi-domain VNE, resiliency, and disaster recovery. The convergence of VNE and the digital twins has tremendous potential for enabling seamless operations, boosting resilience, and improving the overall QoS. Though these technologies continue evolving, they will inevitably play a vital role in managing complex modern networks.

F. VNE for Software-Defined Agile and Intent-Driven Networks

SDN has significantly eased the burden on network administrators by offering automation, management, flexibility, programmability, and cost-effectiveness [355], [356]. The authors in [164], [319], [357], [358] explored the problem of VNE over SDN-enabled VNRs and SNs, focusing on improving the revenue of SPs, the acceptance ratio, and the response from the switch controller by examining the correlation between VMs. However, these studies lacked scalability, dynamic adaptation, energy efficiency, security, and QoS compliance while overlooking multi-domain integration, fault tolerance, and cost efficiency.

On the other hand, intent-based networking (IBN) takes the automation further by focusing on high-level business policies and objectives, automated network provisioning, continuous monitoring and assurance, adaptive self-optimization, business intent alignment, enhanced security, and extreme scalability [359]. The amalgamation of IBN and VNE can pave the way for highly efficient and adaptable network management. In this regard, the researchers can explore a few directions, including developing formal methods and languages to express the intents, intent inference and mapping algorithms, intent assurance and verification procedures, and intent orchestration tools. Additionally, the research community can look into security, where the networking decisions for VNE based on intent do not jeopardize the network's security. Finally, these

IBN-powered VNE strategies need to be tested for scalability and efficiency in real-world deployment, and standardization needs to be done to avoid interoperability issues.

G. VNE for VR/XR Technologies

VNE over VR/XR technologies is an upcoming field with significant potential to elevate immersive experiences and connectivity in virtual environments. The current research trends focus on optimizing the QoE by considering embedding latency, resource assignment, and adaptive network configuration [360]. Moreover, with the adoption of contemporary technologies such as 5G, edge computing, and AI/ML, the real-time performance of such applications is expected to achieve a significant boost. Some research works in this direction have been discussed in [152], [207], [293], [361], [362]. Additionally, the users of VR/XR may span across different locations and domains, requiring multi-domain embedding, and some proposals for the same have been presented in [300], [363]. Though VNE for VR/XR technologies is still in its infancy, some research directions that are likely to be uninterrupted attention include developing a mixed reality with holographic displays with extraordinarily low latency and high bandwidth requirements. With the growing popularity of augmented reality/virtual reality, researchers can look into embedding techniques supporting large-scale, real-time, and immersive social interactions. Additionally, the current research on VR/XR lacks standardization, thereby raising issues of interoperability and ethical and regulatory concerns. The research community should look into data privacy, content moderation, and safety for VNs, and quantum computing with secure key distribution and quantum communication can be able choice [364]. For far too long, the researchers have focused on developing SP-centric solutions catering to cost, profit, utilization, etc. However, the trends must shift to creating more user-centric solutions, considering preferences, user mobility, and dynamic network adaptation to provide a personalized and adaptive virtual experience.

H. Security and Privacy Concerns

As NV enables the users to share the common infrastructure of the SPs, security concerns are central to be considered [365], [366]. The SP must ensure that the executing VNR is secured and does not perform any malicious activity. For instance, a user may request the SP not to co-host the VNR components on the same servers/physical links to avoid cross and side-channel attacks [367], [368]. These issues cannot be packaged as software and must be integrated with VNE schemes, which calls for more secure embedding strategies. Moreover, blockchain as an enabler has tremendous scope to facilitate seamless access control and authenticate users within a shared virtual environment. It should be looked into by carefully considering the related open research issues regarding scalability and overhead reduction [369], [370]. The advancements in AI/ML can assist the SPs in detecting traffic patterns and anomalies and provide a suitable response in real-time [371]. FL can play a significant role in this context, especially when parties do not want to share sensitive

data [372]. Other security aspects that should be considered are threat intelligence sharing, standardization, interoperability concerns, vulnerability assessment, compliance regulations, ethical considerations, and user awareness. Overall, the future of security and privacy preservation for VNE will require a multi-faceted approach, incorporating advanced technologies, regulatory compliance, user empowerment, and ethical considerations to ensure the integrity and privacy of VNs.

VII. CONCLUSIVE REMARKS AND FINAL DISCUSSION

VNE has been a central research and industrial issue in network virtualization in recent years. It has been demonstrated to be a computationally intractable problem comprising two closely related sub-problems, i.e., VME and VLE, which are unsolvable in polynomial time. Therefore, developing a pseudo-optimal solution is challenging, and the research community has spent considerable effort addressing the problem and proposing multiple innovative solutions. This paper presents an exhaustive survey of recent VNE solutions, explicitly focusing on the current research trends in the area. We start the study by formally introducing the problem, identifying a novel taxonomy to shed light on the existing solutions in the literature, and subsequently providing a thorough comparative analysis of them.

After carefully examining the literature, we found multiple additional opportunities for the research community to work on. For instance, firstly, the research on distributed VNE has only received limited attention, and issues of secure communication between SPs can be further investigated. Secondly, topics such as utilizing renewable sources to reduce carbon footprints (sustainability), developing large-scale VM migration plans, developing emulated setups, and providing end-to-end security services for end-users have some unexplored aspects and have colossal research potential. Moreover, integrating NV into less traditional and/or less core networking deployments, such as WSNs, Optical and satellite networks, AR/XR, digital twins, and SAGIN, has opened up additional technical challenges requiring significant attention and provides exciting opportunities for future research. Finally, the future of security and privacy preservation for VNEs requires a multi-faceted solution, incorporating advanced technologies, regulatory compliance, user empowerment, and ethical considerations, which the research community can explore.

REFERENCES

- [1] M. F. Bari et al., "Data center network virtualization: A survey," *IEEE Commun. Surveys Tuts.*, vol. 15, no. 2, pp. 909–928, 2nd Quart., 2013.
- [2] S. Chaisiri, B.-S. Lee, and D. Niyato, "Optimization of resource provisioning cost in cloud computing," *IEEE Trans. Services Comput.*, vol. 5, no. 2, pp. 164–177, Apr.–Jun. 2012.
- [3] "Amazon data centers." Accessed: Mar. 2023. [Online]. Available: <https://aws.amazon.com/compliance/data-center/data-centers/>
- [4] "Google data centers." Accessed: Mar. 2023. [Online]. Available: <https://www.google.com/about/datacenters/>
- [5] "Azure data centers." Accessed: Mar. 2023. [Online]. Available: <https://azure.microsoft.com/en-in/global-infrastructure/geographies/#overview>
- [6] "Facebook data centers." Accessed: Mar. 2023. [Online]. Available: <https://datacenters.fb.com/>
- [7] "VMware." Accessed: Mar. 2023. [Online]. Available: <http://www.vmware.com>
- [8] "Xen." Accessed: Mar. 2023. [Online]. Available: <http://xen.org>
- [9] R. P. Goldberg, "Architecture of virtual machines," in *Proc. Workshop Virtual Comput. Syst.*, 1973, pp. 74–112.
- [10] S. T. King, G. W. Dunlap, and P. M. Chen, "Operating system support for virtual machines," in *Proc. USENIX Annu. Tech. Conf. General Track*, 2003, pp. 71–84.
- [11] J. Smith and R. Nair, "The architecture of virtual machines," *Computer*, vol. 38, no. 5, pp. 32–38, 2005.
- [12] N. Ogino, T. Kitahara, S. Arakawa, and M. Murata, "Virtual network embedding with multiple priority classes sharing substrate resources," *Comput. Netw.*, vol. 112, pp. 52–66, Jan. 2017.
- [13] A. Shieh, S. Kandula, A. G. Greenberg, C. Kim, and B. Saha, "Sharing the data Center network," in *Proc. NSDI*, vol. 11, 2011, pp. 23–23.
- [14] A. Belbekkouche, M. M. Hasan, and A. Karmouch, "Resource discovery and allocation in network virtualization," *IEEE Commun. Surveys Tuts.*, vol. 14, no. 4, pp. 1114–1128, 4th Quart., 2012.
- [15] A. Fischer, J. F. Botero, M. T. Beck, H. de Meer, and X. Hesselbach, "Virtual network embedding: A survey," *IEEE Commun. Surveys Tuts.*, vol. 15, no. 4, pp. 1888–1906, 4th Quart., 2013.
- [16] H. Cao, L. Yang, Z. Liu, and M. Wu, "Exact solutions of VNE: A survey," *China Commun.*, vol. 13, no. 6, pp. 48–62, 2016.
- [17] C. Yu, W. Hou, and L. Guo, "A survey on virtual network embedding in optical cloud data center networks," in *Proc. Int. Conf. Softw. Netw. (ICSN)*, 2016, pp. 1–5.
- [18] H. Cao, H. Hu, Z. Qu, and L. Yang, "Heuristic solutions of virtual network embedding: A survey," *China Commun.*, vol. 15, no. 3, pp. 186–219, 2018.
- [19] H. Cao, S. Wu, Y. Hu, Y. Liu, and L. Yang, "A survey of embedding algorithm for virtual network embedding," *China Commun.*, vol. 16, no. 12, pp. 1–33, 2019.
- [20] W. Jiang, H. Han, M. He, and W. Gu, "When game theory meets satellite communication networks: A survey," *Comput. Commun.*, vol. 217, pp. 208–229, Mar. 2024.
- [21] S. Wu, N. Chen, A. Xiao, P. Zhang, C. Jiang, and W. Zhang, "AI-empowered virtual network embedding: A comprehensive survey," *IEEE Commun. Surveys Tuts.*, early access, Jul. 8, 2024, doi: 10.1109/COMST.2024.3424533.
- [22] T. Anderson, L. Peterson, S. Shenker, and J. Turner, "Overcoming the Internet impasse through virtualization," *Computer*, vol. 38, no. 4, pp. 34–41, 2005.
- [23] F. Baroncelli, B. Martini, and P. Castoldi, "Network virtualization for cloud computing," *Annales Des TéléCommun.*, vol. 65, pp. 713–721, Mar. 2010.
- [24] H. Alshaer, "An overview of network virtualization and cloud network as a service," *Int. J. Netw. Manag.*, vol. 25, no. 1, pp. 1–30, 2015.
- [25] X. Liu, B. Wang, and Z. Yang, "Virtual network embedding based on topology potential," *Entropy*, vol. 20, no. 12, p. 941, 2018.
- [26] R. P. Esteves, L. Z. Granville, and R. Boutaba, "On the management of virtual networks," *IEEE Commun. Mag.*, vol. 51, no. 7, pp. 80–88, Jul. 2013.
- [27] N. C. Fernandes et al., "Virtual networks: Isolation, performance, and trends," *Annales Des TéléCommun.*, vol. 66, no. 5, pp. 339–355, 2011.
- [28] S. Peng, R. Nejabati, and D. Simeonidou, "Role of optical network virtualization in cloud computing," *J. Opt. Commun. Netw.*, vol. 5, no. 10, pp. A162–A170, 2013.
- [29] J. Ramakrishnan, M. S. Shabbir, N. M. Kassim, P. T. Nguyen, and D. Mavaluru, "A comprehensive and systematic review of the network virtualization techniques in the IoT," *Int. J. Commun. Syst.*, vol. 33, no. 7, 2020, Art. no. e4331.
- [30] N. M. M. K. Chowdhury and R. Boutaba, "Network virtualization: State of the art and research challenges," *IEEE Commun. Mag.*, vol. 47, no. 7, pp. 20–26, Jul. 2009.
- [31] K. Tutschku, T. Zinner, A. Nakao, and P. Tran-Gia, "Network virtualization: Implementation steps towards the future Internet," *Electron. Commun. EASST*, vol. 17, p. 219, Jan. 2009.
- [32] A. Berl, A. Fischer, and H. de Meer, "Using system virtualization to create virtualized networks," *Electron. Commun. EASST*, vol. 17, p. 218, Jan. 2009.
- [33] H. Cui, W. Gao, J. Liu, and Y. Liu, "A virtual network embedding algorithm based on virtual topology connection feature," in *Proc. 16th Int. Symp. Wireless Pers. Multimedia Commun. (WPMC)*, 2013, pp. 1–5.
- [34] X. Cheng et al., "Virtual network embedding through topology-aware node ranking," *ACM SIGCOMM Comput. Commun. Rev.*, vol. 41, no. 2, pp. 38–47, 2011.

- [35] I. Hamzaoui, B. Duthil, V. Courboulay, and H. Medromi, "Proactive and power efficient hybrid virtual network embedding: An AWS cloud case study," *IEEE Access*, vol. 10, pp. 57499–57513, 2022.
- [36] P. Zhang, Y. Su, J. Wang, C. Jiang, C.-H. Hsu, and S. Shen, "Reinforcement learning assisted bandwidth aware virtual network resource allocation," *IEEE Trans. Netw. Service Manag.*, vol. 19, no. 4, pp. 4111–4123, Dec. 2022.
- [37] W. Zhang, D. Wang, S. Yu, H. He, and Y. Wang, "Repeatable multi-dimensional virtual network embedding in cloud service platform," *IEEE Trans. Services Comput.*, vol. 15, no. 6, pp. 3499–3512, Nov./Dec. 2022.
- [38] L. Li, J. Chen, T. Huang, and H. Cui, "An efficient virtual network embedding algorithm based on subgraph," in *Proc. Int. Symp. Wireless Pers. Multimedia Commun. (WPMC)*, 2014, pp. 23–28.
- [39] S. Haeri and L. Trajković, "Virtual network embedding via Monte Carlo tree search," *IEEE Trans. Cybern.*, vol. 48, no. 2, pp. 510–521, Feb. 2018.
- [40] H. T. Dinh, C. Lee, D. Niyato, and P. Wang, "A survey of mobile cloud computing: Architecture, applications, and approaches," *Wireless Commun. Mobile Comput.*, vol. 13, no. 18, pp. 1587–1611, 2013.
- [41] P. Zhang, "Incorporating energy and load balance into virtual network embedding process," *Comput. Commun.*, vol. 129, pp. 80–88, Sep. 2018.
- [42] M. P. Gilesh, S. M. Kumar, and L. Jacob, "Resource availability—Aware adaptive provisioning of virtual data center networks," *Int. J. Netw. Manag.*, vol. 29, no. 2, 2019, Art. no. e2066.
- [43] F. Ansah and H. de Meer, "DBvLEA: A demand-based approach to virtual link mapping for multi-service industrial applications," in *Proc. IEEE 15th Int. Conf. Netw. Service Manag. (CNSM)*, 2019, pp. 1–9.
- [44] D. G. Andersen (Carnegie Mellon University, Pittsburgh, PA, USA). *Theoretical Approaches to Node Assignment*. 2018. [Online]. Available: <https://doi.org/10.1184/R1/6610829.v1>
- [45] E. Amaldi, S. Coniglio, A. M. Koster, and M. Tieves, "On the computational complexity of the virtual network embedding problem," *Electron. Notes Discr. Math.*, vol. 52, no. 1, pp. 213–220, 2016.
- [46] L. D. Nguyen, N. Kim, S. Kim, and C.-K. Kim, "RT-VNE: A real-time strategy for virtual network embedding towards resource efficiency," in *Proc. Int. Conf. Inf. Netw. (ICOIN)*, 2017, pp. 185–190.
- [47] M. Shen, X. Ke, F. Li, F. Li, L. Zhu, and L. Guan, "Availability-aware virtual network embedding for multi-tier applications in cloud networks," in *Proc. IEEE 17th Int. Conf. High Perform. Comput. Commun. IEEE 7th Int. Symp. Cyberspace Safety Security IEEE 12th Int. Conf. Embedded Softw. Syst.*, 2015, pp. 1–6.
- [48] Q. Zhong, Y. Wang, L. Meng, A. Xiao, and H. Zhang, "A max-flow/min-cut theory based multi-domain virtual network splitting mechanism," in *Proc. 17th Asia-Pac. Netw. Oper. Manag. Symp. (APNOMS)*, 2015, pp. 392–395.
- [49] E. S. Correa, L. A. Fletscher, and J. F. Botero, "Virtual data Center embedding: A survey," *IEEE Latin America Trans.*, vol. 13, no. 5, pp. 1661–1670, May 2015.
- [50] W. Xia, Y. Wen, C. H. Foh, D. Niyato, and H. Xie, "A survey on software-defined networking," *IEEE Commun. Surveys Tuts.*, vol. 17, no. 1, pp. 27–51, 1st Quart., 2015.
- [51] B. A. A. Nunes, M. Mendonca, X.-N. Nguyen, K. Obraczka, and T. Turetli, "A survey of software-defined networking: Past, present, and future of programmable networks," *IEEE Commun. Surveys Tuts.*, vol. 16, no. 3, pp. 1617–1634, 3rd Quart., 2014.
- [52] K. Benzekki, A. E. Fergougui, and A. E. Elalaoui, "Software-defined networking (SDN): A survey," *Security Commun. Netw.*, vol. 9, no. 18, pp. 5803–5833, 2016.
- [53] R. Mijumbi, J. Serrat, J.-L. Gorricho, N. Bouten, F. De Turck, and R. Boutaba, "Network function virtualization: State-of-the-art and research challenges," *IEEE Commun. Surveys Tuts.*, vol. 18, no. 1, pp. 236–262, 1st Quart., 2015.
- [54] B. Yi et al., "A comprehensive survey of network function virtualization," *Comput. Netw.*, vol. 133, pp. 212–262, Mar. 2018.
- [55] B. Han, V. Gopalakrishnan, L. Ji, and S. Lee, "Network function virtualization: Challenges and opportunities for innovations," *IEEE Commun. Mag.*, vol. 53, no. 2, pp. 90–97, Feb. 2015.
- [56] T. Wang and M. Hamdi, "PRESTO: Towards efficient online virtual network embedding in virtualized cloud data centers," *Comput. Netw.*, vol. 106, pp. 196–208, Sep. 2016.
- [57] T. Chen, J. Liu, Q. Tang, T. Huang, and R. Huo, "Virtual network embedding algorithm for location-based identifier allocation," *IEEE Access*, vol. 7, pp. 31159–31169, 2019.
- [58] J. Zhao and Z. Wang, "Energy-efficient multi-domain virtual network embedding algorithm based on reinforcement learning," in *Proc. Adv. Nat. Comput. Fuzzy Syst. Knowl. Disc. (ICNC-FSKD)*, 2022, pp. 1132–1143.
- [59] A. Blenk, P. Kalmbach, P. van der Smagt, and W. Kellerer, "Boost online virtual network embedding: Using neural networks for admission control," in *Proc. 12th Int. Conf. Netw. Service Manag. (CNSM)*, 2016, pp. 10–18.
- [60] P. Shantharama, A. S. Thyagaturu, and M. Reisslein, "Hardware-accelerated platforms and infrastructures for network functions: A survey of enabling technologies and research studies," *IEEE Access*, vol. 8, pp. 132021–132085, 2020.
- [61] "Cisco CRS-X." Accessed: Jul. 2024. [Online]. Available: <https://fq0o.short.gy/KellGg>
- [62] "Juniper networks PTX series." Accessed: Jul. 2024. [Online]. Available: <https://fq0o.short.gy/fr0wHf>
- [63] "Arista 7500R series." Accessed: Jul. 2024. [Online]. Available: <https://fq0o.short.gy/UypROQ>
- [64] "Huawei NE series." Accessed: Jul. 2024. [Online]. Available: <https://fq0o.short.gy/MT6BwX>
- [65] H. Shafiei, A. Khonsari, and P. Mousavi, "Serverless computing: A survey of opportunities, challenges, and applications," *ACM Comput. Surveys*, vol. 54, no. 11s, pp. 1–32, 2022.
- [66] J. Wen, Z. Chen, X. Jin, and X. Liu, "Rise of the planet of serverless computing: A systematic review," *ACM Trans. Softw. Eng. Methodol.*, vol. 32, no. 5, pp. 1–61, 2023.
- [67] "Dell EMC PowerEdge." Accessed: Jul. 2024. [Online]. Available: <https://fq0o.short.gy/oEvhPN>
- [68] "HPE ProLiant." Accessed: Jul. 2024. [Online]. Available: <https://fq0o.short.gy/qRGXQF>
- [69] "Broadcom NetXtreme." Accessed: Jul. 2024. [Online]. Available: <https://fq0o.short.gy/yfacRd>
- [70] "Intel Wi-Fi 6 AX200." Accessed: Jul. 2024. [Online]. Available: <https://fq0o.short.gy/DLKB1R>
- [71] "SMF-28 ultra." Accessed: Jul. 2024. [Online]. Available: <https://fq0o.short.gy/whCyLP>
- [72] "TrueWave RS." Accessed: Jul. 2024. [Online]. Available: <https://fq0o.short.gy/BuvZwv>
- [73] "Cisco IOS." Accessed: Jul. 2024. [Online]. Available: <https://fq0o.short.gy/70Gq1>
- [74] "Juniper Junos." Accessed: Jul. 2024. [Online]. Available: <https://fq0o.short.gy/i8tzGT>
- [75] "Arista EOS." Accessed: Jul. 2024. [Online]. Available: <https://www.arista.com/en/products/eos>
- [76] S. Badotra and J. Singh, "Open daylight as a controller for software defined networking," *Int. J. Adv. Res. Comput. Sci.*, vol. 8, no. 5, p. 10, 2017.
- [77] P. Berde et al., "ONOS: Towards an open, distributed SDN OS," in *Proc. 3rd Workshop Hot Topics Softw. Defined Netw.*, 2014, pp. 1–6.
- [78] J. Pettit, B. Pfaff, J. Stringer, C.-C. Tu, B. Blanco, and A. Tessmer, "Bringing platform harmony to VMware NSX," *ACM SIGOPS Oper. Syst. Rev.*, vol. 52, no. 1, pp. 123–128, 2018.
- [79] O. Sefraoui et al., "OpenStack: Toward an open-source solution for cloud computing," *Int. J. Comput. Appl.*, vol. 55, no. 3, pp. 38–42, 2012.
- [80] T. Szigeti, D. Zacks, M. Falkner, and S. Arena, *Cisco Digital Network Architecture: Intent-Based Networking for the Enterprise*. Indianapolis, IN, USA: Cisco Press, 2018.
- [81] J. Turnbull, *Monitoring With Prometheus*. London, U.K.: Turnbull Press, 2018.
- [82] M. Chakraborty and A. P. Kundan, "Grafana," in *Monitoring Cloud-Native Applications: Lead Agile Operations Confidently Using Open Source Software*. Berkeley, CA, USA: Apress, 2021, pp. 187–240.
- [83] A. Satpathy, S. K. Addya, A. K. Turuk, B. Majhi, and G. Sahoo, "Crow search based virtual machine placement strategy in cloud data centers with live migration," *Comput. Elect. Eng.*, vol. 69, pp. 334–350, Jul. 2018.
- [84] F. Xu, F. Liu, H. Jin, and A. V. Vasilakos, "Managing performance overhead of virtual machines in cloud computing: A survey, state of the art, and future directions," *Proc. IEEE*, vol. 102, no. 1, pp. 11–31, Jan. 2014.
- [85] W. Huang, J. Liu, B. Abali, and D. K. Panda, "A case for high performance computing with virtual machines," in *Proc. ICS*, 2006, pp. 125–134.

- [86] J. Carapinha and J. Jiménez, "Network virtualization: A view from the bottom," in *Proc. 1st ACM Workshop Virtualized Infrastruct. Syst. Archit.*, 2009, pp. 73–80.
- [87] G. Schaffrath et al., "Network virtualization architecture: Proposal and initial prototype," in *Proc. 1st ACM Workshop Virtualized Infrastruct. Syst. Archit.*, 2009, pp. 63–72.
- [88] A. Satpathy, M. N. Sahoo, L. Behera, and C. Swain, "ReMatch: An efficient virtual data center re-matching strategy based on matching theory," *IEEE Trans. Services Comput.*, vol. 16, no. 2, pp. 1373–1386, Mar./Apr. 2023.
- [89] M. F. Bari, S. R. Chowdhury, R. Ahmed, and R. Boutaba, "On orchestrating virtual network functions," in *Proc. 11th Int. Conf. Netw. Service Manag. (CNSM)*, 2015, pp. 50–56.
- [90] J. F. Riera, E. Escalona, J. Batalle, E. Grasa, and J. A. Garcia-Espin, "Virtual network function scheduling: Concept and challenges," in *Proc. Int. Conf. Smart Commun. Netw. Technol. (SaCoNeT)*, 2014, pp. 1–5.
- [91] A. Laghrissi and T. Taleb, "A survey on the placement of virtual resources and virtual network functions," *IEEE Commun. Surveys Tuts.*, vol. 21, no. 2, pp. 1409–1434, 2nd Quart., 2018.
- [92] A. Sekiguchi, T. Kondo, K. Mori, K. Kumakura, L. Zhang, and F. Teraoka, "Poster: A container orchestration method considering shared containers in an MEC environment," in *Proc. 22nd Annu. Int. Conf. Mobile Syst. Appl. Services*, 2024, pp. 698–699.
- [93] Q. He, F. Zhang, G. Bian, W. Zhang, Z. Li, and D. Duan, "Real-time network virtualization based on SDN and docker container," *Clust. Comput.*, vol. 26, no. 3, pp. 2069–2083, 2023.
- [94] C. Boettiger, "An introduction to docker for reproducible research," *ACM SIGOPS Oper. Syst. Rev.*, vol. 49, no. 1, pp. 71–79, 2015.
- [95] Y. Zhang, R. Meredith, W. Reeves, J. Coriolano, M. A. Babar, and A. Rahman, "Does generative AI generate smells related to container orchestration? An exploratory study with kubernetes manifests," in *Proc. IEEE/ACM 21st Int. Conf. Min. Softw. Rep. (MSR)*, 2024, pp. 192–196.
- [96] S. D'Oro, L. Galluccio, S. Palazzo, and G. Schembra, "Exploiting congestion games to achieve distributed service chaining in NFV networks," *IEEE J. Sel. Areas Commun.*, vol. 35, no. 2, pp. 407–420, Feb. 2017.
- [97] W. John et al., "Research directions in network service chaining," in *Proc. IEEE SDN Future Netw. Services (SDN4FNS)*, 2013, pp. 1–7.
- [98] D. Bhamare, R. Jain, M. Samaka, and A. Erbad, "A survey on service function chaining," *J. Netw. Comput. Appl.*, vol. 75, pp. 138–155, Nov. 2016.
- [99] N. C. Thang and M. Park, "Detecting compromised switches and middlebox-bypass attacks in service function chaining," in *Proc. IEEE 29th Int. Telecommun. Netw. Appl. Conf. (ITNAC)*, 2019, pp. 1–6.
- [100] X. Foukas, G. Patounas, A. Elmokashfi, and M. K. Marina, "Network slicing in 5G: Survey and challenges," *IEEE Commun. Mag.*, vol. 55, no. 5, pp. 94–100, May 2017.
- [101] H. Gamage, N. Rajatheva, and M. Latva-Aho, "Channel coding for enhanced mobile broadband communication in 5G systems," in *Proc. Eur. Conf. Netw. Commun. (EuCNC)*, 2017, pp. 1–6.
- [102] P. Popovski, K. F. Trillingsgaard, O. Simeone, and G. Durisi, "5G wireless network slicing for eMBB, URLLC, and mMTC: A communication-theoretic view," *IEEE Access*, vol. 6, pp. 55765–55779, 2018.
- [103] I. Jovović, I. Forenbacher, and M. Periša, "Massive machine-type communications: An overview and perspectives towards 5G," in *Proc. 3rd Int. Virtual Res. Conf. Tech. Disciplines*, vol. 3, 2015, pp. 1–8.
- [104] W. Webb, "Broadband fixed wireless access as a key component of the future integrated communications environment," *IEEE Commun. Mag.*, vol. 39, no. 9, pp. 115–121, Sep. 2001.
- [105] M. Masdari, S. S. Nabavi, and V. Ahmadi, "An overview of virtual machine placement schemes in cloud computing," *J. Netw. Comput. Appl.*, vol. 66, pp. 106–127, May 2016.
- [106] F. L. Pires and B. Barán, "A virtual machine placement taxonomy," in *Proc. 15th IEEE/ACM Int. Symp. Clust. Cloud Grid Comput.*, 2015, pp. 159–168.
- [107] F. Zhang, G. Liu, X. Fu, and R. Yahyapour, "A survey on virtual machine migration: Challenges, techniques, and open issues," *IEEE Commun. Surveys Tuts.*, vol. 20, no. 2, pp. 1206–1243, 2nd Quart., 2018.
- [108] Z. Li, X. Yu, L. Yu, S. Guo, and V. Chang, "Energy-efficient and quality-aware VM consolidation method," *Future Gener. Comput. Syst.*, vol. 102, pp. 789–809, Jan. 2020.
- [109] T. Hidayat, K. Ramli, N. Thereza, A. Daulay, R. Rushendra, and R. Mahardiko, "Machine learning to estimate workload and balance resources with live migration and VM placement," *Informatics*, vol. 11, no. 3, p. 50, 2024.
- [110] S. Shahryari, F. Tashtarian, and S.-A. Hosseini-Seno, "CoPaM: Cost-aware VM placement and migration for mobile services in multi-cloudlet environment: An SDN-based approach," *Comput. Commun.*, vol. 191, pp. 257–273, Jul. 2022.
- [111] O. Biran et al., "A stable network-aware vm placement for cloud systems," in *Proc. 12th IEEE/ACM Int. Symp. Clust. Cloud Grid Comput. (CCGRID)*, 2012, pp. 498–506.
- [112] F. F. Moges and S. L. Abebe, "Energy-aware VM placement algorithms for the OpenStack neat consolidation framework," *J. Cloud Comput.*, vol. 8, no. 1, p. 2, 2019.
- [113] G. P. Alkimi, D. M. Batista, and N. L. Da Fonseca, "Mapping virtual networks onto substrate networks," *J. Internet Services Appl.*, vol. 4, pp. 1–15, Jan. 2013.
- [114] E. Rodriguez, G. P. Alkimi, N. L. da Fonseca, and D. M. Batista, "Energy-aware mapping and live migration of virtual networks," *IEEE Syst. J.*, vol. 11, no. 2, pp. 637–648, Jun. 2017.
- [115] R. Buyya, M. Pathan, and A. Vakali, Eds., *Content Delivery Networks*, vol. 9, Heidelberg, Germany: Springer, 2008.
- [116] H. Yoo, Y.-C. Lee, K. Shin, and S.-W. Kim, "Directed network embedding with virtual negative edges," in *Proc. 15th ACM Int. Conf. Web Search Data Min.*, 2022, pp. 1291–1299.
- [117] A. Satpathy, M. N. Sahoo, A. K. Sangaiah, C. Swain, and S. Bakshi, "CoMap: An efficient virtual network re-mapping strategy based on coalitional matching theory," *Comput. Netw.*, vol. 216, Oct. 2022, Art. no. 109248.
- [118] A. Satpathy, M. N. Sahoo, C. Swain, M. Bilal, S. Bakshi, and H. Song, "GAMap: A genetic algorithm-based effective virtual data Center re-embedding strategy," *IEEE Trans. Green Commun. Netw.*, vol. 8, no. 2, pp. 791–801, Jun. 2024.
- [119] C. K. Dehury and P. K. Sahoo, "DYVINE: Fitness-based dynamic virtual network embedding in cloud computing," *IEEE J. Sel. Areas Commun.*, vol. 37, no. 5, pp. 1029–1045, May 2019.
- [120] A. Satpathy, M. N. Sahoo, L. Behera, C. Swain, and A. Mishra, "Vmatch: A matching theory based VDC reconfiguration strategy," in *Proc. IEEE 13th Int. Conf. Cloud Comput. (CLOUD)*, 2020, pp. 133–140.
- [121] A. Satpathy, M. N. Sahoo, L. Chottray, B. Majhi, A. Mishra, and S. Bakshi, "VRMap: A cost and time aware remapping of virtual data centres over a geo-distributed infrastructure," in *Proc. Int. Conf. Commun. Syst. Netw. (COMSNETS)*, 2020, pp. 427–434.
- [122] K. K. T.G., S. K. Addya, A. Satpathy, and S. Koolagudi, "NORD: NNode ranking-based efficient virtual network embedding over single domain substrate networks," *Comput. Netw.*, vol. 225, Apr. 2023, Art. no. 109661.
- [123] H. Xu and B. Li, "Anchor: A versatile and efficient framework for resource management in the cloud," *IEEE Trans. Parallel Distrib. Syst.*, vol. 24, no. 6, pp. 1066–1076, Jun. 2013.
- [124] T. Wood et al., "Black-box and gray-box strategies for virtual machine migration," in *Proc. NSDI*, vol. 7, 2007, pp. 17–17, doi: 10.5555/1973430.1973447.
- [125] P. Zhang, H. Yao, and Y. Liu, "Virtual network embedding based on computing, network, and storage resource constraints," *IEEE Internet Things J.*, vol. 5, no. 5, pp. 3298–3304, Oct. 2018.
- [126] J. Zhang, C. Zhao, H. Wu, M. Lin, and R. Duan, "Virtual network embedding based on graph entropy," *Entropy*, vol. 20, no. 5, p. 315, 2018.
- [127] Y. Liang and S. Zhang, "Embedding parallelizable virtual networks," *Comput. Commun.*, vol. 102, pp. 47–57, Apr. 2017.
- [128] G. Sun, D. Liao, D. Zhao, Z. Xu, and H. Yu, "Live migration for multiple correlated virtual machines in cloud-based data centers," *IEEE Trans. Services Comput.*, vol. 11, no. 2, pp. 279–291, Mar./Apr. 2018.
- [129] B. Dab, I. Fajjari, N. Aitsaadi, and G. Pujolle, "VNR-GA: Elastic virtual network reconfiguration algorithm based on genetic metaheuristic," in *Proc. IEEE Global Commun. Conf. (GLOBECOM)*, 2013, pp. 2300–2306.
- [130] M. He, L. Zhuang, S. Yang, Z. Xu, W. Li, and J. Lu, "An energy-efficient VNE algorithm based on bidirectional long short-term memory," *J. Netw. Syst. Manag.*, vol. 30, no. 3, p. 45, 2022.
- [131] T. G. K. Kumar, A. Srivastava, A. Satpathy, S. K. Addya, and S. G. Koolagudi, "MatchVNE: A stable virtual network embedding strategy based on matching theory," in *Proc. 15th Int. Conf. Commun. Syst. Netw. (COMSNETS)*, 2023, pp. 355–359.

- [132] J. M. Kleinberg, "Approximation algorithms for disjoint paths problems," Ph.D. dissertation, Dept. Comput. Sci., Massachusetts Inst. Technol., Cambridge, MA, USA, 1996.
- [133] S. G. Kolliopoulos and C. Stein, "Improved approximation algorithms for unsplittable flow problems," in *Proc. IEEE 38th Annu. Symp. Found. Comput. Sci.*, 1997, pp. 426–436.
- [134] J. Li, W. Shi, Q. Ye, S. Zhang, W. Zhuang, and X. Shen, "Joint virtual network topology design and embedding for cybertwin-enabled 6G core networks," *IEEE Internet Things J.*, vol. 8, no. 22, pp. 16313–16325, Nov. 2021.
- [135] R. S. Luis et al., "Multicore fiber interconnects for multi-terabit spine-leaf datacenter network topologies," *J. Opt. Commun. Netw.*, vol. 15, no. 7, pp. C41–C47, 2023.
- [136] P. J. Roig, S. Alcaraz, K. Gilly, and C. Juiz, "Modelling a leaf and spine topology for VM migration in fog computing," in *Proc. IEEE 24th Int. Conf. Electron.*, 2020, pp. 1–6.
- [137] Y. Zhou, X. Yang, Y. Li, D. Jin, L. Su, and L. Zeng, "Incremental re-embedding scheme for evolving virtual network requests," *IEEE Commun. Lett.*, vol. 17, no. 5, pp. 1016–1019, May 2013.
- [138] H. Lu and F. Zhang, "Resource fragmentation-aware embedding in dynamic network virtualization environments," *IEEE Trans. Netw. Service Manag.*, vol. 19, no. 2, pp. 936–948, Jun. 2022.
- [139] M. T. Beck, A. Fischer, J. F. Botero, C. Linnhoff-Popien, and H. de Meer, "Distributed and scalable embedding of virtual networks," *J. Netw. Comput. Appl.*, vol. 56, pp. 124–136, Oct. 2015.
- [140] V. Lira, E. Tavares, M. Oliveira, E. Sousa, and B. Nogueira, "Virtual network mapping considering energy consumption and availability," *Computing*, vol. 101, pp. 937–967, May 2019.
- [141] C. K. Dehury and P. K. Sahoo, "Failure aware semi-centralized virtual network embedding in cloud computing fat-tree data center networks," *IEEE Trans. Cloud Comput.*, vol. 10, no. 2, pp. 1156–1172, Apr.–Jun. 2022.
- [142] A. Fischer, J. F. B. Vega, M. Duelli, D. Schlosser, X. H. Serra, and H. De Meer, "ALEVIN—A framework to develop, compare, and analyze virtual network embedding algorithms," *J. Electron. Commun. EASST*, vol. 37, pp. 1–12, Aug. 2011.
- [143] D. Stezenbach, M. Hartmann, and K. Tutschku, "Parameters and challenges for virtual network embedding in the future Internet," in *Proc. IEEE Netw. Oper. Manag. Symp.*, 2012, pp. 1272–1278.
- [144] J. Tian, X. Zeng, and X. Zhu, "DDS: A delay-based differentiated service virtual network embedding algorithm," *Appl. Sci.*, vol. 12, no. 19, p. 9897, 2022.
- [145] A. Tizghadam and A. Leon-Garcia, "On robust traffic engineering in transport networks," in *Proc. IEEE GLOBECOM Global Telecommun. Conf.*, 2008, pp. 1–6.
- [146] A. Fischer, M. T. Beck, and H. De Meer, "An approach to energy-efficient virtual network embeddings," in *Proc. IFIP/IEEE Int. Symp. Integr. Netw. Manag. (IM)*, 2013, pp. 1142–1147.
- [147] X. Cheng et al., "Virtual network embedding through topology awareness and optimization," *Comput. Netw.*, vol. 56, no. 6, pp. 1797–1813, 2012.
- [148] P. Lv, Z. Cai, J. Xu, and M. Xu, "Multicast service-oriented virtual network embedding in wireless mesh networks," *IEEE Commun. Lett.*, vol. 16, no. 3, pp. 375–377, Mar. 2012.
- [149] I. Fajjari, N. Aitsaadi, G. Pujolle, and H. Zimmermann, "VNE-AC: Virtual network embedding algorithm based on ant colony metaheuristic," in *Proc. IEEE Int. Conf. Commun. (ICC)*, 2011, pp. 1–6.
- [150] X. Zhang, X. Chen, and C. Phillips, "Achieving effective resilience for QoS-aware application mapping," *Comput. Netw.*, vol. 56, no. 14, pp. 3179–3191, 2012.
- [151] S. B. Masti and S. V. Raghavan, "VNA: An enhanced algorithm for virtual network embedding," in *Proc. IEEE 21st Int. Conf. Comput. Commun. Netw. (ICCCN)*, 2012, pp. 1–9.
- [152] P. Zhang, N. Chen, S. Li, K.-K. R. Choo, C. Jiang, and S. Wu, "Multi-domain virtual network embedding algorithm based on horizontal federated learning," *IEEE Trans. Inf. Forensics Security*, vol. 18, pp. 3363–3375, 2023.
- [153] G. Sun, H. Yu, V. Anand, L. Li, and H. Di, "Optimal provisioning for virtual network request in cloud-based data centers," *Photon. Netw. Commun.*, vol. 24, pp. 118–131, Feb. 2012.
- [154] GNU General Public License. "GNU linear programming kit, free software foundation, Boston, MA." Accessed: Jul. 23, 2023. [Online]. Available: <https://www.gnu.org/software/glpk/>
- [155] E. Zegura, K. Calvert, and S. Bhattacharjee, "How to model an Internet network," in *Proc. IEEE INFOCOM Conf. Comput. Commun.*, vol. 2, 1996, pp. 594–602.
- [156] A. Hagberg, P. Swart, and D. S. Chult, *Exploring Network Structure, Dynamics, and Function Using NetworkX*, Los Alamos Nat. Lab., Los Alamos, NM, USA, 2008.
- [157] A. A. Nasiri, F. Derakhshan, and S. S. Heydari, "Distributed virtual network embedding for software-defined networks using multiagent systems," *IEEE Access*, vol. 9, pp. 12027–12043, 2021.
- [158] X. Guan, B.-Y. Choi, and S. Song, "Energy efficient virtual network embedding for green data centers using data center topology and future migration," *Comput. Commun.*, vol. 69, pp. 50–59, Sep. 2015.
- [159] "Simulation platform for Scotfield Cao." Accessed: May 2023. [Online]. Available: <https://pan.baidu.com>
- [160] M. Pham, D. B. Hoang, and Z. Chaczko, "Congestion-aware and energy-aware virtual network embedding," *IEEE/ACM Trans. Netw.*, vol. 28, no. 1, pp. 210–223, Feb. 2020.
- [161] H. Cao, Y. Zhu, G. Zheng, and L. Yang, "A novel optimal mapping algorithm with less computational complexity for virtual network embedding," *IEEE Trans. Netw. Service Manag.*, vol. 15, no. 1, pp. 356–371, Mar. 2018.
- [162] H. Cao, L. Yang, and H. Zhu, "Novel node-ranking approach and multiple topology attributes-Based embedding algorithm for single-domain virtual network embedding," *IEEE Internet Things J.*, vol. 5, no. 1, pp. 108–120, Apr. 2018.
- [163] H. Cao, Y. Guo, Z. Qu, S. Wu, H. Zhu, and L. Yang, "ER-VNE: A joint energy and revenue embedding algorithm for embedding virtual networks," *IEEE Access*, vol. 6, pp. 47815–47827, 2018.
- [164] M. K. Haghani, B. Bakhshi, and A. Capone, "Multi-objective embedding of software-defined virtual networks," *Comput. Commun.*, vol. 129, pp. 32–42, Sep. 2018.
- [165] W. Fan, F. Xiao, X. Chen, L. Cui, and S. Yu, "Efficient virtual network embedding of cloud-based data center networks into optical networks," *IEEE Trans. Parallel Distrib. Syst.*, vol. 32, no. 11, pp. 2793–2808, Nov. 2021.
- [166] R. Buyya, R. Ranjan, and R. N. Calheiros, "Modeling and simulation of scalable cloud computing environments and the CloudSim toolkit: Challenges and opportunities," in *Proc. Int. Conf. High Perform. Comput. Simulat.*, 2009, pp. 1–11.
- [167] A. Medina, A. Lakhina, I. Matta, and J. Byers, "BRITE: An approach to universal topology generation," in *Proc. IEEE 9th Int. Symp. Model. Anal. Simulat. Comput. Telecommun. Syst. (MASCOTS)*, 2001, pp. 346–353.
- [168] H. Soroush and T. Ljiljana. "VNE-SIM." Accessed: Jul. 23, 2013. [Online]. Available: <https://www.sfu.ca/~ljilja/cnl/projects/VNE-Sim/vne-sim-web/>
- [169] S. Haeri and L. Trajković, "VNE-Sim: A virtual network embedding simulator," in *Proc. 9th EAI Int. Conf. Simulat. Tools Techn.*, 2016, pp. 112–117.
- [170] Z. Yan, J. Ge, Y. Wu, H. Zheng, L. Li, and T. Li, "Automatic virtual network embedding based on deep reinforcement learning," in *Proc. IEEE 21st Int. Conf. High Perform. Comput. Commun.*, 2019, pp. 625–631.
- [171] Z. Yan, J. Ge, Y. Wu, L. Li, and T. Li, "Automatic virtual network embedding: A deep reinforcement learning approach with graph convolutional networks," *IEEE J. Sel. Areas Commun.*, vol. 38, no. 6, pp. 1040–1057, Jun. 2020.
- [172] R. Matthias, E. Alexander, and D. Elias, "Alib." 2020. [Online]. Available: <https://github.com/vnep-approx/alib>
- [173] S. Knight, H. X. Nguyen, N. Falkner, R. Bowden, and M. Roughan, "The Internet Topology Zoo," *IEEE J. Sel. Areas Commun.*, vol. 29, no. 9, pp. 1765–1775, Oct. 2011.
- [174] S. Knight, N. Falkner, H. X. Nguyen, P. Tune, and M. Roughan, "I can see for miles: Re-visualizing the Internet," *IEEE Netw.*, vol. 26, no. 6, pp. 26–32, Nov./Dec. 2012.
- [175] M. Rost and S. Schmid, "Virtual network embedding approximations: Leveraging randomized rounding," *IEEE/ACM Trans. Netw.*, vol. 27, no. 5, pp. 2071–2084, Oct. 2019.
- [176] B. Németh, Y.-A. Pignolet, M. Rost, S. Schmid, and B. Vass, "Cost-efficient embedding of virtual networks with and without routing flexibility," in *Proc. IFIP Netw. Conf. (Netw.)*, 2020, pp. 476–484.
- [177] M. Park, Y. Lee, I. Yeom, and H. Woo, "GEMMA: Reinforcement learning-based graph embedding and mapping for virtual network applications," *IEEE Access*, vol. 9, pp. 105463–105476, 2021.
- [178] C. Wang et al., "Modeling on virtual network embedding using reinforcement learning," *Concurrency Comput. Pract. Exp.*, vol. 32, no. 23, 2020, Art. no. e6020.
- [179] A. Paszke et al., "Pytorch: An imperative style, high-performance deep learning library," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019, p. 12.

- [180] Q. Hu, Y. Wang, and X. Cao, "Resolve the virtual network embedding problem: A column generation approach," in *Proc. IEEE INFOCOM*, 2013, pp. 410–414.
- [181] Y. Wang, Q. Hu, and X. Cao, "A branch-and-price framework for optimal virtual network embedding," *Comput. Netw.*, vol. 94, pp. 318–326, Jan. 2016.
- [182] N. Kokash, *An Introduction to Heuristic Algorithms*, Dept. Inform. Telecommun., Univ. Trento, Trento, Italy, 2005.
- [183] R. Anand, D. Aggarwal, and V. Kumar, "A comparative analysis of optimization solvers," *J. Stat. Manag. Syst.*, vol. 20, no. 4, pp. 623–635, 2017.
- [184] J. L. Gearhart, K. L. Adair, J. D. Durfee, K. A. Jones, N. Martin, and R. J. Detry, "Comparison of open-source linear programming solvers," Sandia Nat. Lab., Albuquerque, NM, USA, Rep. SAND2013-8847, 2013.
- [185] M. Melo, S. Sargento, U. Killat, A. Timm-Giel, and J. Carapinha, "Optimal virtual network embedding: Node-link formulation," *IEEE Trans. Netw. Service Manag.*, vol. 10, no. 4, pp. 356–368, Dec. 2013.
- [186] Q. Hu, Y. Wang, and X. Cao, "Virtual network embedding: An optimal decomposition approach," in *Proc. 23rd Int. Conf. Comput. Commun. Netw. (ICCCN)*, 2014, pp. 1–6.
- [187] R. Mijumbi, J. Serrat, J.-L. Gorricho, and R. Boutaba, "A path generation approach to embedding of virtual networks," *IEEE Trans. Netw. Service Manag.*, vol. 12, no. 3, pp. 334–348, Sep. 2015.
- [188] M. Zhang and X. Tang, "Hop-limit path mapping algorithm for virtual network embedding," *Wireless Pers. Commun.*, vol. 95, pp. 2033–2048, Jan. 2017.
- [189] V. Campos, M. Laguna, R. Martí, "Context-independent scatter and tabu search for permutation problems," *INFORMS J. Comput.*, vol. 17, no. 1, pp. 111–122.
- [190] F. Glover, E. Taillard, and E. Taillard, "A user's guide to Tabu search," *Ann. Oper. Res.*, vol. 41, no. 1, pp. 1–28, 1993.
- [191] A. Hertz, E. Taillard, and D. de Werra, "A tutorial on tabu search," in *Proc. Giornate di Lavoro AIRO*, vol. 95, 1995, pp. 121–136.
- [192] M. Diallo, A. Quintero, and S. Pierre, "A tabu search approach for a virtual networks splitting strategy across multiple cloud providers," *Int. J. Metaheuristics*, vol. 7, no. 3, pp. 197–238, 2020.
- [193] S. Ayoubi, C. Assi, K. Shaban, and L. Narayanan, "MINTED: Multicast virtual network embedding in cloud data centers with delay constraints," *IEEE Trans. Commun.*, vol. 63, no. 4, pp. 1291–1305, Apr. 2015.
- [194] D. E. Goldberg and J. H. Holland, "Genetic algorithms and machine learning," *Mach. Learn.*, vol. 3, no. 2, pp. 95–99, 1988. doi: [10.1023/A:1022602019183](https://doi.org/10.1023/A:1022602019183).
- [195] I. Pathak and D. P. Vidyarthi, "A model for virtual network embedding across multiple infrastructure providers using genetic algorithm," *Sci. China Inf. Sci.*, vol. 60, no. 4, 2017, Art. no. 40308.
- [196] J. Rubio-Loyola and C. Aguilar-Fuster, "Novel initialization functions for metaheuristic-based online virtual network embedding," *J. Netw. Syst. Manag.*, vol. 32, no. 3, p. 46, 2024.
- [197] M. C. Çavdar, I. Korpoglu, and Ö. Ulusoy, "A utilization based genetic algorithm for virtual machine placement in cloud systems," *Comput. Commun.*, vol. 214, pp. 136–148, Jan. 2024.
- [198] Z. Ding, Y.-C. Tian, Y.-G. Wang, W.-Z. Zhang, and Z.-G. Yu, "Accelerated computation of the genetic algorithm for energy-efficient virtual machine placement in data centers," *Neural Comput. Appl.*, vol. 35, no. 7, pp. 5421–5436, 2023.
- [199] V. Barthwal, M. Rauthan, R. Varma, and S. Gaur, "Efficient virtual machine placement strategy based on enhanced genetic approach," *SN Comput. Sci.*, vol. 5, no. 5, p. 466, 2024.
- [200] H. Gu et al., "Towards virtual machine scheduling research based on multi-decision AHP method in the cloud computing platform," *PeerJ Comput. Sci.*, vol. 9, no. 2, 2023, Art. no. e1675.
- [201] A. K. Singh, S. R. Swain, D. Saxena, and C.-N. Lee, "A bio-inspired virtual machine placement toward sustainable cloud resource management," *IEEE Syst. J.*, vol. 17, no. 3, pp. 3894–3905, Sep. 2023.
- [202] P. Zhang, H. Yao, M. Li, and Y. Liu, "Virtual network embedding based on modified genetic algorithm," *Peer-to-Peer Netw. Appl.*, vol. 12, no. 2, pp. 481–492, 2019.
- [203] P. Zhang, X. Pang, G. Kibalya, N. Kumar, S. He, and B. Zhao, "GCMD: Genetic correlation multi-domain virtual network embedding algorithm," *IEEE Access*, vol. 9, pp. 67167–67175, 2021.
- [204] P. Martinez-Julia, V. P. Kaffle, and H. Asaeda, "A genetic approach to continuous optimization of virtual network embedding," in *Proc. 24th Conf. Innov. Clouds Internet Netw. Workshops (ICIN)*, 2021, pp. 70–74.
- [205] Q. Lu and C. Huang, "Performance modeling and joint resource allocation algorithms for online virtual network embedding," *IEEE Trans. Netw. Service Manag.*, vol. 21, no. 1, pp. 1048–1066, Feb. 2024.
- [206] P. Martinez-Julia, V. P. Kaffle, and H. Asaeda, "Achieving optimized embedding of virtual networks by incremental evolution," in *Proc. 25th Conf. Innov. Clouds Internet Netw. (ICIN)*, 2022, pp. 121–125.
- [207] Q. Lu, K. Nguyen, and C. Huang, "Distributed parallel algorithms for online virtual network embedding applications," *Int. J. Commun. Syst.*, vol. 36, no. 1, 2020, Art. no. e4325.
- [208] K. Nguyen, Q. Lu, and C. Huang, "Efficient virtual network embedding with node ranking and intelligent link mapping," in *Proc. IEEE 9th Int. Conf. Cloud Netw. (CloudNet)*, 2020, pp. 1–5.
- [209] K. Nguyen and C. Huang, "Toward adaptive joint node and link mapping algorithms for embedding virtual networks: A conciliation strategy," *IEEE Trans. Netw. Service Manag.*, vol. 19, no. 3, pp. 3323–3340, Sep. 2022.
- [210] X. Chang, X. Mi, and J. Muppala, "Performance evaluation of artificial intelligence algorithms for virtual network embedding," *Eng. Appl. Artif. Intell.*, vol. 26, no. 10, pp. 2540–2550, 2013.
- [211] Z. Zhou, X. Chang, Y. Yang, and L. Li, "Resource-aware virtual network parallel embedding based on genetic algorithm," in *Proc. 17th Int. Conf. Parallel Distrib. Comput. Appl. Technol. (PDCAT)*, 2016, pp. 81–86.
- [212] H. Afifi, K. Horbach, and H. Karl, "A genetic algorithm framework for solving wireless virtual network embedding," in *Proc. Int. Conf. Wireless Mobile Comput. Netw. Commun. (WiMob)*, 2019, pp. 1–6.
- [213] L. Shoosharian and F. Safaei, "A maximally robustness embedding algorithm in virtual data centers with multi-attribute node ranking based on TOPSIS," *J. Supercomput.*, vol. 75, pp. 8059–8093, Aug. 2019.
- [214] E. Guler, M. Karakus, and F. Ayaz, "Genetic algorithm enabled virtual multicast tree embedding in software-defined networks," *J. Netw. Comput. Appl.*, vol. 209, Jan. 2023, Art.no. 103538.
- [215] R. G. M. Báez, D. L. Correa, J. Colbes, and D. P. Pinto-Roa, "Virtual multicast tree embedding and protection over elastic optical networks based on genetic algorithms," in *Proc. XLIX Latin Amer. Comput. Conf. (CLEI)*, 2023, pp. 1–10.
- [216] G. Sun, D. Liao, S. Bu, H. Yu, Z. Sun, and V. Chang, "The efficient framework and algorithm for provisioning evolving VDC in federated data centers," *Future Gener. Comput. Syst.*, vol. 73, pp. 79–89, Aug. 2017.
- [217] I. Fajjari, N. Aitsaadi, G. Pujolle, and H. Zimmermann, "VNR algorithm: A greedy approach for virtual networks reconfigurations," in *Proc. IEEE Global Telecommun. Conf. (GLOBECOM)*, 2011, pp. 1–6.
- [218] H. Cao, S. Wu, Q. Wang, and L. Yang, "Dynamic mapping and quality of service driven re-embedding in virtualization environment," in *Proc. IEEE Global Commun. Conf. (GLOBECOM)*, 2019, pp. 1–6.
- [219] J. Duan, Z. Guo, and Y. Yang, "Cost efficient and performance guaranteed virtual network embedding in multicast fat-tree DCNs," in *Proc. IEEE Conf. Comput. Commun. (INFOCOM)*, 2015, pp. 136–144.
- [220] M. Gilesh, S. M. Kumar, L. Jacob, and U. Bellur, "Towards a complete virtual data Center embedding algorithm using hybrid strategy," in *Proc. IEEE 37th Int. Conf. Distrib. Comput. Syst. (ICDCS)*, 2017, pp. 2616–2617.
- [221] A. Hashmi and C. Gupta, "VNE-NR: A node-ranking method for performing topology-aware and resource-driven virtual network embedding," in *Proc. 11th Int. Conf. Comput. Commun. Netw. Technol. (ICCCNT)*, 2020, pp. 1–6.
- [222] J. Ding, T. Huang, J. Liu, and Y.-J. Liu, "Virtual network embedding based on real-time topological attributes," *Front. Inf. Technol. Electron. Eng.*, vol. 16, no. 2, pp. 109–118, 2015.
- [223] E. Guler, G. Luo, K. Koneru, and X. Cao, "Closeness-centrality based multicast-aware virtual network embedding," in *Proc. IEEE Global Commun. Conf. (GLOBECOM)*, 2016, pp. 1–6.
- [224] M. Feng, J. Liao, J. Wang, S. Qing, and Q. Qi, "Topology-aware virtual network embedding based on multiple characteristics," in *Proc. IEEE Int. Conf. Commun. (ICC)*, 2014, pp. 2956–2962.
- [225] P. Zhang, H. Yao, C. Qiu, and Y. Liu, "Virtual network embedding using node multiple metrics based on simplified ELECTRE method," *IEEE Access*, vol. 6, pp. 37314–37327, 2018.
- [226] M. Lu, Y. Lian, Y. Chen, and M. Li, "Collaborative dynamic virtual network embedding algorithm based on resource importance measures," *IEEE Access*, vol. 6, pp. 55026–55042, 2018.
- [227] H. Wu, F. Zhou, Y. Chen, and R. Zhang, "On virtual network embedding: Paths and cycles," *IEEE Trans. Netw. Service Manag.*, vol. 17, no. 3, pp. 1487–1500, Oct. 2020.

- [228] X. Xiao, X. Zheng, J. Bian, C. Ji, and X. Cui, "A dynamic and resource sharing virtual network mapping algorithm," *Digit. Commun. Netw.*, vol. 9, no. 5, pp. 1101–1112, 2023.
- [229] A. Amokrane, M. F. Zhani, R. Langar, R. Boutaba, and G. Pujolle, "Greenhead: Virtual data center embedding across distributed infrastructures," *IEEE Trans. Cloud Comput.*, vol. 1, no. 1, pp. 36–49, Jan.–Jun. 2013.
- [230] L. Gong, Y. Wen, Z. Zhu, and T. Lee, "Toward profit-seeking virtual network embedding algorithm via global resource capacity," in *Proc. IEEE Conf. Comput. Commun. (IEEE INFOCOM)*, 2014, pp. 1–9.
- [231] D. Liao, G. Sun, V. Anand, and H. Yu, "Survivable provisioning for multicast service oriented virtual network requests in cloud-based data centers," *Opt. Switch. Netw.*, vol. 14, pp. 260–273, Aug. 2014.
- [232] X. Chen and C. Li, "Energy efficient virtual network embedding for path splitting," in *Proc. 16th Asia-Pac. Netw. Oper. Manag. Symp.*, 2014, pp. 1–4.
- [233] N. Triki, N. Kara, M. E. Barachi, and S. Hadjres, "A green energy-aware hybrid virtual network embedding approach," *Comput. Netw.*, vol. 91, pp. 712–737, Nov. 2015.
- [234] Z. Zhang, S. Su, J. Zhang, K. Shuang, and P. Xu, "Energy aware virtual network embedding with dynamic demands: Online and offline," *Comput. Netw.*, vol. 93, pp. 448–459, Dec. 2015.
- [235] S. Shanbhag, A. Reddy Kandoor, C. Wang, R. Mettu, and T. Wolf, "VHub: Single-stage virtual network mapping through hub location," *Comput. Netw.*, vol. 77, pp. 169–180, Feb. 2015.
- [236] E. Ghazisaeei and C. Huang, "Energy-aware node and link reconfiguration for virtualized network environments," *Comput. Netw.*, vol. 93, pp. 460–479, Dec. 2015.
- [237] Y. Jiang, J. Lan, Z. Wang, and Y. Deng, "Embedding and reconfiguration algorithms for service aggregation in network virtualization," *Int. J. Commun. Syst.*, vol. 29, no. 1, pp. 33–46, 2016.
- [238] S. R. Chowdhury et al., "Dedicated protection for survivable virtual network embedding," *IEEE Trans. Netw. Service Manag.*, vol. 13, no. 4, pp. 913–926, Dec. 2016.
- [239] T. M. Nam, N. H. Thanh, H. T. Hieu, N. T. Manh, N. V. Huynh, and H. D. Tuan, "Joint network embedding and server consolidation for energy-efficient dynamic data center virtualization," *Comput. Netw.*, vol. 125, pp. 76–89, Oct. 2017.
- [240] X. Liu, X. Wang, D. Yuan, and M. Chen, "A scheme of survival virtual network embedding based on robust topology design," in *Proc. IEEE 9th Int. Conf. Commun. Softw. Netw. (ICCSN)*, 2017, pp. 993–998.
- [241] M. H. Dahir, H. Alizadeh, and D. Gözüpek, "Energy efficient virtual network embedding in federated software defined networks," in *Proc. 25th Signal Process. Commun. Appl. Conf. (SIU)*, 2017, pp. 1–4.
- [242] P. Zhang, S. Wu, M. Wang, H. Yao, and Y. Liu, "Topology based reliable virtual network embedding from a QoE perspective," *China Commun.*, vol. 15, no. 10, pp. 38–50, 2018.
- [243] L. Gao and G. N. Rouskas, "Virtual network reconfiguration with load balancing and migration cost considerations," in *Proc. IEEE Conf. Comput. Commun. (IEEE INFOCOM)*, 2018, pp. 2303–2311.
- [244] H. Cao, Y. Guo, S. Wu, H. Zhu, and L. Yang, "An energy efficient embedding algorithm for multi-domain virtual network embedding," in *Proc. IEEE Globecom Workshops (GC Wkshps)*, 2018, pp. 1–5.
- [245] X. Zheng, J. Tian, X. Xiao, X. Cui, and X. Yu, "A heuristic survivable virtual network mapping algorithm," *Soft Comput.*, vol. 23, pp. 1453–1463, Mar. 2019.
- [246] S. Liu and S. Liang, "A sleeping substrate network approach for wireless virtual network embedding," in *Proc. IEEE 5th Int. Conf. Comput. Commun. (ICCC)*, 2019, pp. 1203–1207.
- [247] X. Cong, K. Shuang, and L. Zi, "Energy optimized virtual network embedding with location constraint in the enterprise network," *IEEE Access*, vol. 8, pp. 56170–56180, 2020.
- [248] N. Shahriar et al., "Virtual network embedding with guaranteed connectivity under multiple substrate link failures," *IEEE Trans. Commun.*, vol. 68, no. 2, pp. 1025–1043, Feb. 2020.
- [249] L. Li, W. Wei, W. Ren, K. Wang, and S. Wang, "Multi-service virtual network embedding in wireless network," in *Proc. 31st Int. Telecommun. Netw. Appl. Conf. (ITNAC)*, 2021, pp. 107–112.
- [250] S. Gong, J. Chen, X. Yin, and Q. Zhu, "Energy-efficient virtual network embedding for heterogeneous networks," in *Proc. 1st IEEE Int. Conf. Comput. Commun. Internet (ICCCI)*, 2016, pp. 85–90.
- [251] G. Sun, H. Yu, L. Li, V. Anand, and H. Di, "The framework and algorithms for the survivable mapping of virtual network onto a substrate network," *IETE Tech. Rev.*, vol. 28, no. 5, pp. 381–391, 2011.
- [252] N. Farooq Butt, M. Chowdhury, and R. Boutaba, "Topology-awareness and reoptimization mechanism for virtual network embedding," in *Proc. NETWORKING*, 2010, pp. 27–39.
- [253] C. Swain et al., "METO: Matching-theory-based efficient task offloading in IoT-fog interconnection networks," *IEEE Internet Things J.*, vol. 8, no. 16, pp. 12705–12715, Aug. 2020.
- [254] Y.-J. Lai, T.-Y. Liu, and C.-L. Hwang, "Topsis for MODM," *Eur. J. Oper. Res.*, vol. 76, no. 3, pp. 486–500, 1994.
- [255] M. Behzadian, S. K. Otaghara, M. Yazdani, and J. Ignatius, "A state-of-the-art survey of TOPSIS applications," *Exp. Syst. Appl.*, vol. 39, no. 17, pp. 13051–13069, 2012.
- [256] I. Fajjari, N. Aitsaadi, B. Dab, and G. Pujolle, "Novel adaptive virtual network embedding algorithm for cloud's private backbone network," *Comput. Commun.*, vol. 84, pp. 12–24, Jun. 2016.
- [257] K. Metwally, A. Jarray, and A. Karmouch, "A distributed auction-based framework for scalable IaaS provisioning in geo-data centers," *IEEE Trans. Cloud Comput.*, vol. 8, no. 3, pp. 647–659, Jul.–Sep. 2020.
- [258] F. Esposito, D. Di Paola, and I. Matta, "On distributed virtual network embedding with guarantees," *IEEE/ACM Trans. Netw.*, vol. 24, no. 1, pp. 569–582, Feb. 2016.
- [259] M. Dorigo, M. Birattari, and T. Stutzle, "Ant colony optimization," *IEEE Comput. Intell. Mag.*, vol. 1, no. 4, pp. 28–39, Nov. 2006.
- [260] Z. Zhang, X. Cheng, S. Su, Y. Wang, K. Shuang, and Y. Luo, "A unified enhanced particle swarm optimization-based virtual network embedding algorithm," *Int. J. Commun. Syst.*, vol. 26, no. 8, pp. 1054–1073, 2013.
- [261] A. K. Sangaiah, A. Javadpour, P. Pinto, S. Rezaei, and W. Zhang, "Enhanced resource allocation in distributed cloud using fuzzy meta-heuristics optimization," *Comput. Commun.*, vol. 209, pp. 14–25, Sep. 2023.
- [262] I. Fajjari, N. Aitsaadi, M. Pióro, and G. Pujolle, "A new virtual network static embedding strategy within the cloud's private backbone network," *Comput. Netw.*, vol. 62, pp. 69–88, Apr. 2014.
- [263] F. Zhu and H. Wang, "A modified ACO algorithm for virtual network embedding based on graph decomposition," *Comput. Commun.*, vol. 80, pp. 1–15, Apr. 2016.
- [264] Z. Qiang, W. Qiang, F. Sheng, and L. Wu, "Heuristic survivable virtual network embedding based on node migration and link remapping," in *Proc. IEEE 7th Joint Int. Inf. Technol. Artif. Intell. Conf.*, 2014, pp. 181–185.
- [265] H.-K. Zheng et al., "Link mapping-oriented ant colony system for virtual network embedding," in *Proc. IEEE Congr. Evol. Comput. (CEC)*, 2017, pp. 1223–1230.
- [266] O. B. Bayram and A. Ozcan, "Determining optimal paths of virtual links in avionics full-duplex switched ethernet networks using modified ant colony optimization algorithm," *Exp. Syst. Appl.*, vol. 229, Nov. 2023, Art. no. 120433.
- [267] A. Ikoma, Y. Ohsita, and M. Murata, "Resource allocation considering impact of network on performance in a disaggregated data center," *IEEE Access*, vol. 12, pp. 67600–67618, 2024.
- [268] K. Guo, Y. Wang, X. Qiu, W. Li, and A. Xiao, "Particle swarm optimization based multi-domain virtual network embedding," in *Proc. IFIP/IEEE Int. Symp. Integr. Netw. Manag. (IM)*, 2015, pp. 798–801.
- [269] P. Zhang, H. Yao, C. Fang, and Y. Liu, "Multi-objective enhanced particle swarm optimization in virtual network embedding," *EURASIP J. Wireless Commun. Netw.*, vol. 2016, no. 1, pp. 1–9, 2016.
- [270] X. Wang, Q. Chen, and H. Qiu, "A effective two-step strategy of multi-domain virtual network embedding in 5G network slicing," in *Proc. 3rd IEEE Int. Conf. Comput. Commun. (ICCC)*, 2017, pp. 1174–1179.
- [271] A. Song, W.-N. Chen, Y.-J. Gong, X. Luo, and J. Zhang, "A divide-and-conquer evolutionary algorithm for large-scale virtual network embedding," *IEEE Trans. Evol. Comput.*, vol. 24, no. 3, pp. 566–580, Jun. 2019.
- [272] L. Boyang, W. Muqing, and Z. Haosen, "Virtual network embedding based on hybrid adaptive genetic algorithm," in *Proc. IEEE 5th Int. Conf. Comput. Commun. (ICCC)*, 2019, pp. 1197–1202.
- [273] A. Song, W.-N. Chen, T. Gu, H. Zhang, and J. Zhang, "A constructive particle swarm optimizer for virtual network embedding," *IEEE Trans. Netw. Sci. Eng.*, vol. 7, no. 3, pp. 1406–1420, Jul.–Sep. 2019.
- [274] A. Song, W.-N. Chen, T. Gu, H. Yuan, S. Kwong, and J. Zhang, "Distributed virtual network embedding system with historical archives and set-based particle swarm optimization," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 51, no. 2, pp. 927–942, Feb. 2021.
- [275] R. Chai, D. Xie, L. Luo, and Q. Chen, "Multi-objective optimization-based virtual network embedding algorithm for software-defined networking," *IEEE Trans. Netw. Service Manag.*, vol. 17, no. 1, pp. 532–546, Mar. 2020.
- [276] Y. Chiang, Y.-H. Chao, C.-H. Hsu, C.-T. Chou, and H.-Y. Wei, "Virtual network embedding with dynamic speed switching orchestration in fog/edge network," *IEEE Access*, vol. 8, pp. 84753–84768, 2020.

- [277] E. Guler, F. Ayaz, M. Karakus, and D. Hanbay, "Particle swarm optimization-enhanced virtual multicast trees embedding in SDNs," in *Proc. IEEE Int. Black Sea Conf. Commun. Netw. (BlackSeaCom)*, 2023, pp. 265–270.
- [278] J. Kennedy and R. Eberhart, "Particle swarm optimization," in *Proc. Int. Conf. Neural Netw. (ICNN)*, vol. 4, 1995, pp. 1942–1948.
- [279] D. Wang, D. Tan, and L. Liu, "Particle swarm optimization algorithm: An overview," *Soft Comput.*, vol. 22, pp. 387–408, Jun. 2018.
- [280] M. Clerc, *Particle Swarm Optimization*, vol. 93. Hoboken, NJ, USA: Wiley, 2010.
- [281] I. Pathak and D. P. Vidyarthi, "An interactive artificial bee colony based virtual network embedding," in *Proc. IEEE UP Sect. Conf. Elect. Comput. Electron. (UPCON)*, 2015, pp. 1–6.
- [282] X. Liu, Z. Zhang, X. Li, and S. Su, "Optimal virtual network embedding based on artificial bee colony," *EURASIP J. Wireless Commun. Netw.*, vol. 2016, no. 1, pp. 1–9, 2016.
- [283] J. Inführ and G. Raidl, "A memetic algorithm for the virtual network mapping problem," *J. Heuristics*, vol. 22, pp. 475–505, Aug. 2016.
- [284] I. Pathak, A. Tripathi, and D. P. Vidyarthi, "A model for virtual network embedding using artificial bee colony," *Int. J. Commun. Syst.*, vol. 31, no. 10, 2018, Art. no. e3573.
- [285] C. Yu, Q. Lian, D. Zhang, and C. Wu, "PAME: Evolutionary membrane computing for virtual network embedding," *J. Parallel Distrib. Comput.*, vol. 111, pp. 136–151, Jan. 2018.
- [286] W. Yi, W. Muqing, and H. Xiaolan, "An effective strategy of centralized multi-domain virtual network embedding," in *Proc. IEEE 5th Int. Conf. Comput. Commun. (ICCC)*, 2019, pp. 1186–1191.
- [287] P. Zhang, Y. Hong, X. Pang, and C. Jiang, "VNE-HPSO: Virtual network embedding algorithm based on hybrid particle swarm optimization," *IEEE Access*, vol. 8, pp. 213389–213400, 2020.
- [288] N. Balamurugan, J. Raja, and R. Pitchai, "Multicriteria dragonfly graph theory based resource optimized virtual network mapping technique for home medical care service provisioning in cloud," *Peer-to-Peer Netw. Appl.*, vol. 13, no. 1, pp. 1872–1885, 2020.
- [289] Z. Li, F. Yuan, and L. Ma, "A load balancing algorithm for solving multi-objective virtual network embedding," *Trans. Emerg. Telecommun. Technol.*, vol. 33, no. 6, 2022, Art. no. e4066.
- [290] Z. Xu et al., "An edge-based approach for virtual network embedding based on the graph edit distance," *PLoS ONE*, vol. 18, no. 7, 2023, Art. no. e0288037.
- [291] N. Balamurugan, J. Raja, and R. Pitchai, "Graph Modeled oppositional whale optimization for efficient cloud service provision using virtual network mapping," *Wireless Pers. Commun.*, vol. 135, no. 2, pp. 675–695, 2024.
- [292] L. Lahlou, C. Tata, N. Kara, A. Leivadeas, and A. Gherbi, "Edge-cloud online joint placement of virtual network functions and allocation of compute and network resources using meta-heuristics," *J. Ambient Intell. Humanized Comput.*, vol. 14, no. 6, pp. 7531–7558, 2023.
- [293] R. Gomes, D. Vieira, M. B. Pereira, and M. F. de Castro, "Artificial algae optimization for virtual network embedding problems in 5G network slicing scenarios," *Exp. Syst. Appl.*, vol. 239, Apr. 2024, Art. no. 122436.
- [294] A. Bouchair, B. Yagoubi, and S. A. Makhlof, "HS-GWO: A hybrid approach for virtual network embedding in SDN-enabled distributed cloud," in *Proc. Future Inf. Commun. Conf.*, 2022, pp. 594–610.
- [295] H. Yao, X. Chen, M. Li, P. Zhang, and L. Wang, "A novel reinforcement learning algorithm for virtual network embedding," *Neurocomputing*, vol. 284, pp. 1–9, Apr. 2018.
- [296] G. Sun, D. Liao, D. Zhao, Z. Sun, and V. Chang, "Towards provisioning hybrid virtual networks in federated cloud data centers," *Future Gener. Comput. Syst.*, vol. 87, pp. 457–469, Oct. 2018.
- [297] A. Jahani, L. M. Khanli, M. T. Hagh, and M. A. Badamchizadeh, "Green virtual network embedding with supervised self-organizing map," *Neurocomputing*, vol. 351, pp. 60–76, Jul. 2019.
- [298] Z. Yan, N. Wei, Q. Jin, and X. Zhou, "Latency-aware resource-efficient virtual network embedding in software defined networking," in *Proc. 28th Wireless Opt. Commun. Conf. (WOCC)*, 2019, pp. 1–5.
- [299] X. Zheng, Y. Zhang, H. Zhang, and Q. Xue, "An RBF neural network-based dynamic virtual network embedding algorithm," *Concurrency Comput. Pract. Exp.*, vol. 31, no. 23, 2019, Art. no. e4516.
- [300] D. Andreoletti, T. Velichkova, G. Verticale, M. Tornatore, and S. Giordano, "A privacy-preserving reinforcement learning algorithm for multi-domain virtual network embedding," *IEEE Trans. Netw. Service Manag.*, vol. 17, no. 4, pp. 2291–2304, Dec. 2020.
- [301] H. K. Thakkar, C. K. Dehury, and P. K. Sahoo, "MUVINE: Multi-stage virtual network embedding in cloud data centers using reinforcement learning-based predictions," *IEEE J. Sel. Areas Commun.*, vol. 38, no. 6, pp. 1058–1074, Jun. 2020.
- [302] M. Lu, Y. Gu, and D. Xie, "A dynamic and collaborative multi-layer virtual network embedding algorithm in SDN based on reinforcement learning," *IEEE Trans. Netw. Service Manag.*, vol. 17, no. 4, pp. 2305–2317, Dec. 2020.
- [303] H. Yao, S. Ma, J. Wang, P. Zhang, C. Jiang, and S. Guo, "A continuous-decision virtual network embedding scheme relying on reinforcement learning," *IEEE Trans. Netw. Service Manag.*, vol. 17, no. 2, pp. 864–875, Jun. 2020.
- [304] H. Yao, B. Zhang, P. Zhang, S. Wu, C. Jiang, and S. Guo, "RDAM: A reinforcement learning based dynamic attribute matrix representation for virtual network embedding," *IEEE Trans. Emerg. Topics Comput.*, vol. 9, no. 2, pp. 901–914, Apr.–Jun. 2021.
- [305] P. Zhang, C. Wang, C. Jiang, and A. Benslimane, "Security-aware virtual network embedding algorithm based on reinforcement learning," *IEEE Trans. Netw. Sci. Eng.*, vol. 8, no. 2, pp. 1095–1105, Apr.–Jun. 2021.
- [306] J. Cheng, Y. Wu, Y. Lin, Y. E, F. Tang, and J. Ge, "VNE-HRL: A proactive virtual network embedding algorithm based on hierarchical reinforcement learning," *IEEE Trans. Netw. Service Manag.*, vol. 18, no. 4, pp. 4075–4087, Dec. 2021.
- [307] M. Elkael, M. A. Aba, A. Araldo, H. Castel-Taleb, and B. Jouaber, "Monkey business: Reinforcement learning meets neighborhood search for virtual network embedding," *Comput. Netw.*, vol. 216, Oct. 2022, Art. no. 109204.
- [308] M. He, L. Zhuang, S. Tian, G. Wang, and K. Zhang, "DROI: Energy-efficient virtual network embedding algorithm based on dynamic regions of interest," *Comput. Netw.*, vol. 166, Jan. 2020, Art. no. 106952.
- [309] G. Li, C. Xi, and R. Zhu, "Multi-agent deep reinforced virtual network embedding in elastic optical networks," in *Proc. 20th Int. Conf. Opt. Commun. Netw. (ICOON)*, 2022, pp. 1–3.
- [310] G. Zheng et al., "A single-player Monte Carlo tree search method combined with node importance for virtual network embedding," *Ann. Telecommun.*, vol. 76, pp. 297–312, Jun. 2021.
- [311] Q. Gao, N. Lyu, J. Miao, and W. Pan, "Wireless virtual network embedding algorithm based on deep reinforcement learning," *Electronics*, vol. 11, no. 14, p. 2243, 2022.
- [312] S. Ma et al., "Graph convolutional network aided virtual network embedding for Internet of Things," *IEEE Trans. Netw. Sci. Eng.*, vol. 10, no. 1, pp. 265–274, Jan./Feb. 2023.
- [313] H.-K. Lim, I. Ullah, J.-B. Kim, and Y.-H. Han, "Virtual network embedding based on hierarchical cooperative Multiagent reinforcement learning," *IEEE Internet Things J.*, vol. 11, no. 5, pp. 8552–8568, Mar. 2024.
- [314] M. Mohammadi, A. Al-Fuqaha, S. Sorour, and M. Guizani, "Deep learning for IoT big data and streaming analytics: A survey," *IEEE Commun. Surveys Tuts.*, vol. 20, no. 4, pp. 2923–2960, 4th Quart., 2018.
- [315] M. Dolati, S. B. Hassanpour, M. Ghaderi, and A. Khonsari, "DeepViNE: Virtual network embedding with deep reinforcement learning," in *Proc. IEEE INFOCOM Conf. Comput. Commun. Workshops (INFOCOM WKSHPS)*, 2019, pp. 879–885.
- [316] Y. Yuan, Z. Tian, C. Wang, F. Zheng, and Y. Lv, "A Q-learning-based approach for virtual network embedding in data center," *Neural Comput. Appl.*, vol. 32, no. 7, pp. 1995–2004, 2020.
- [317] P. Zhang, C. Wang, N. Kumar, and L. Liu, "Space-air-ground integrated multi-domain network resource orchestration based on virtual network architecture: A DRL method," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 3, pp. 2798–2808, Mar. 2022.
- [318] Z. Duan and T. Wang, "Towards learning-based energy-efficient online coordinated virtual network embedding framework," *Comput. Netw.*, vol. 239, Feb. 2024, Art. no. 110139.
- [319] Y. Zhang, Y. Wang, M. Zhang, and B. Lu, "VDC embedding scheme based on virtual nodes combination in software defined data center," in *Proc. IEEE Adv. Inf. Manag. Commun. Electron. Autom. Control Conf. (IMCEC)*, 2016, pp. 931–935.
- [320] D. Yang, J. Liu, R. Zhang, and T. Huang, "Multi-constraint virtual network embedding algorithm for satellite networks," in *Proc. IEEE Global Commun. Conf. (GLOBECOM)*, 2020, pp. 1–6.
- [321] W. Fan et al., "Dynamic virtual network embedding of mobile cloud system based on global resources in Internet of Vehicles," *IEEE Trans. Veh. Technol.*, vol. 70, no. 8, pp. 8161–8174, Aug. 2021.

- [322] Y. Yuan, Y. Yang, and C. Wang, "Virtual network embedding with virtual nodes ranking and multi points sampling," in *Proc. 10th Int. Conf. Adv. Cloud Big Data (CBD)*, 2022, pp. 48–53.
- [323] W. Fan, F. Xiao, M. Lv, L. Han, J. Wang, and X. He, "Node essentiality assessment and distributed collaborative virtual network embedding in datacenters," *IEEE Trans. Parallel Distrib. Syst.*, vol. 34, no. 4, pp. 1265–1280, Apr. 2023.
- [324] M. Pilkington, "Blockchain technology: Principles and applications," in *Research Handbook on Digital Transformations*. London, U.K.: Edward Elgar, 2016, pp. 225–253.
- [325] J. Yli-Huumo, D. Ko, S. Choi, S. Park, and K. Smolander, "Where is current research on blockchain technology—A systematic review," *PLoS ONE*, vol. 11, no. 10, 2016, Art. no. e0163477.
- [326] L. Nonde, T. E. Elgorashi, and J. M. Elmigahni, "Virtual network embedding employing renewable energy sources," in *Proc. IEEE Global Commun. Conf. (GLOBECOM)*, 2016, pp. 1–6.
- [327] W. Wu, W. Wang, X. Fang, J. Luo, and A. V. Vasilakos, "Electricity price-aware consolidation algorithms for time-sensitive VM services in cloud systems," *IEEE Trans. Services Comput.*, vol. 14, no. 6, pp. 1726–1738, Nov./Dec. 2021.
- [328] M. Lin, A. Wierman, L. L. Andrew, and E. Thereska, "Dynamic right-sizing for power-proportional data centers," *IEEE/ACM Trans. Netw.*, vol. 21, no. 5, pp. 1378–1391, Jun. 2013.
- [329] P. G. J. Leelipushpam and J. Sharmila, "Live VM migration techniques in cloud environment—A survey," in *Proc. IEEE Conf. Inf. Commun. Technol.*, 2013, pp. 408–413.
- [330] M. F. Bari, M. F. Zhani, Q. Zhang, R. Ahmed, and R. Boutaba, "CQNCr: Optimal VM migration planning in cloud data centers," in *Proc. IFIP Netw. Conf.*, 2014, pp. 1–9.
- [331] G. Smith, A. Chaturvedi, A. Mishra, and S. Banerjee, "Wireless virtualization on commodity 802.11 hardware," in *Proc. 2nd ACM Int. Workshop Wireless Netw. Testbeds Exp. Eval. Characterization*, 2007, pp. 75–82.
- [332] S. Perez, J. M. Cabero, and E. Miguel, "Virtualization of the wireless medium: A simulation-based study," in *Proc. VTC Spring IEEE 69th Veh. Technol. Conf.*, 2009, pp. 1–5.
- [333] R. Katona, V. Cionca, D. O'Shea, and D. Pesch, "Virtual network embedding for wireless sensor networks time-efficient QoS/QoI-aware approach," *IEEE Internet Things J.*, vol. 8, no. 2, pp. 916–926, May 2021.
- [334] P. Zhang, X. Pang, Y. Bi, H. Yao, H. Pan, and N. Kumar, "DSCD: Delay sensitive cross-domain virtual network embedding algorithm," *IEEE Trans. Netw. Sci. Eng.*, vol. 7, no. 4, pp. 2913–2925, Oct.–Dec. 2020.
- [335] D. Li and J. Wu, "On data center network architectures for inter-connecting dual-port servers," *IEEE Trans. Comput.*, vol. 64, no. 11, pp. 3210–3222, Nov. 2015.
- [336] N. Shahriar et al., "Achieving a fully-flexible virtual network embedding in elastic optical networks," in *Proc. IEEE Conf. Comput. Commun. (IEEE INFOCOM)*, 2019, pp. 1756–1764.
- [337] S. Madanantha, T. Truong-Huu, and M. Gurusamy, "Adaptive bandwidth allocation for virtual network embedding in optical data center networks," in *Proc. IEEE 41st Conf. Local Comput. Netw. (LCN)*, 2016, pp. 316–324.
- [338] A. Al-Fuqaha, M. Guizani, M. Mohammadi, M. Aledhari, and M. Ayyash, "Internet of Things: A survey on enabling technologies, protocols, and applications," *IEEE Commun. Surveys Tuts.*, vol. 17, no. 4, pp. 2347–2376, 3rd Quart., 2015.
- [339] Y. Guan, Q. Song, W. Qi, L. Guo, K. Li, and A. Jamalipour, "Multidimensional resource fragmentation-aware virtual network embedding for IoT applications in MEC networks," *IEEE Internet Things J.*, vol. 10, no. 24, pp. 22223–22232, Dec. 2023.
- [340] K. Nguyen, W. Shi, and M. St-Hilaire, "Cost-aware node ranking algorithm for embedding virtual networks in Internet of Vehicles," in *Proc. Int. Conf. Artif. Intell. Commun. Netw.*, 2022, pp. 3–19.
- [341] J. Li, W. Shi, H. Wu, S. Zhang, and X. Shen, "Cost-aware dynamic SFC mapping and scheduling in SDN/NFV-enabled space-air-ground-integrated networks for Internet of Vehicles," *IEEE Internet Things J.*, vol. 9, no. 8, pp. 5824–5838, Apr. 2021.
- [342] Y. Zhang, P. Zhang, C. Jiang, S. Wang, H. Zhang, and C. Rong, "QoS aware virtual network embedding in space-air-ground-ocean integrated network," *IEEE Trans. Services Comput.*, vol. 17, no. 4, pp. 1712–1723, Jul./Aug. 2024.
- [343] L. P. Kaelbling, M. L. Littman, and A. W. Moore, "Reinforcement learning: A survey," *J. Artif. Intell. Res.*, vol. 4, pp. 237–285, May 1996.
- [344] K. Arulkumaran, M. P. Deisenroth, M. Brundage, and A. A. Bharath, "Deep reinforcement learning: A brief survey," *IEEE Signal Process. Mag.*, vol. 34, no. 6, pp. 26–38, Nov. 2017.
- [345] S. Dong, P. Wang, and K. Abbas, "A survey on deep learning and its applications," *Comput. Sci. Rev.*, vol. 40, May 2021, Art. no. 100379.
- [346] S. Pouyanfar et al., "A survey on deep learning: Algorithms, techniques, and applications," *ACM Comput. Surveys*, vol. 51, no. 5, pp. 1–36, 2018.
- [347] P. M. Mammen, "Federated learning: Opportunities and challenges," 2021, *arXiv:2101.05428*.
- [348] L. Li, Y. Fan, M. Tse, and K.-Y. Lin, "A review of applications in federated learning," *Comput. Ind. Eng.*, vol. 149, Nov. 2020, Art. no. 106854.
- [349] C. Zhang, Y. Xie, H. Bai, B. Yu, W. Li, and Y. Gao, "A survey on federated learning," *Knowl.-Based Syst.*, vol. 216, Jun. 2021, Art. no. 106775.
- [350] S. Wu, N. Chen, G. Wen, L. Xu, P. Zhang, and H. Zhu, "Virtual network embedding for task offloading in IIoT: A DRL-assisted federated learning scheme," *IEEE Trans. Ind. Informat.*, vol. 20, no. 4, pp. 6814–6824, Apr. 2024.
- [351] X. Huang, H. Yang, S. Hu, and X. Shen, "Digital twin-driven network architecture for video streaming," *IEEE Netw.*, vol. 38, no. 6, pp. 334–341, Nov. 2024.
- [352] H. Xu, J. Wu, Q. Pan, X. Guan, and M. Guizani, "A survey on digital twin for Industrial Internet of Things: Applications, technologies and tools," *IEEE Commun. Surveys Tuts.*, vol. 25, no. 4, pp. 2569–2598, 4th Quart., 2023.
- [353] D. Jones, C. Snider, A. Nassehi, J. Yon, and B. Hicks, "Characterising the digital twin: A systematic literature review," *CIRP J. Manuf. Sci. Technol.*, vol. 29, pp. 36–52, May 2020.
- [354] F. Tao, H. Zhang, A. Liu, and A. Y. C. Nee, "Digital twin in industry: State-of-the-art," *IEEE Trans. Ind. Informat.*, vol. 15, no. 4, pp. 2405–2415, Apr. 2019.
- [355] D. Kreutz, F. M. Ramos, P. E. Verissimo, C. E. Rothenberg, S. Azodolmolky, and S. Uhlig, "Software-defined networking: A comprehensive survey," *Proc. IEEE*, vol. 103, no. 1, pp. 14–76, Jan. 2015.
- [356] E. Zeydan and Y. Turk, "Recent advances in intent-based networking: A survey," in *Proc. IEEE 91st Veh. Technol. Conf. (VTC-Spring)*, 2020, pp. 1–5.
- [357] J. Ma, "Resource management framework for virtual data center embedding based on software defined networking," *Comput. Elect. Eng.*, vol. 60, pp. 76–89, May 2017.
- [358] A. Bouchair, B. Yagoubi, and S. A. Makhoulouf, "Joint policy for virtual network embedding in distributed SDN-enabled cloud," *J. Netw. Syst. Manag.*, vol. 31, no. 4, p. 69, 2023.
- [359] Y. Han, J. Li, D. Hoang, J.-H. Yoo, and J. W.-K. Hong, "An intent-based network virtualization platform for SDN," in *Proc. 12th Int. Conf. Netw. Service Manage. (CNSM)*, 2016, pp. 353–358.
- [360] J. Ratcliffe, F. Soave, N. Bryan-Kinns, L. Tokarchuk, and I. Farkhatdinov, "Extended reality (XR) remote research: A survey of drawbacks and opportunities," in *Proc. CHI Conf. Human Factors Comput. Syst.*, 2021, pp. 1–13.
- [361] A. Samar and K. M. Sivalingam, "RL-based virtual network embedding using VNF sharing for network slicing in 5G networks," in *Proc. IEEE/IFIP Netw. Oper. Manag. Symp. (NOMS)*, 2023, pp. 1–7.
- [362] I. Ullah, H.-K. Lim, Y.-J. Seok, M. S. Khan, and Y.-H. Han, "Efficient virtual network embedding with hierarchical and cooperative multi-agent reinforcement learning," in *Proc. Int. Conf. Commun. Comput. Digit. Syst. (C-CODE)*, 2023, pp. 1–6.
- [363] D. Thakur and M. Khatua, "Multi-domain virtual network embedding with dynamic flow migration in software-defined networks," *J. Netw. Comput. Appl.*, vol. 162, Jul. 2020, Art. no. 102639.
- [364] *Quantum Computing: Progress and Prospects*. Cambridge, MA, USA: Academic, 2019.
- [365] N. A. Khan and M. A. Qurashi, "Security tradeoff in network virtualization and their countermeasures," in *Proc. ICICIT*, 2023, pp. 741–749.
- [366] P. Rezaeimoghaddam and I. Al-Anbagi, "Cost-efficient and trust-aware virtual network embedding for dense industrial IoT systems using multiagent systems," *IEEE Trans. Netw. Service Manag.*, vol. 21, no. 1, pp. 1100–1114, Feb. 2024.
- [367] F. Liu et al., "Catalyst: Defeating last-level cache side channel attacks in cloud computing," in *Proc. IEEE Int. Symp. High Perform. Comput. Archit. (HPCA)*, 2016, pp. 406–418.
- [368] M. Godfrey and M. Zulkernine, "Preventing cache-based side-channel attacks in a cloud environment," *IEEE Trans. Cloud Comput.*, vol. 2, no. 4, pp. 395–408, Oct.–Dec. 2014.

- [369] Z. Zheng, S. Xie, H.-N. Dai, X. Chen, and H. Wang, "Blockchain challenges and opportunities: A survey," *Int. J. Web Grid Services*, vol. 14, no. 4, pp. 352–375, 2018.
- [370] D. Yaga, P. Mell, N. Roby, and K. Scarfone, "Blockchain technology overview," 2019, *arXiv:1906.11078*.
- [371] M. A. Al-Garadi, A. Mohamed, A. K. Al-Ali, X. Du, I. Ali, and M. Guizani, "A survey of machine and deep learning methods for Internet of Things (IoT) security," *IEEE Commun. Surveys Tuts.*, vol. 22, no. 3, pp. 1646–1685, 3rd Quart., 2020.
- [372] P. Bellavista, L. Foschini, and A. Mora, "Decentralised learning in federated deployment environments: A system-level survey," *ACM Comput. Surveys*, vol. 54, no. 1, pp. 1–38, 2021.



Anurag Satpathy received the Ph.D. degree in computer science and engineering from the National Institute of Technology at Rourkela, Rourkela, India. He is a Postdoctoral Fellow with the Department of Computer Science, Missouri University of Science and Technology, Rolla, MO, USA. His research findings have been published in prestigious conferences, such as IEEE ICWS, IEEE ICC, IEEE Cloud, IEEE SCC, IEEE SMARTCOMP, and IEEE LCN, as well as in renowned journals, including IEEE TRANSACTIONS ON CONSUMER ELECTRONICS,

IEEE TRANSACTIONS ON SERVICES COMPUTING, IEEE INTERNET OF THINGS JOURNAL, IEEE TRANSACTIONS ON CLOUD COMPUTING, IEEE TRANSACTIONS ON GREEN COMMUNICATIONS AND NETWORKING, and *Computer Networks*. His research interests encompass cloud computing, the Internet of Things, and distributed systems. His publications at IEEE Cloud and IEEE COMSNETS have received the YESC Student Paper Participation Award and the Best Poster Award, respectively. He serves as a TPC Member of top-tier conferences and workshops, including IEEE PerCom, IEEE/ACM ICDCN, and IEEE COMSNETS. He also serves as a Reviewer for high-quality peer-reviewed journals like IEEE/ACM TRANSACTIONS ON NETWORKING, IEEE TRANSACTIONS ON SERVICES COMPUTING, IEEE TRANSACTIONS ON MOBILE COMPUTING, *ACM Transactions on Sensor Networks*, *Future Generation Computer Systems*, *Expert Systems with Applications*, *Computer Networks* (Elsevier), *The Journal of Supercomputing*, and *Cluster Computing* (Springer).



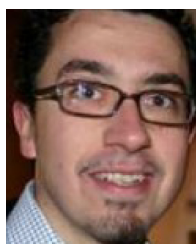
Manmath Narayan Sahoo (Member, IEEE) received the Ph.D. degree in computer science from the National Institute of Technology at Rourkela, Rourkela, India, in 2014, where he is an Associate Professor with the Department of Computer Science and Engineering. He has published his research outcomes in many peer-reviewed journals and flagship conferences, including IEEE ICWS, IEEE ICC, IEEE Cloud, IEEE TRANSACTIONS ON SERVICES COMPUTING, IEEE INTERNET OF THINGS JOURNAL, and *Computer Networks*. His

research interests include mobile ad hoc networks, sensor networks, and cloud computing. He is also an active reviewer, a guest editor, the track chair, and the program chair in many journals and conferences of international repute.



Chittaranjan Swain received the Ph.D. degree from the Department of Computer Science and Engineering, National Institute of Technology at Rourkela, Rourkela, India. He is currently working as an Assistant Professor with the Department of Computer Science and Engineering, Atal Bihari Vajpayee Indian Institute of Information Technology and Management, Gwalior. Before this, he worked as an Assistant Professor with the Department of Computer Science and Engineering, Siksha 'O' Anusandhan, Bhubaneswar and Kalasalingam

University, India. His current research includes developing efficient resource allocation strategies for real-time services in IoT-Fog networks. Some of his research findings have been published in reputed conferences (IEEE ICWS, IEEE ICC, IEEE Cloud) and journals [IEEE TRANSACTIONS ON CONSUMER ELECTRONICS, IEEE TRANSACTIONS ON SERVICES COMPUTING, IEEE INTERNET OF THINGS JOURNAL, IEEE TRANSACTIONS ON GREEN COMMUNICATIONS AND NETWORKING, and *Computer Networks* (Elsevier)]. He is a Reviewer of IEEE TRANSACTIONS ON MOBILE COMPUTING, IEEE INTERNET OF THINGS JOURNAL, *Journal of Super Computing* (Springer), *Journal of Cluster Computing* (Springer), *Cloud Computing* (Springer), *Telecommunication Systems* (Springer), and *Automatica* (Talyor and Francis). His research interests include fog computing, edge computing, cloud computing, networking, Internet of Things, and distributed systems.



Paolo Bellavista (Senior Member, IEEE) received the Ph.D. degree in computer science engineering from the University of Bologna, Italy, in 2001, where he is currently a Full Professor. He served as the Scientific Coordinator for the H2020 IoTwins Project. His research interests include middleware for mobile computing, QoS management in the cloud continuum, infrastructures for big data processing in industrial environments, and performance optimization in wide-scale and latency-sensitive deployment environments. He serves on the editorial

boards for IEEE COMMUNICATIONS SURVEYS AND TUTORIALS, *ACM Computing Surveys*, IEEE TRANSACTIONS ON NETWORK AND SERVICE MANAGEMENT, ACM TRANSACTIONS ON INTERNET TECHNOLOGY, and *Pervasive and Mobile Computing* (Elsevier).



Mohsen Guizani (Fellow, IEEE) received the B.S. (with distinction), M.S., and Ph.D. degrees in electrical and computer engineering from Syracuse University, Syracuse, NY, USA. He is currently a Professor of Machine Learning with the Mohamed Bin Zayed University of Artificial Intelligence, Abu Dhabi, UAE. Previously, he worked in different institutions in USA. He is the author of 11 books, more than 1000 publications and several U.S. patents. His research interests include applied machine learning and artificial intelligence, smart city, Internet of

Things, intelligent autonomous systems, and cybersecurity. He has won several research awards, including the "2015 IEEE Communications Society Best Survey Paper Award," the Best ComSoc Journal Paper Award in 2021 as well as five Best Paper Awards from ICC and Globecom Conferences. He is also the recipient of the 2017 IEEE Communications Society Wireless Technical Committee Recognition Award, the 2018 AdHoc Technical Committee Recognition Award, and the 2019 IEEE Communications and Information Security Technical Recognition Award. He was listed as a Clarivate Analytics Highly Cited Researcher in Computer Science from 2019 to 2022. He served as the Editor-in-Chief of IEEE NETWORK and is currently serving on the editorial boards of many IEEE Transactions and Magazines. He was the Chair of the IEEE Communications Society Wireless Technical Committee and the Chair of the TAOS Technical Committee. He served as the IEEE Computer Society Distinguished Speaker and is currently the IEEE ComSoc Distinguished Lecturer.



Khan Muhammad (Senior Member, IEEE) received the Ph.D. degree in digital content from Sejong University, Republic of Korea, in February 2019, where he served as a Non-Tenure-Track Assistant Professor with the Department of Software from March 2019 to February 2022. From February 2022 to August 2024, he was a Tenure-Track Assistant Professor with the Department of Applied AI, School of Convergence, College of Computing and Informatics, Sungkyunkwan University, Seoul, where he is currently an Associate Professor and the Director of the Visual Analytics for Knowledge Laboratory. He holds ten patents and has authored over 250 papers in peer-reviewed journals and conferences. His research interests include intelligent video surveillance, medical image analysis, information security, and multimedia data analysis. He serves on the editorial boards of over ten journals and has been consecutively recognized as a Highly Cited Researcher by Web of Science (Clarivate) from 2021 to 2024.



Sambit Bakshi (Senior Member, IEEE) is currently with the Department of Computer Science and Engineering, National Institute of Technology Rourkela, India, where he also serves as the Head of the Centre for Automation Technology. He has published widely in more than 100 journals and conferences. His area of interest includes surveillance, biometric security, digital forensics, and social security analytics. He currently serves as an Associate Editor of multiple journals including, IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY, IEEE SIGNAL PROCESSING LETTERS, IEEE TRANSACTIONS ON CONSUMER ELECTRONICS, IEEE TRANSACTIONS ON AUTOMATION SCIENCE AND ENGINEERING, *Computers and Electrical Engineering*, IEEE TRANSACTIONS ON COMPUTATIONAL SOCIAL SYSTEMS, IEEE TRANSACTIONS ON TECHNOLOGY AND SOCIETY, and IEEE TRANSACTIONS ON EDUCATION. He previously served as an Associate Editor for IEEE SYSTEMS JOURNAL, IEEE IT Professional Magazine, IEEE ACCESS, *Innovations in Systems and Software Engineering Springer: A NASA Journal*, *Expert Systems with Applications*, *Engineering Applications of Artificial Intelligence*, *Image and Vision Computing*, *Multimedia Systems*, *Multimedia Tools and Applications*, *Neural Processing Letters*, *PLoS ONE*, and *Expert Systems* in the past. He is the Founding Chair of IEEE Rourkela Subsection and IEEE Kolkata Section Biometrics Council Chapter. He is a Distinguished Lecturer of IEEE Systems Council from 2023 to 2025 and a Distinguished Lecturer of IEEE Women in Engineering from 2024 to 2025.